Concepts

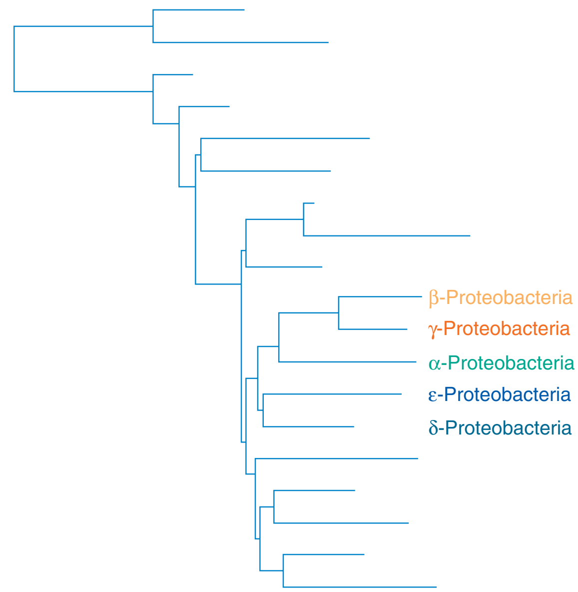
1. The proteobacteria of the second edition of *Bergey's Manual* come from volumes 1 and 3 of the first edition. In the first edition bacteria are placed in a particular section based on a few major phenotypic properties such as general shape, nutritional type, motility, oxygen relationships, and so forth. The second edition uses nucleic acid sequences, particularly 16S rRNA sequence comparisons, to place bacteria in phylogenetic groupings.
2. Many of these gram-negative bacteria are of considerable importance, either as disease agents or because of their effects on the habitat. Others, such as *E. coli,* are major experimental organisms studied in many laboratories.
3. Although many of these bacteria do not vary drastically in general appearance, they often are very diverse in their metabolism and life-styles, which range from obligately intracellular parasitism to a free-living existence in soil and aquatic habitats.
4. Bacteria do not always have simple, unsophisticated morphology but may produce prosthecae, stalks, buds, sheaths, or complex fruiting bodies.
5. Chemolithotrophic bacteria obtain energy and electrons by oxidizing inorganic compounds rather than the organic nutrients employed by most bacteria. They often have substantial ecological impact because of their ability to oxidize many forms of inorganic nitrogen and sulfur.
6. Many bacteria that specialize in predatory or parasitic modes of existence, such as *Bdellovibrio* and the rickettsias, have relinquished some of their metabolic independence through the loss of metabolic pathways. They depend on the prey's or host's energy supply and/or cell constituents.

Microbes is a vigitable, an' ivry man is like a conservatory full iv millyons iv these potted plants.

—*Finley Peter Dunne*

Chapters 20 and 21 described many of the groups found in volumes 1 and 5 of the second edition of *Bergey's Manual.* Chapter 20 was devoted to the *Archaea* and chapter 21 surveyed the bacterial groups described in volumes 1 and 5. This chapter introduces the bacteria that are covered in volume 2 of the second edition of *Bergey's Manual.* These gram-negative bacteria are located in volumes 1 and 3 of the first edition. The general organization of the second edition will be followed here. This chapter depicts the major biological features of each group and a few selected representative forms of particular interest.

Volume 2 of the second edition of *Bergey's Manual* is devoted entirely to the **proteobacteria,** sometimes called the purple bacteria because of the purple photosynthetic bacteria scattered through several of its subgroups. The proteobacteria are the largest and most diverse group of bacteria; currently there are over 380 genera and 1,300 species. Although the 16S rRNA studies show that they are phylogenetically related, proteobacteria vary markedly in many respects. The morphology of these gram-negative bacteria ranges from simple rods and cocci to genera with prosthecae, buds, and even fruiting bodies. Physiologically these bacteria are just as diverse. Photoautotrophs, chemolithotrophs, and chemoheterotrophs are all well represented. There is no obvious overall pattern in metabolism, morphology, or reproductive strategy of the proteobacteria.



Comparison of 16S rRNA sequences has revealed what appears to be five subgroups of proteobacteria (see the diagram at the top of this page). The second edition of *Bergey's Manual* places all these procaryotes in the phylum *Proteobacteria,* which has five classes: *Alphaproteobacteria, Betaproteobacteria, Gammaproteobacteria, Deltaproteobacteria,* and *Epsilonproteobacteria.* Members of the purple photosynthetic bacteria are found among the α-, β-, and γ-proteobacteria. This has led to the proposal that the proteobacteria arose from a photosynthetic ancestor, presumably similar to the purple bacteria. Subsequently photosynthesis would have been lost by various lines, and new metabolic capacities would have been acquired as these bacteria adapted to different ecological niches.

22.1 Class *Alphaproteobacteria*

The **α-proteobacteria** include most of the oligotrophic proteobacteria (those capable of growing at low nutrient levels). Some have unusual metabolic modes such as methylotrophy (*Methylobacterium*), chemolithotrophy (*Nitrobacter*), and the ability to fix nitrogen (*Rhizobium*). Members of genera such as *Rickettsia* and *Brucella* are important pathogens; in fact, *Rickettsia* has become an obligately intracellular parasite. Many genera are characterized by distinctive morphology such as prosthecae.

The class *Alphaproteobacteria* has six orders and 18 families. **Figure 22.1** illustrates the phylogenetic relationships between some typical α-proteobacteria, and table 22.1 summarizes the general characteristics of many of the bacteria discussed in the following sections.

The Purple Nonsulfur Bacteria

All the purple bacteria use anoxygenic photosynthesis, possess bacteriochlorophylls *a* or *b,* and have their photosynthetic apparatus in membrane systems that are continuous with the plasma membrane.

Most are motile by polar flagella. With the exception of *Rhodocyclus* (β-proteobacteria), the purple nonsulfur bacteria are located among the α-proteobacteria. Photosynthetic Bacteria (pp. 468–76)

The **purple nonsulfur bacteria** are exceptionally flexible in their choice of an energy source. Normally they grow anaerobically as photoorganoheterotrophs; they trap light energy and employ organic molecules as both electron and carbon sources (*see table 21.1*).

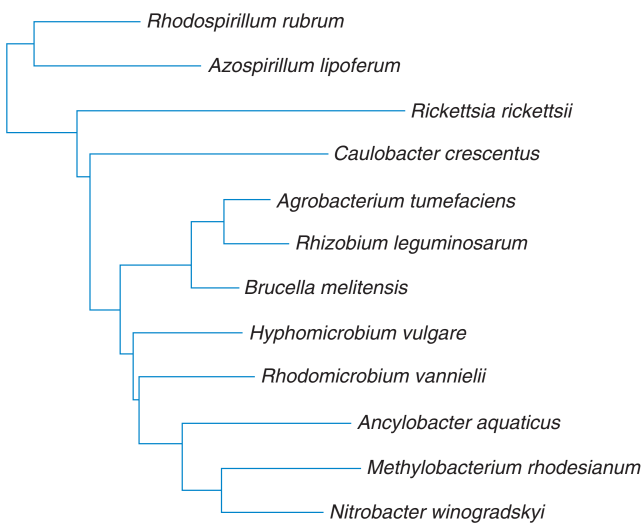


Figure 22.1 Phylogenetic Relationships Among Selected α-Proteobacteria. The relationships of a few species based on 16S rRNA sequence data are shown here. *Source: The Ribosomal Database Project.*

Although they are called nonsulfur bacteria, some species can oxidize very low, nontoxic levels of sulfide to sulfate, but they do not oxidize elemental sulfur to sulfate. In the absence of light, most purple nonsulfur bacteria can grow aerobically as chemoorganoheterotrophs, but some species carry out fermentations and grow anaerobically. Oxygen inhibits bacteriochlorophyll and carotenoid synthesis so that cultures growing aerobically in the dark are colorless.

Purple nonsulfur bacteria vary considerably in morphology (**figure 22.2**). They may be spirals (*Rhodospirillum*), rods (*Rhodopseudomonas*), half circles or circles (*Rhodocyclus*), or they may even form prosthecae and buds (*Rhodomicrobium*). Because of their metabolism, they are most prevalent in the mud and water of lakes and ponds with abundant organic matter and low sulfide levels. There also are marine species.

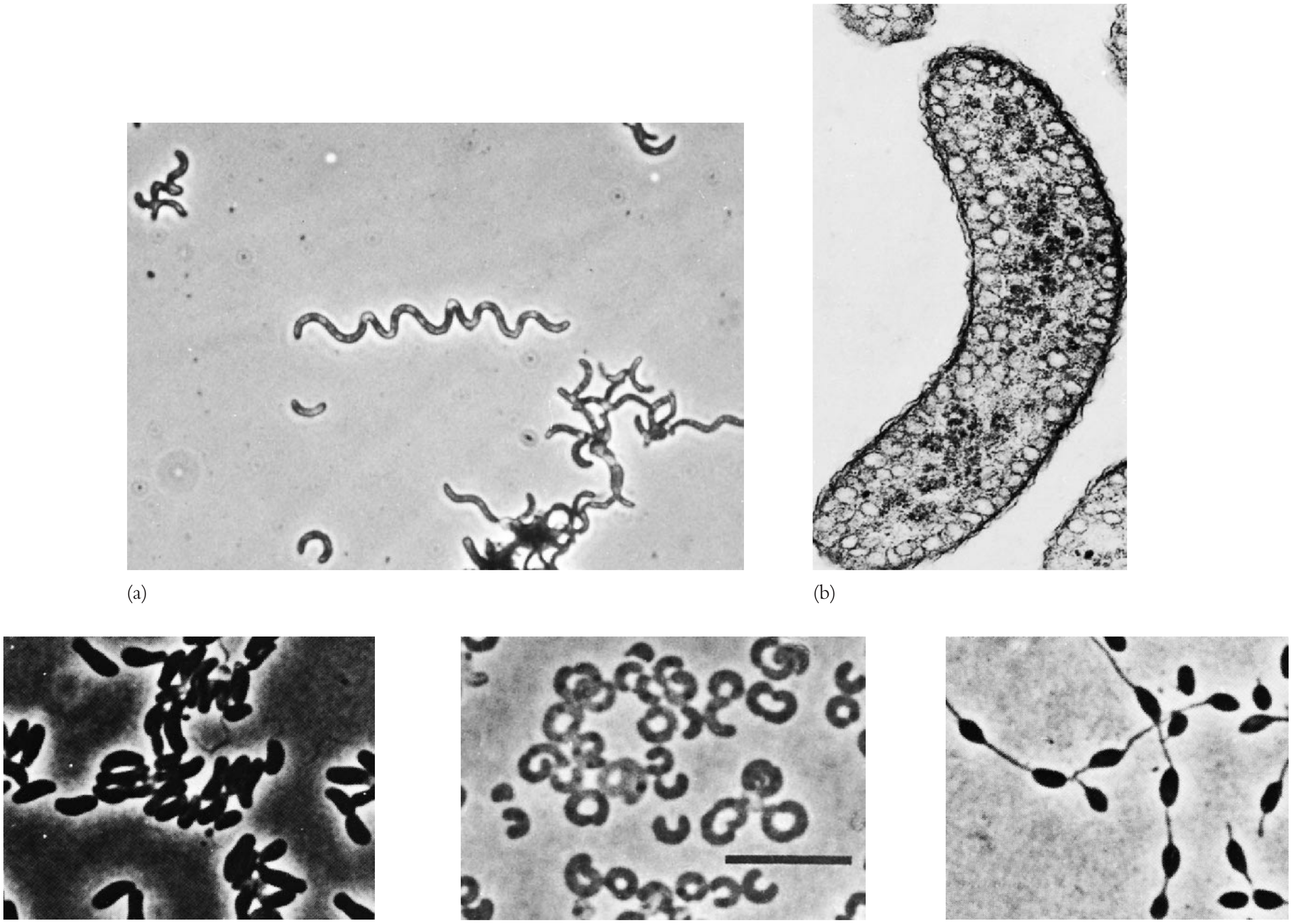
Rickettsia and *Coxiella*

In the second edition of *Bergey's Manual,* the genus *Rickettsia* will be located in the order *Rickettsiales* and family *Rickettsiaceae* of the α-proteobacteria, whereas *Coxiella* will be in the order *Legionellales* and family *Coxiellaceae* of the γ-proteobacteria. The first edition places these two genera and other bacteria that grow intracellularly in a separate section, and we also will discuss *Rickettsia* and *Coxiella* together because of their similarity in life-style, despite their apparent phylogenetic distance.

These bacteria are rod-shaped, coccoid, or pleomorphic with typical gram-negative walls and no flagella. Although their size varies, they tend to be very small. For example, *Rickettsia* is 0.3 to 0.5 μm in diameter and 0.8 to 2.0 μm long; *Coxiella* is 0.2 to 0.4 μm by 0.4 to 1.0 μm. All species are parasitic or mutualistic. The parasitic forms grow in vertebrate erythrocytes, macrophages (*see figure 31.4*), and vascular endothelial cells. Often they also live in

Table 22.1 Characteristics of Selected α-Proteobacteria

Genus	Dimensions (μm) and Morphology	G + C Content (mol%)	Oxygen Requirement	Other Distinctive Characteristics
Agrobacterium	0.6–1.0 × 1.5–3.0; motile, nonsporing rods with peritrichous flagella	57–63	Aerobic	Chemoorganotroph that can invade plants and cause tumors
Caulobacter	0.4–0.6 × 1–2; rod- or vibrioid-shaped with a flagellum and prostheca and holdfast	62–67	Aerobic	Heterotrophic and oligotrophic; asymmetric cell division
Hyphomicrobium	0.3–1.2 × 1–3; rod-shaped or oval with polar prosthecae	59–65	Aerobic	Reproduces by budding; methylotrophic
Nitrobacter	0.5–0.8 × 1.0–2.0; rod- or pear-shaped, sometimes motile by flagella	60–62	Aerobic	Chemolithotroph, oxidizes nitrite to nitrate
Rhizobium	0.5–0.9 × 1.2–3.0; motile rods with flagella	59–64	Aerobic	Invades leguminous plants to produce nitrogen-fixing root nodules
Rhodospirillum	0.7–1.5 wide; spiral cells with polar flagella	62–64	Anaerobic, microaerobic, aerobic	Photoheterotroph under anaerobic conditions
Rickettsia	0.3–0.5 × 0.8–2.0; short nonmotile rods	29–33	Aerobic	Obligately intracellular parasite



(a) (b) (c) (d) (e)

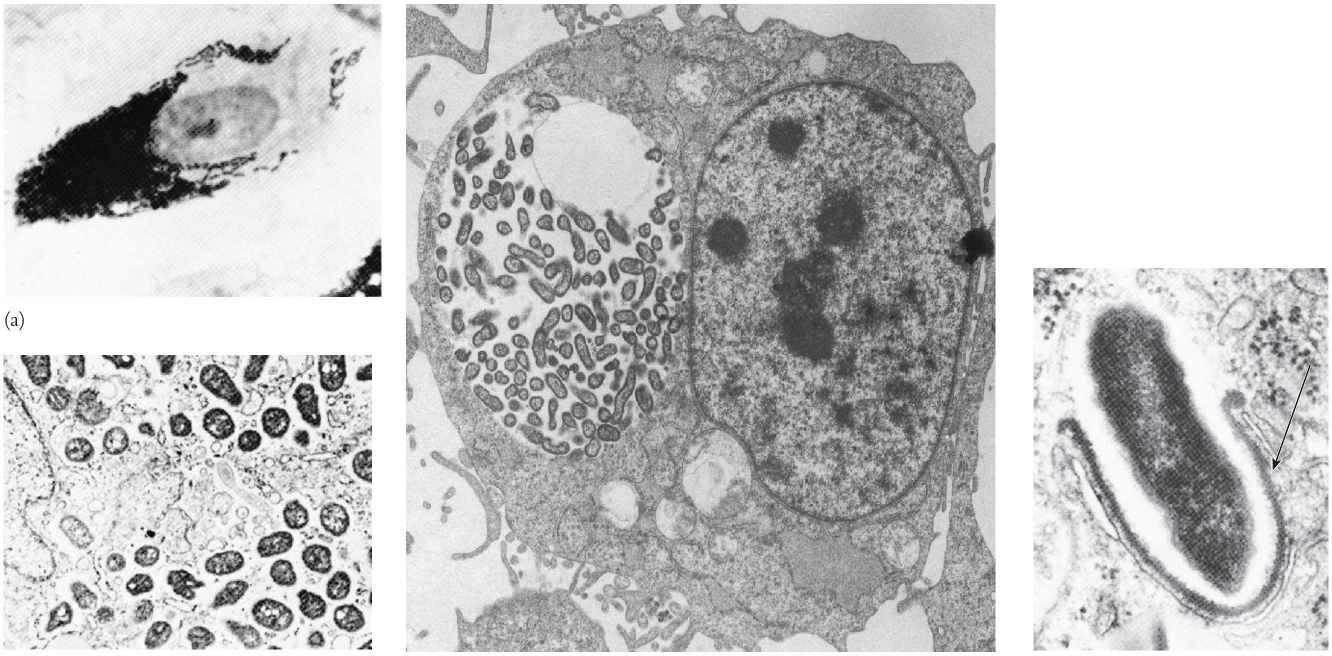
Figure 22.2 Typical Purple Nonsulfur Bacteria. (**a**) *Rhodospirillum rubrum;* phase contrast (×410). (**b**) *R. rubrum* grown anaerobically in the light. Note vesicular invaginations of the cytoplasmic membranes; transmission electron micrograph (×51,000). (**c**) *Rhodopseudomonas acidophila;* phase contrast. (**d**) *Rhodocyclus purpureus;* phase contrast. Bar = 10 μm. (**e**) *Rhodomicrobium vannielii* with vegetative cells and buds; phase contrast.

blood-sucking arthropods such as fleas, ticks, mites, or lice, which serve as vectors (*see section 37.6*) or primary hosts.

Because these genera contain important human pathogens, their reproduction and metabolism have been intensively studied. Rickettsias enter the host cell by inducing phagocytosis. Members of the genus *Rickettsia* immediately escape the phagosome and reproduce by binary fission in the cytoplasm (**figure 22.3**). In contrast, *Coxiella* remains within the phagosome after it has fused with a lysosome and actually reproduces within the phagolysosome. Eventually the host cell bursts, releasing new organisms. Besides incurring damage from cell lysis, the host is harmed by the toxic effects of rickettsial cell walls (wall toxicity appears related to the mechanism of penetration into host cells).

Rickettsias are very different from most other bacteria in physiology and metabolism. They lack the glycolytic pathway and do not use glucose as an energy source, but rather oxidize glutamate and tricarboxylic acid cycle intermediates such as succinate. The rickettsial plasma membrane has carrier-mediated transport systems, and host cell nutrients and coenzymes are absorbed and directly used. For example, rickettsias take up both NAD and uridine diphosphate glucose. Their membrane also has an adenylate exchange carrier that exchanges ADP for external ATP. Thus host ATP may provide much of the energy needed for growth. This metabolic dependence explains why many of these organisms must be cultivated in the yolk sacs of chick embryos or in tissue culture cells. Results from genome sequencing show that *R. prowazekii* is similar in many ways to mitochondria (*see p. 351*). Possibly mitochondria arose from an endosymbiotic association with an ancestor of *Rickettsia.*

This order contains many important pathogens. *Rickettsia prowazekii* and *R. typhi* are associated with typhus fever, and *R. rickettsii,* with Rocky Mountain spotted fever. *Coxiella burnetii* causes Q fever in humans. These diseases are discussed later in some detail (*see chapter 39*). It should be noted that rickettsias are



(a) (b) (c) (d)

Figure 22.3 *Rickettsia* and *Coxiella*. Rickettsial morphology and reproduction. (**a**) A human fibroblast filled with *Rickettsia prowazekii* (×1,200). (**b**) A chicken embryo fibroblast late in infection with free cytoplasmic *R. prowazekii* (×13,600). (**c**) *Coxiella burnetti* growing within fibroblast vacuoles (×9,000). (**d**) *R. prowazekii* leaving a disrupted phagosome (arrow) and entering the cytoplasmic matrix (×46,000).

also important pathogens of domestic animals such as dogs, horses, sheep, and cattle.

The *Caulobacteraceae* and *Hyphomicrobiaceae*

A number of the proteobacteria are not simple rods or cocci but have some sort of appendage. These bacteria can have at least one of three different features: a prostheca, a stalk, or reproduction by budding. A **prostheca** (pl., prosthecae) is an extension of the cell, including the plasma membrane and cell wall, that is narrower than the mature cell. A **stalk** is a nonliving appendage produced by the cell and extending from it (*see figure 3.2f*). **Budding** is distinctly different from the **binary fission** normally used by bacteria (*see p. 223*). The bud first appears as a small protrusion at a single point and enlarges to form a mature cell. Most or all of the bud's cell envelope is newly synthesized. In contrast, portions of the parental cell envelope are shared with the progeny cells during binary fission. Finally, the parental cell retains its identity during budding, and the new cell is often smaller than its parent. In binary fission the parental cell disappears as it forms the progeny. The families *Caulobacteraceae* and *Hyphomicrobiaceae* of the α-proteobacteria contain two of the best studied of prosthecate genera: *Caulobacter* and *Hyphomicrobium.*

The genus *Hyphomicrobium* contains chemoheterotrophic, aerobic, budding bacteria that frequently attach to solid objects in freshwater, marine, and terrestrial environments. (They even grow in laboratory water baths.) The vegetative cell measures about 0.5 to 1.0 by 1 to 3 μm (**figure 22.4**). At the beginning of the reproductive cycle, the mature cell produces a hypha or prostheca, 0.2 to 0.3 μm in diameter, that grows to several μm in length (**figure 22.5**). The nucleoid divides, and a copy moves into the hypha while a bud forms at its end. As the bud matures, it produces one to three flagella, and a septum divides the bud from the hypha. The bud is finally released as an oval- to pear-shaped swarmer cell, which

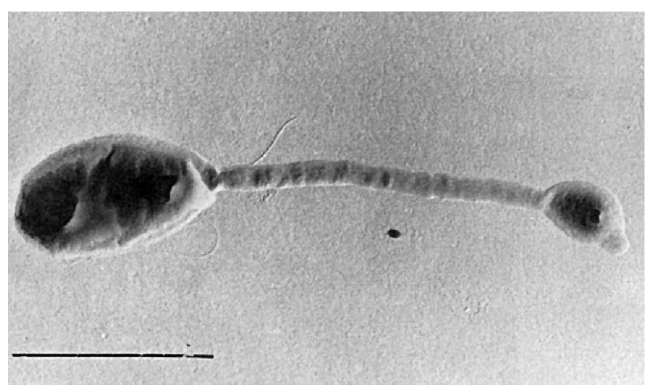


Figure 22.4 Prosthecate, Budding Bacteria. *Hyphomicrobium* morphology. *Hyphomicrobium facilis* with hypha and young bud. Bar = 1 μm.

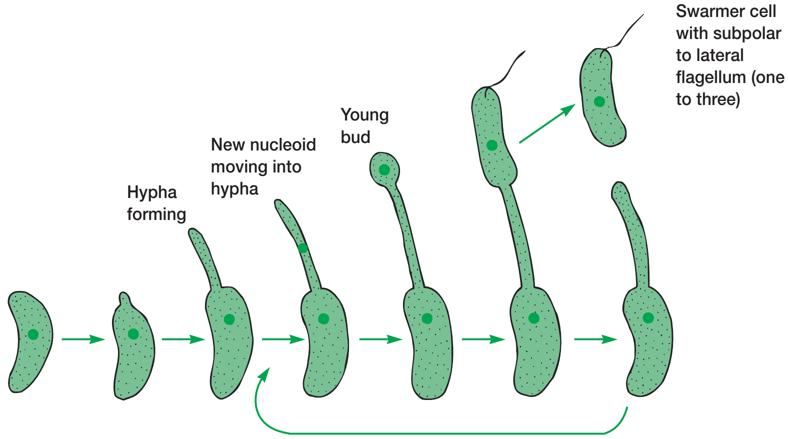


Figure 22.5 The Life Cycle of *Hyphomicrobium*. See text for details.

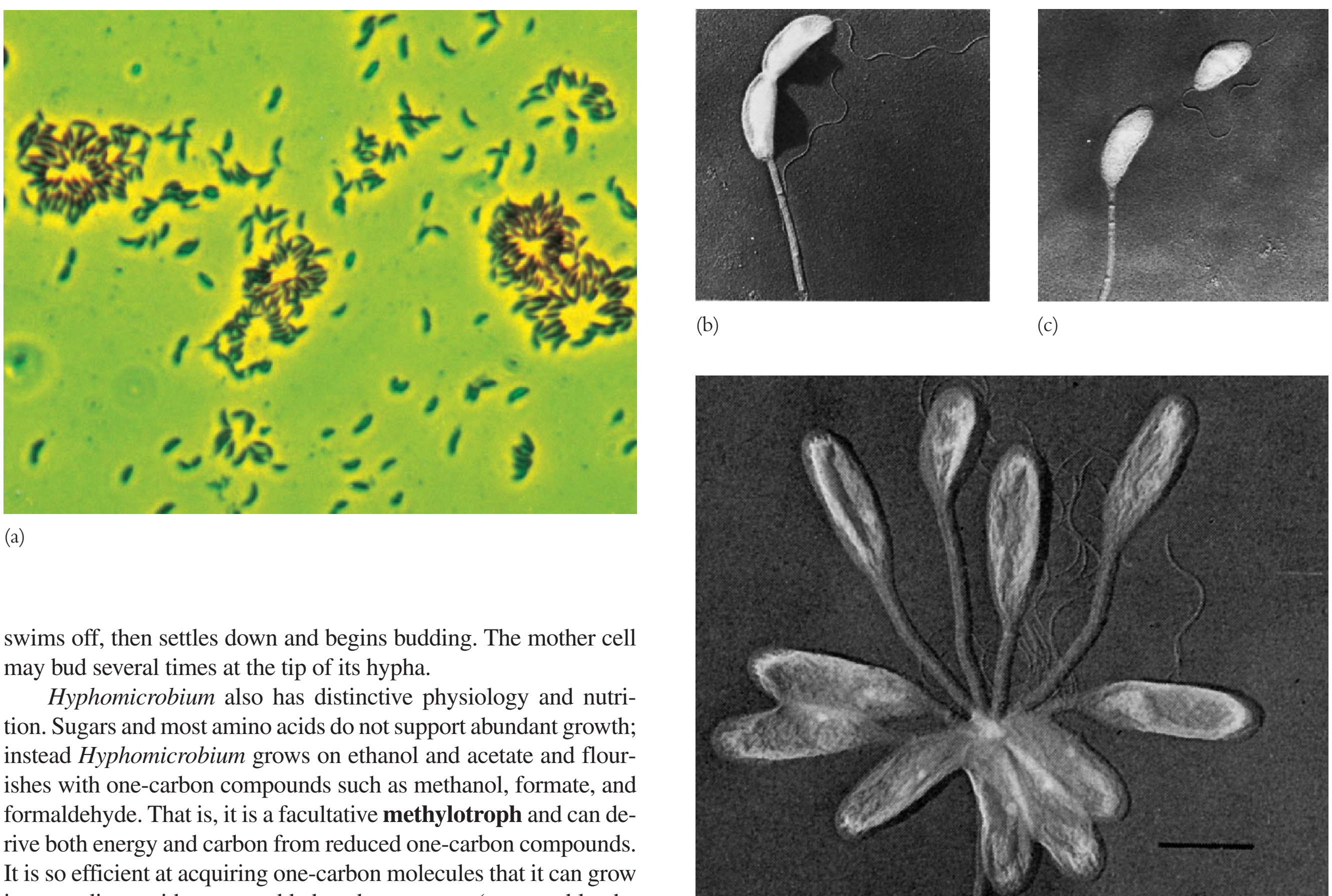

(a)
(b)
(c)
(d)

Figure 22.6 *Caulobacter* Morphology and Reproduction. (**a**) Rosettes of cells adhering to each other by their prosthecae; phase contrast (×600). (**b**) A cell dividing to produce a swarmer (×6,030). Note prostheca and flagellum. (**c**) A cell with a flagellated swarmer (×6,030). (**d**) A rosette as seen in the electron microscope. Bar = 1 μm.

swims off, then settles down and begins budding. The mother cell may bud several times at the tip of its hypha.

Hyphomicrobium also has distinctive physiology and nutrition. Sugars and most amino acids do not support abundant growth; instead *Hyphomicrobium* grows on ethanol and acetate and flourishes with one-carbon compounds such as methanol, formate, and formaldehyde. That is, it is a facultative **methylotroph** and can derive both energy and carbon from reduced one-carbon compounds. It is so efficient at acquiring one-carbon molecules that it can grow in a medium without an added carbon source (presumably the medium absorbs sufficient atmospheric carbon compounds). *Hyphomicrobium* may comprise up to 25% of the total bacterial population in oligotrophic or nutrient-poor freshwater habitats.

Bacteria in the genus *Caulobacter* may be polarly flagellated rods or possess a prostheca and **holdfast,** by which they attach to solid substrata (**figure 22.6**). Caulobacters are usually isolated

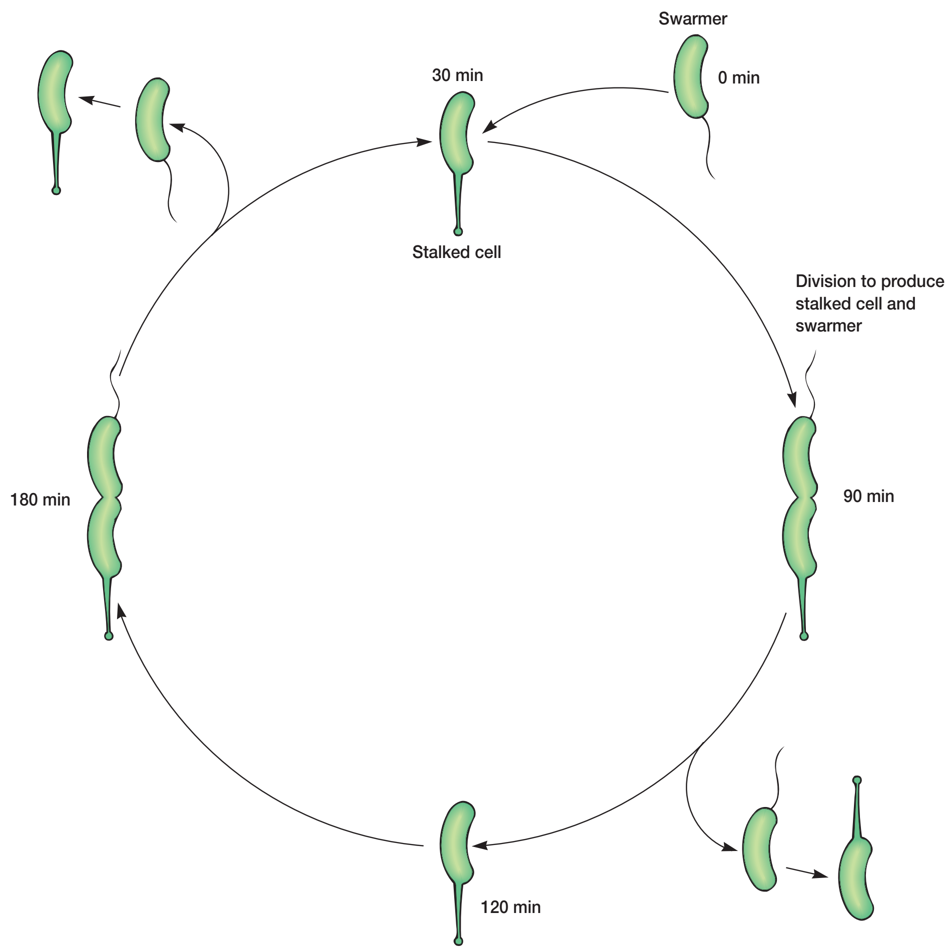


Figure 22.7 ***Caulobacter* Life Cycle.** A cycle starting with a swarmer cell at zero minutes is pictured. The cycle timing is for *Caulobacter* growing at 30°C in glucose minimal medium. See text for details.

from freshwater and marine habitats with low nutrient levels, but they also are present in the soil. They often adhere to bacteria, algae, and other microorganisms and may absorb nutrients released by their hosts. The prostheca differs from that of *Hyphomicrobium* in that it lacks cytoplasmic components and is composed almost totally of the plasma membrane and cell wall. It grows longer in nutrient-poor media and can reach more than 10 times the length of the cell body. The prostheca may improve the efficiency of nutrient uptake from dilute habitats by increasing surface area; it also gives the cell extra buoyancy.

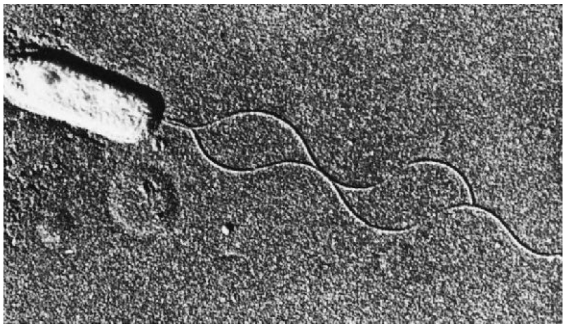
The life cycle of *Caulobacter* is unusual (**figure 22.7**). When ready to reproduce, the cell elongates and a single polar flagellum forms at the end opposite the prostheca. The cell then undergoes asymmetric transverse binary fission to produce a flagellated swarmer cell that swims off. The swarmer, which cannot reproduce, eventually comes to rest and forms a new prostheca on the flagellated end while its flagellum disappears. The new stalked form then begins to form swarmers. This cycle takes only about two hours for completion.

Family *Rhizobiaceae*

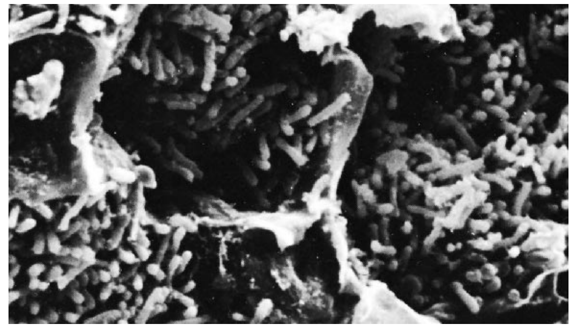
In the second edition of *Bergey's Manual,* the order *Rhizobiales* of the α-proteobacteria will contain 10 families with a great variety of phenotypes. The family *Hyphomicrobiaceae* has already been discussed. The first family in this order is *Rhizobiaceae,* in which are located the gram-negative, aerobic genera *Rhizobium* and *Agrobacterium.*

Members of the genus *Rhizobium* are 0.5 to 0.9 by 1.2 to 3.0 μm motile rods, often containing poly-β-hydroxybutyrate granules, that become pleomorphic under adverse conditions (**figure 22.8**). They grow symbiotically within root nodule cells of legumes as nitrogen-fixing bacteroids (figure 22.8*b*). In contrast, *Azotobacter* is a free-living soil genus and fixes atmospheric nitrogen nonsymbiotically (p. 504). The biochemistry of nitrogen fixation (pp. 212–14); The Rhizobium-legume symbiosis (pp. 675–78)

The genus *Agrobacterium* is placed in the family *Rhizobiaceae* but differs from *Rhizobium* in not stimulating root nodule formation or fixing nitrogen. Instead agrobacteria invade the crown, roots, and



(a)



(b)

Figure 22.8 ***Rhizobium.*** (**a**) *Rhizobium leguminosarum* with two polar flagella (×14,000). (**b**) Scanning electron micrograph of bacteroids in alfalfa root (×640).

stems of many plants and transform plant cells into autonomously proliferating tumor cells. The best-studied species is *A. tumefaciens,* which enters many broad-leaved plants through wounds and causes crown gall disease (**figure 22.9**). The ability to produce tumors is dependent on the presence of a large Ti (for tumor-inducing) plasmid. Tumor production by *Agrobacterium* is discussed in greater detail in Box 14.2 and section 30.4. Plasmids (pp. 294–97)

Figure 22.9 ***Agrobacterium.*** Crown gall tumor of a tomato plant caused by *Agrobacterium tumefaciens.*

Nitrifying Bacteria

Section 20 of the first edition of *Bergey's Manual* is devoted to aerobic chemolithotrophic bacteria, those bacteria that derive energy and electrons from reduced inorganic compounds. Normally they employ CO_2 as their carbon source and thus chemolithoautotrophs, but some can function as chemolithoheterotrophs and use reduced organic carbon sources. These bacteria are divided into three groups based on the inorganic compounds they prefer to oxidize: nitrifiers, colorless sulfur bacteria (sulfur oxidizers), and iron- and manganese-oxidizing bacteria. In the second edition, the chemolithotrophs are distributed between the α-, β-, and γ-proteobacteria. The nitrifying bacteria are α-, β-, and γ-proteobacteria; the sulfur-oxidizing bacteria are in the β and γ subgroups. Nitrifying bacteria will be discussed here. The colorless sulfur-oxidizing bacteria are covered in the section on β-proteobacteria. Chemolithotrophic metabolism is described in chapter 9, and the focus here is on the biology of these organisms. Their ecological impact is discussed further in the context of microbial ecology. The biochemistry of chemolithotrophy (pp. 193–94)

The **nitrifying bacteria** are a very diverse collection of bacteria. The second edition of *Bergey's Manual* places nitrifying genera in three classes and several families: *Nitrobacter* in the *Bradyrhizobiaceae,* α-proteobacteria; *Nitrosomonas* and *Nitrosospira* in the *Nitrosomonadaceae,* β-proteobacteria; *Nitrococcus* in the *Ectothiorhodospiraceae,* γ-proteobacteria; and *Nitrosococcus* in the *Chromatiaceae,* γ-proteobacteria. Although all are aerobic, gram-negative organisms without endospores and able to oxidize either ammonia or nitrite, they differ considerably in other properties (**table 22.2**). Nitrifiers may be rod-shaped, ellipsoidal, spherical, spirillar or lobate, and they may possess either polar or peritrichous flagella (**figure 22.10**). Often they have extensive membrane complexes lying in their cytoplasm. Identification is based on properties such as their preference for nitrite or ammonia, their general shape, and the nature of any cytomembranes present.

Table 22.2 Selected Characteristics of Representative Nitrifying Bacteria

Species	Cell Morphology and Size (μm)	Reproduction	Motility	Cytomembranes	G + C (Mol%)	Habitat
Ammonia-Oxidizing Bacteria						
Nitrosomonas europaea	Rod; 0.8–1.0 × 1.0–2.0	Binary fission	+ or –; 1 or 2 subpolar flagella	Peripheral, lamellar	47.4–51.0	Soil, sewage, freshwater, marine
Nitrosococcus oceanus	Coccoid; 1.8–2.2 in diameter	Binary fission	+; 1 or more subpolar flagella	Centrally located parallel bundle, lamellar	50.5–51.0	Obligately marine
Nitrosospira briensis	Spiral; 0.3–0.4 in diameter	Binary fission	+ or –; 1 to 6 peritrichous flagella	Lacking	54.1 (1 strain)	Soil
Nitrite-Oxidizing Bacteria						
Nitrobacter winogradskyi	Rod; 0.6–0.8 × 1.0–2.0	Budding	+ or –; 1 polar flagellum	Polar cap of flattened vesicles in peripheral region of the cell	60.7–61.7	Soil, freshwater, marine
Nitrococcus mobilis	Coccoid; 1.5–1.8 in diameter	Binary fission	+; 1 or 2 subpolar flagella	Tubular cytomembranes randomly arranged throughout cytoplasm	61.2 (1 strain)	Marine

From S. W. Watson, et al., "The Family Nitrobacteraceae" in *The Prokaryotes,* Vol. 1, edited by M. P. Starr et al., 1981. Copyright © 1981 Springer-Verlag New York, Inc., New York, NY. Reprinted by permission.

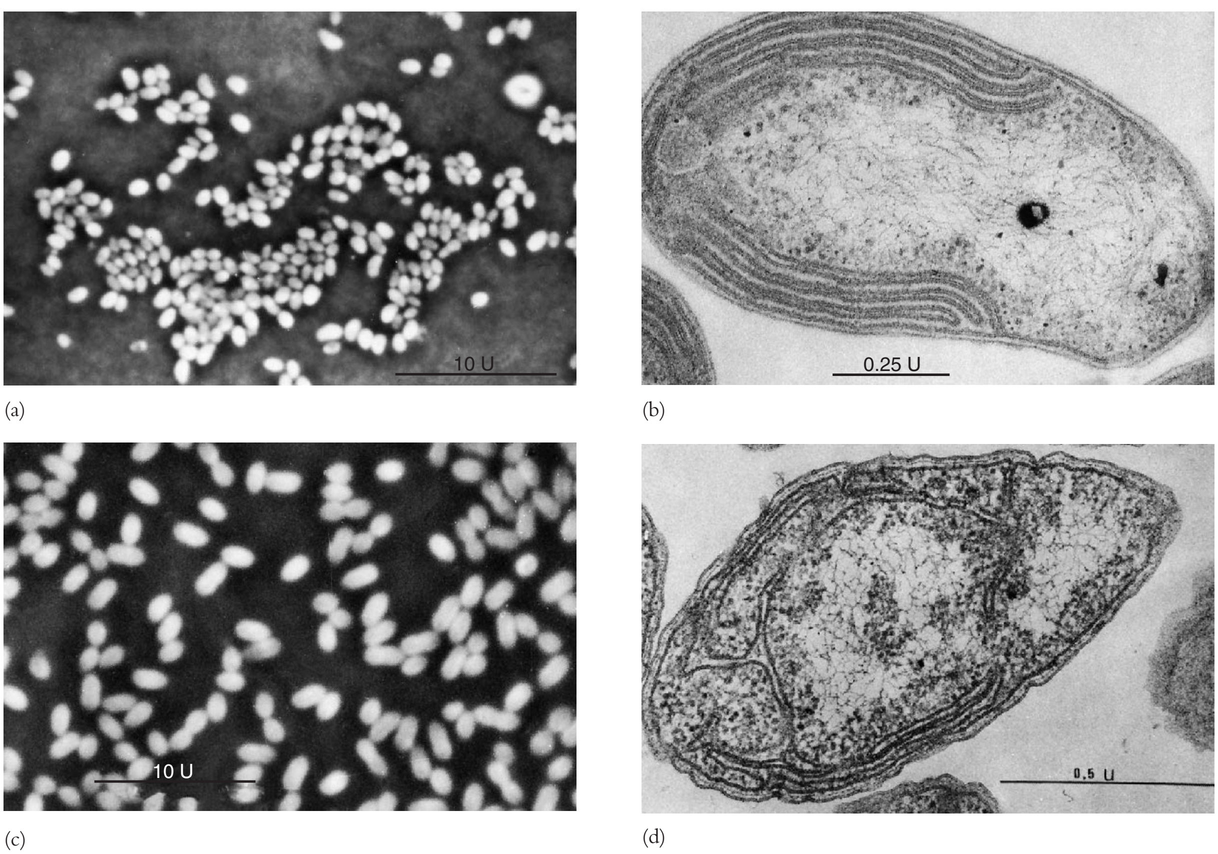


Figure 22.10 Representative Nitrifying Bacteria. (**a**) *Nitrobacter winogradskyi;* phase contrast (×2,500). (**b**) *N. winogradskyi.* Note the polar cap of cytomembranes (×213,000). (**c**) *Nitrosomonas europaea;* phase contrast (×2,500). (**d**) *N. europaea* with extensive cytoplasmic membranes (×81,700).

Nitrifying bacteria are very important ecologically and can be isolated from soil, sewage disposal systems, and freshwater and marine habitats. The genera *Nitrobacter* and *Nitrococcus* oxidize nitrite to nitrate; *Nitrosomonas, Nitrosospira,* and *Nitrosococcus* oxidize ammonia to nitrite. When two genera such as *Nitrobacter* and *Nitrosomonas* grow together in a niche, ammonia is converted to nitrate, a process called **nitrification** (*see section 9.10*). Nitrification occurs rapidly in soils treated with fertilizers containing ammonium salts. Nitrate nitrogen is readily used by plants, but it is also rapidly lost through leaching of the water soluble nitrate and by denitrification to nitrogen gas. Thus nitrification is a mixed blessing. The nitrogen cycle and microorganisms (pp. 615–16)

1. Describe the general properties of the α-proteobacteria.
2. Discuss the characteristics and physiology of the purple nonsulfur bacteria. Where would one expect to find them growing?
3. Briefly describe the characteristics and life cycle of the genus *Rickettsia.*
4. In what way does the physiology and metabolism of the rickettsias differ from that of other bacteria?
5. Name some important rickettsial diseases.
6. Define the following terms: prostheca, stalk, budding, swarmer cell, methylotroph, and holdfast.
7. Briefly describe the morphology and life cycles of *Hyphomicrobium* and *Caulobacter.*
8. What is unusual about the physiology of *Hyphomicrobium?* How does this influence its ecological distribution?
9. How do *Agrobacterium* and *Rhizobium* differ in life-style? What effect does *Agrobacterium* have on plant hosts?
10. What are chemolithotrophic bacteria?
11. Give the major characteristics of the nitrifying bacteria and discuss their ecological importance. How does the metabolism of *Nitrobacter* differ from that of *Nitrosomonas?*

22.2 Class *Betaproteobacteria*

The **β-proteobacteria** overlap the α-proteobacteria metabolically but tend to use substances that diffuse from organic decomposition in the anaerobic zone of habitats. Some of these bacteria use hydrogen, ammonia, methane, volatile fatty acids, and similar substances. As with the α-proteobacteria, there is considerable metabolic diversity; the β-proteobacteria may be chemoheterotrophs, photolithotrophs, methylotrophs, and chemolithotrophs. The subgroup contains two genera with important human pathogens: *Neisseria* and *Bordetella.*

The class *Betaproteobacteria* has six orders and 12 families. **Figure 22.11** shows the phylogenetic relationships between some species of β-proteobacteria, and **table 22.3** summarizes the general characteristics of many of the bacteria discussed in this section.

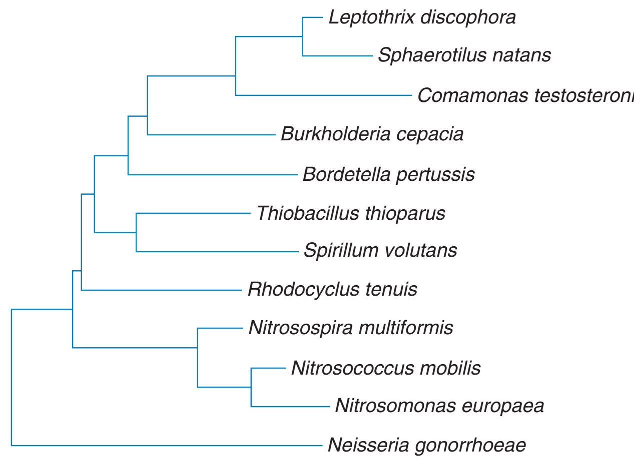


Figure 22.11 Phylogenetic Relationships Among Selected β-Proteobacteria. The relationships of a few species based on 16S rRNA sequence data are shown here. *Source: The Ribosomal Database Project.*

Order *Neisseriales*

The second edition places one family, *Neisseriaceae,* within the order and assigns 14 genera to it. The best-known and most intensely studied genus is *Neisseria.* Members of this genus are nonmotile, aerobic, gram-negative cocci that most often occur in pairs with adjacent sides flattened (*see figure 39.12*). They may have capsules and fimbriae. The genus is chemoorganotrophic, oxidase positive, and almost always catalase positive. Species are inhabitants of the mucous membranes of mammals, and some are human pathogens. *Neisseria gonorrhoeae* is the causative agent of gonorrhea; *Neisseria meningitidis* is responsible for some cases of bacterial meningitis. Gonorrhea (pp. 915–16)

Order *Burkholderiales*

The order contains five families, three of them with well-known genera. The genus *Burkholderia* is placed in the family *Burkholderiaceae.* This genus was established when *Pseudomonas* was divided into at least seven new genera based on rRNA data: *Acidovorax, Aminobacter, Burkholderia, Comamonas, Deleya, Hydrogenophaga,* and *Methylobacterium.* Members of the genus *Burkholderia* are gram-negative, aerobic, nonfermentative, non–spore-forming, mesophilic straight rods. With the exception of one species, all are motile with a single polar flagellum or a tuft of polar flagella. Catalase is produced and they often are oxidase positive. Most species use poly-β-hydroxybutyrate as their carbon reserve. One of the most important species is *B. cepacia,* which will degrade over 100 different organic molecules and is very active in recycling organic materials in nature. This species also is a plant pathogen and causes disease in hospital patients due to contaminated equipment and medications. It is a particular problem for cystic fibrosis patients.

The family *Alcaligenaceae* contains the genus *Bordetella.* This genus is composed of gram-negative, aerobic coccobacilli, about 0.2 to 0.5 by 0.5 to 2.0 μm in size. *Bordetella* is a chemoorganotroph with respiratory metabolism that requires organic sulfur and nitrogen (amino acids) for growth. It is a mammalian parasite that multiplies in respiratory epithelial cells. *Bordetella pertussis* is a nonmotile, encapsulated species that causes whooping cough.

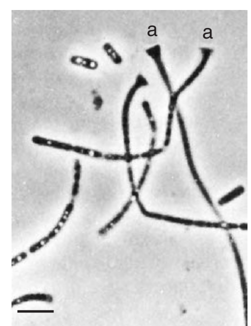
Table 22.3 Characteristics of Selected β-Proteobacteria

Genus	Dimensions (μm) and Morphology	G + C Content (mol%)	Oxygen Requirement	Other Distinctive Characteristics
Bordetella	0.2–0.5 × 0.5–2.0; nonmotile coccobacillus	66–70	Aerobic	Requires organic sulfur and nitrogen; mammalian parasite
Burkholderia	0.3–1.0 × 1–5; straight rods with single flagella or a tuft at the pole	59–69.5	Aerobic	Poly-β-hydroxybutyrate as reserve; can be pathogenic
Leptothrix	0.6–1.4 × 1–12; straight rods in chains with sheath, free cells flagellated	69.5–71	Aerobic	Sheaths encrusted with iron and manganese oxides
Neisseria	0.6–1.0; cocci in pairs with flattened adjacent sides	46–54	Aerobic	Inhabitant of mucous membranes of mammals
Nitrosomonas	Size varies with strain; rod-shaped or ellipsoidal cells with intracytoplasmic membranes	45–54	Aerobic	Chemolithotroph that oxidizes ammonia to nitrite
Sphaerotilus	1.2–2.5 × 2–10; single chains of cells with sheaths, may have holdfasts	70	Aerobic	Sheaths not encrusted with iron and manganese oxides
Thiobacillus	0.5 × 1–4; rods, often with polar flagella	52–68	Aerobic	Chemolithotroph, oxidizes reduced sulfur compounds to sulfate

The family *Comamonadaceae* contains 15 genera with quite diverse characteristics. Some genera (e.g., *Sphaerotilus* and *Leptothrix*) have a **sheath,** a hollow tubelike structure surrounding a chain of cells. Sheaths often are close fitting, but they are never in intimate contact with the cells they enclose and may contain ferric or manganic oxides. They have at least two functions. Sheaths help bacteria attach to solid surfaces and acquire nutrients from slowly running water as it flows past, even if it is nutrient-poor. Sheaths also protect against predators such as protozoa and *Bdellovibrio* (pp. 510–12).

Two well-studied genera are *Sphaerotilus* and *Leptothrix* (**figures 22.12** and **22.13**). *Sphaerotilus* forms long sheathed chains of rods, 0.7 to 2.4 by 3 to 10 μm, attached to submerged plants, rocks, and other solid objects, often by a holdfast (figure 22.12). The sheaths are not usually encrusted by metal oxides. Single swarmer cells with a bundle of subpolar flagella escape the filament and form a new chain after attaching to a solid object at another site. *Sphaerotilus* prefers slowly running freshwater polluted with sewage or industrial waste. It grows so well in activated sewage sludge that it sometimes forms tangled masses of filaments and interferes with the proper settling of sludge (*see section 29.6*). *Leptothrix* characteristically deposits large amounts of iron and manganese oxides in its sheath. This seems to protect it and allow *Leptothrix* to grow in the presence of high concentrations of soluble iron compounds.

(a)



(b)

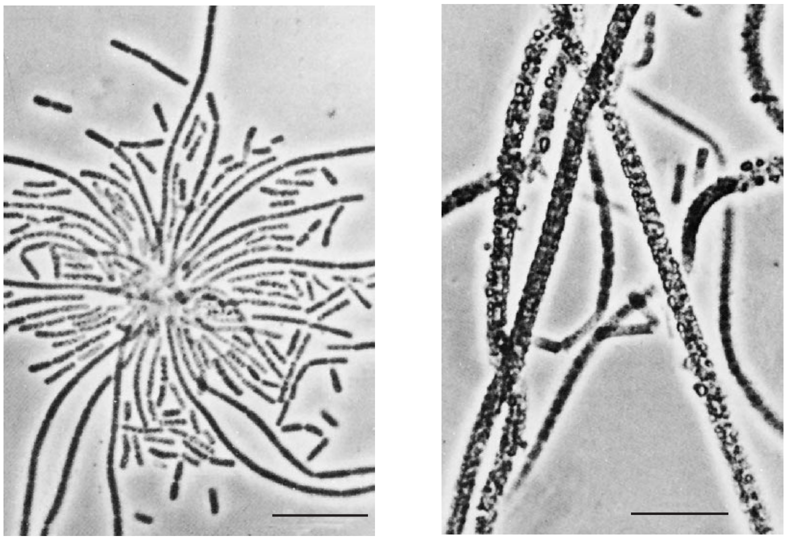
Figure 22.12 Sheathed Bacteria, *Sphaerotilus natans*. (**a**) Sheathed chains of cells and empty sheaths. (**b**) Chains with holdfasts (indicated by the letter a) and individual cells containing poly-β-hydroxybutyrate granules. Bars = 10 μm.

Order *Nitrosomonadales*

A number of chemolithotrophs are found in the order *Nitrosomonadales.* Two genera of nitrifying bacteria (*Nitrosomonas* and *Nitrosospira*) are located here in the family *Nitrosomonadaceae* but were discussed earlier along with other genera of nitrifying bacteria (p. 493). The stalked chemolithotroph *Gallionella* is in this order. The family *Spirillaceae* has one genus, *Spirillum* (**figure 22.14**).

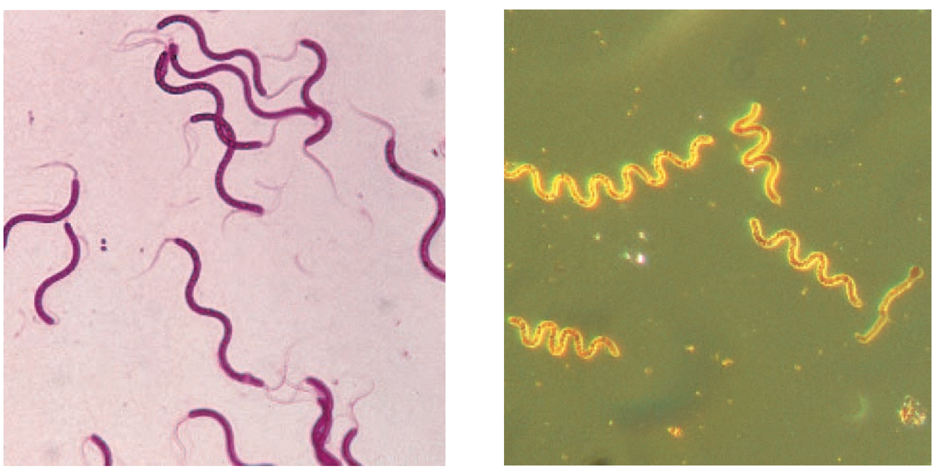
Order *Hydrogenophilales*

This small order contains *Thiobacillus,* one of the best-studied chemolithotrophs and most prominent of the colorless sulfur bacteria. Like the nitrifying bacteria, **colorless sulfur bacteria** are a highly diverse group. Many are unicellular rod-shaped or spiral sulfur-oxidizing bacteria that are nonmotile or motile by flagella (**table 22.4**). The second edition disperses these bacteria between



(a) (b)

Figure 22.13 Sheathed Bacteria, *Leptothrix* Morphology. (**a**) *L. lopholea* trichomes radiating from a collection of holdfasts. (**b**) *L. cholodnii* sheaths encrusted with MnO_2. Bars = 10 μm.



(a) (b)

Figure 22.14 The Genus *Spirillum*. (**a**) *Spirillum volutans* with bipolar flagella visible (×450). (**b**) *Spirillum volutans;* phase contrast (×550).

Table 22.4 Colorless Sulfur-Oxidizing Genera

Genus	Cell Shape	Motility, Flagella	% G + C	Location of Sulfur Deposit[a]	Nutritional Type
Thiobacillus	Rods	+; polar	52–68	Extracellular	Obligate or facultative chemolithotroph
Thiomicrospira	Spirals	+; polar	36–44	Extracellular	Obligate chemolithotroph
Thiobacterium	Rods embedded in gelatinous masses	–	N.A.[b]	Intracellular	Probably chemoorganoheterotroph
Thiospira	Spiral rods, usually with pointed ends	+; polar (single or in tufts)	N.A.	Intracellular	Unknown
Macromonas	Rods, cylindrical or bean shaped	+; polar	67	Intracellular	Probably chemoorganoheterotroph

[a]When hydrogen sulfide is oxidized to elemental sulfur.
[b]N.A., data not available.

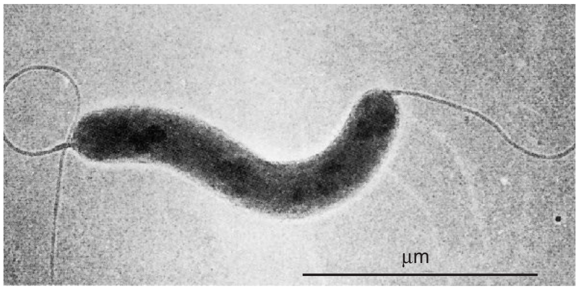


Figure 22.15 Colorless Sulfur Bacteria. *Thiomicrospira pelophila* with polar flagella. Bar = 1 µm.

two classes; for example, *Thiobacillus* is in the β-proteobacteria, whereas *Thiomicrospira, Thiobacterium, Thiospira, Macromonas, Thiothrix, Beggiatoa,* and others are in the γ-proteobacteria. Only some of these bacteria have been isolated and studied in pure culture. Most is known about the genera *Thiobacillus* and *Thiomicrospira. Thiobacillus* is a gram-negative rod, and *Thiomicrospira* is a long spiral cell (**figure 22.15**); both are polarly flagellated. They differ from many of the nitrifying bacteria in that they lack extensive internal membrane systems.

The metabolism of *Thiobacillus* has been intensely studied (*see p. 194*). It grows aerobically by oxidizing a variety of inorganic sulfur compounds (elemental sulfur, hydrogen sulfide, thiosulfate) to sulfate. ATP is produced with a combination of oxidative phosphorylation and substrate-level phosphorylation by means of adenosine 5′-phosphosulfate (*see figure 9.25*). Although *Thiobacillus* normally uses CO_2 as its major carbon source, *T. novellus* and a few other strains can grow heterotrophically. Some species are very flexible metabolically. For example, *Thiobacillus ferrooxidans* also uses ferrous iron as an electron donor and produces ferric iron as well as sulfuric acid. *T. denitrificans* even grows anaerobically by reducing nitrate to nitrogen gas. It should be noted that other sulfur-oxidizing bacteria such as *Thiobacterium* and *Macromonas* probably do not derive energy from sulfur oxidation. They may use the process to detoxify metabolically produced hydrogen peroxide.

Sulfur-oxidizing bacteria have a wide distribution and great practical importance. *Thiobacillus* grows in soil and aquatic habitats, both freshwater and marine. In marine habitats *Thiomicrospira* is more important than *Thiobacillus.* Because of their great acid tolerance (*T. thiooxidans* grows at pH 0.5 and cannot grow above pH 6), these bacteria prosper in habitats they have acidified by sulfuric acid production, even though most other organisms are dying. The production of large quantities of sulfuric acid and ferric iron by *T. ferrooxidans* corrodes concrete and pipe structures. Thiobacilli often cause extensive acid and metal pollution when they release metals from mine wastes. However, sulfur-oxidizing bacteria also are beneficial. They may increase soil fertility when they release elemental sulfur by oxidizing it to sulfate. Thiobacilli are used in processing low-grade metal ores because of their ability to leach metals from ore. The sulfur cycle and microorganisms (pp. 614–15)

1. Describe the general properties of the β-proteobacteria.
2. Briefly describe the following genera and their practical importance: *Neisseria, Burkholderia,* and *Bordetella.*
3. What is a sheath and of what advantage is it?
4. How does *Sphaerotilus* maintain its position in running water? How does it reproduce and disperse its progeny?
5. Characterize the colorless sulfur bacteria and discuss their placement in the second edition of *Bergey's Manual.*
6. How do colorless sulfur bacteria obtain energy by oxidizing sulfur compounds? What is adenosine 5′-phosphosulfate?
7. List several positive and negative impacts sulfur-oxidizing bacteria have on the environment and human activities.

22.3 Class *Gammaproteobacteria*

The **γ-proteobacteria** constitute the largest subgroup of proteobacteria with an extraordinary variety of physiological types. Many important genera are chemoorganotrophic and facultatively anaerobic. Other genera contain aerobic chemoorganotrophs, photolithotrophs, chemolithotrophs, or methylotrophs. According to some DNA-rRNA hybridization studies, the γ-proteobacteria are composed of several deeply branching groups. One consists of the purple sulfur bacteria; a second includes the intracellular parasites *Legionella* and *Coxiella.* The two largest groups contain a wide variety of nonphotosynthetic genera. Ribosomal RNA superfamily I is represented by the families *Vibrionaceae, Enterobacteriaceae,* and *Pasteurellaceae.* These bacteria use the glycolytic and pentose phosphate pathways to catabolize carbohydrates. Most are facultative anaerobes. Ribosomal RNA superfamily II contains mostly aerobes that often use the Entner-Doudoroff and pentose phosphate pathways to catabolize many different kinds of organic molecules. The genera *Pseudomonas, Azotobacter, Moraxella, Xanthomonas,* and *Acinetobacter* belong to this superfamily.

The exceptional diversity of these bacteria is evident from the fact that the second edition of *Bergey's Manual* divides the class *Gammaproteobacteria* into 13 orders, 20 families, and around 160 genera. **Figure 22.16** illustrates the phylogenetic relationships between major groups and selected γ-proteobacteria, and **table 22.5** outlines the general characteristics of some of the bacteria discussed in this section.

The Purple Sulfur Bacteria

As mentioned previously, the purple photosynthetic bacteria are distributed between three subgroups of the proteobacteria. Despite the diversity of these organisms, they do share some general characteristics, which were summarized in table 21.1 (*see p. 469*). Most

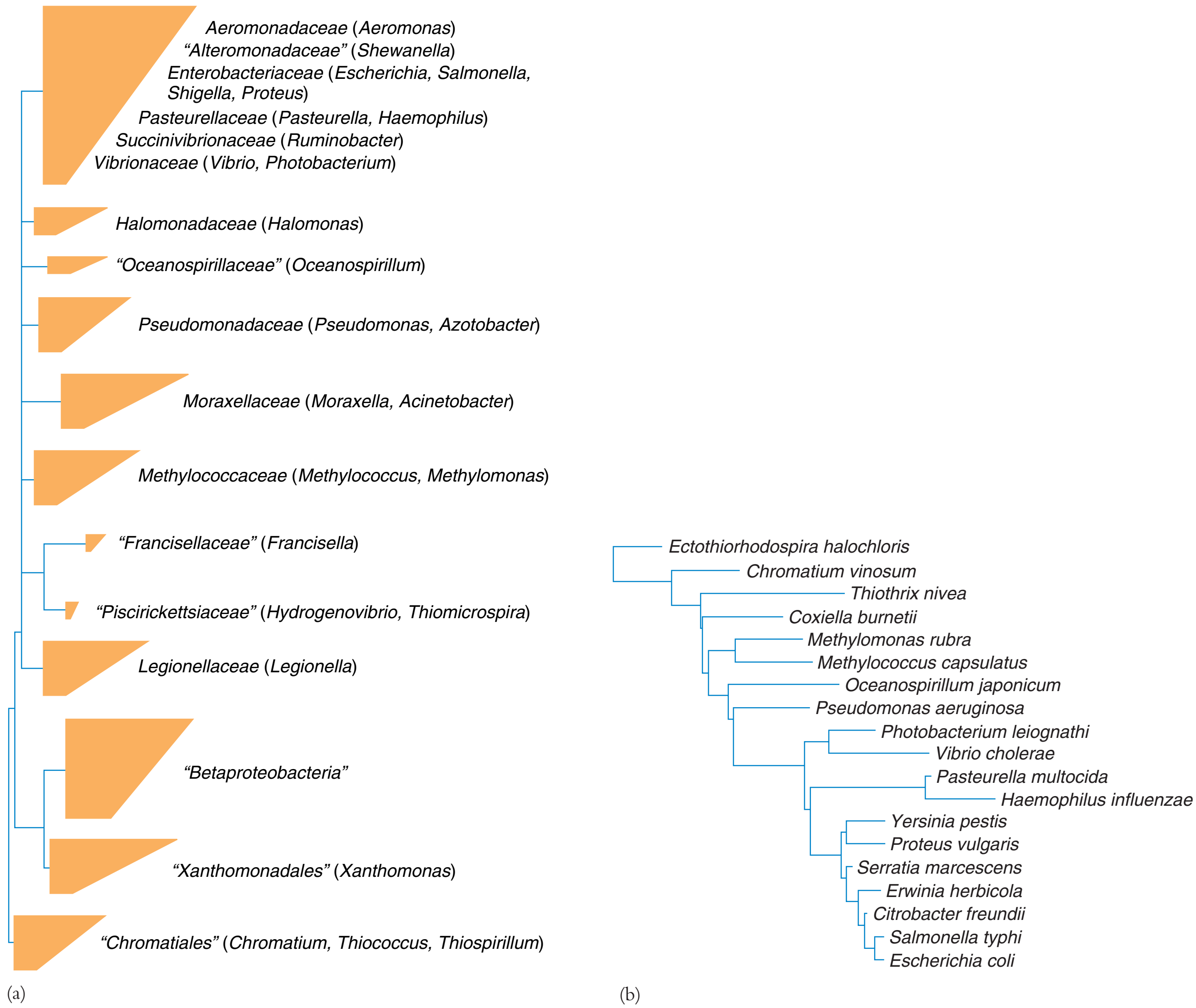


Figure 22.16 Phylogenetic Relationships Among γ-Proteobacteria. (**a**) The major phylogenetic groups based on 16S rRNA sequence comparisons. Representative genera are given in parentheses. Each tetrahedron in the tree represents a group of related organisms; its horizontal edges show the shortest and longest branches in the group. Multiple branching at the same level indicates that the relative branching order of the groups cannot be determined from the data. The quotation marks around some names indicate that they are not formally approved taxonomic names. (**b**) The relationships of a few species based on 16S rRNA sequence data. *Source: The Ribosomal Database Project.*

Table 22.5 Characteristics of Selected γ-Proteobacteria

Genus	Dimensions (μm) and Morphology	G + C Content (mol%)	Oxygen Requirement	Other Distinctive Characteristics
Azotobacter	1.5–2.0; ovoid cells, pleomorphic, peritrichous or nonmotile	63.2–67.5	Aerobic	Can form cysts; fix nitrogen nonsymbiotically
Beggiatoa	≈1–50 × ≈2–10; colorless cells form filaments, either single or in colonies	37–51	Aerobic or microaerophilic	Gliding motility; can form sulfur inclusions with hydrogen sulfide present
Chromatium	1–6 × 1.5–16; rod-shaped or ovoid, straight or slightly curved, polar flagella	48–70	Anaerobic	Photolithoautotroph that can use sulfide; sulfur stored within the cell
Ectothiorhodospira	0.5–1.5 in diameter; vibrioid- or rod-shaped, polar flagella	50.5–69.7	Anaerobic, some aerobic or microaerobic	Internal lamellar stacks of membranes; deposits sulfur granules outside cells
Escherichia	1.1–1.5 × 2–6; straight rods, peritrichous or nonmotile	48–52	Facultatively anaerobic	Mixed acid fermenter; formic acid converted to H_2 and CO_2, lactose fermented, citrate not used
Haemophilus	<1.0 in width; coccobacilli or rods, nonmotile	33–47	Facultative or aerobic	Fermentative; requires growth factors present in blood; parasites on mucous membranes
Leucothrix	Long filaments of short cylindrical cells, usually holdfast is present	46–51	Aerobic	Dispersal by gonidia, filaments don't glide; rosettes formed; heterotrophic
Methylococcus	1.0 in diameter; cocci with capsules, nonmotile	62–63	Aerobic	Can form a cyst; methane, methanol, and formaldehyde are sole carbon and energy sources
Photobacterium	0.8–1.3 × 1.8–2.4; straight, plump rods with polar flagella	40–44	Facultatively anaerobic	Two species can emit blue-green light; Na^+ needed for growth
Pseudomonas	0.5–1.0 × 1.5–5.0; straight or slightly curved rods, polar flagella	58–70	Aerobic	Respiratory metabolism with oxygen as acceptor; some are able to use H_2 or CO as energy source
Vibrio	0.5–0.8 × 1.4–2.6; straight or curved rods with sheathed polar flagella	38–51	Facultatively anaerobic	Fermentative or respiratory metabolism; sodium ions stimulate or are needed for growth; oxidase positive

of the purple nonsulfur bacteria are α-proteobacteria and were discussed earlier in this chapter (pp. 487–88). Because the purple sulfur bacteria are γ-proteobacteria, they will be described here.

Bergey's Manual divides the purple sulfur bacteria into two families: the *Chromatiaceae* and *Ectothiorhodospiraceae.* In the second edition these families are in the order *Chromatiales.* The family *Ectothiorhodospiraceae* contains five genera. *Ectothiorhodospira* has red, spiral-shaped, polarly flagellated cells that deposit sulfur globules externally (**figure 22.17;** *see also figure 3.9*b). Internal photosynthetic membranes are organized as lamellar stacks. The typical purple sulfur bacteria are located in the family *Chromatiaceae,* which is much larger and contains 22 genera.

The **purple sulfur bacteria** are strict anaerobes and usually photolithoautotrophs. They oxidize hydrogen sulfide to sulfur and deposit it internally as sulfur granules (usually within invaginated pockets of the plasma membrane); often they eventually oxidize the sulfur to sulfate. Hydrogen also may serve as an electron donor. *Thiospirillum, Thiocapsa,* and *Chromatium* are typical purple sulfur bacteria (**figure 22.18**). They are found in anaerobic, sulfide-rich zones of lakes (*see chapter 29*). Large blooms of

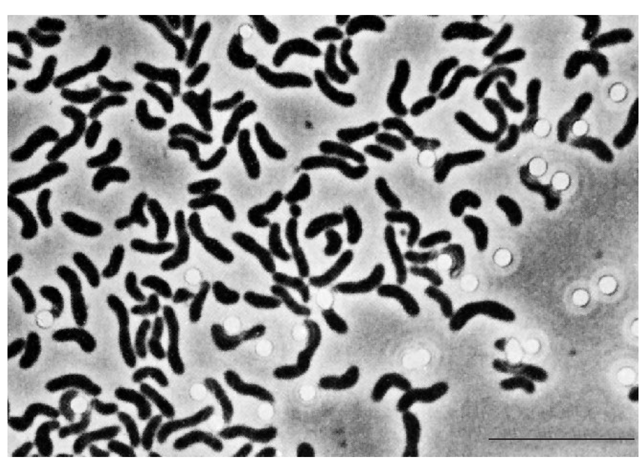
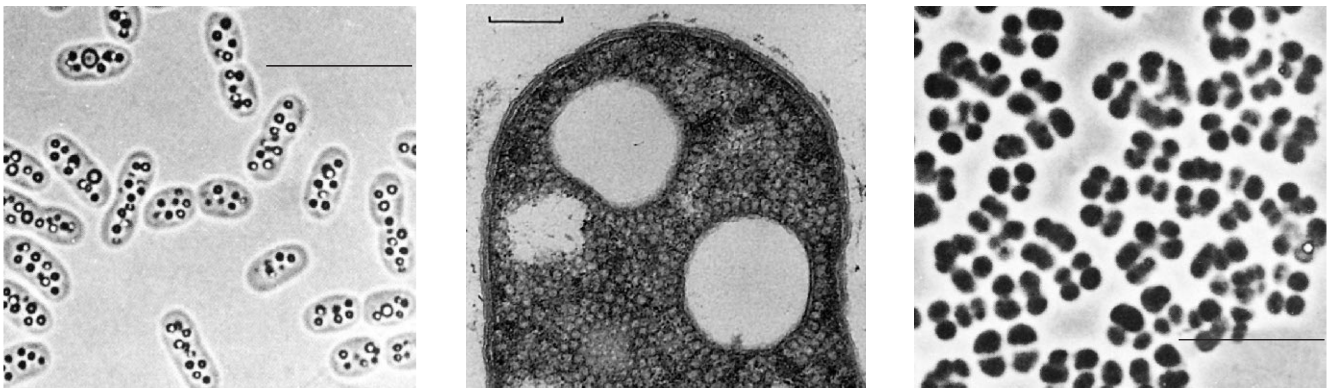
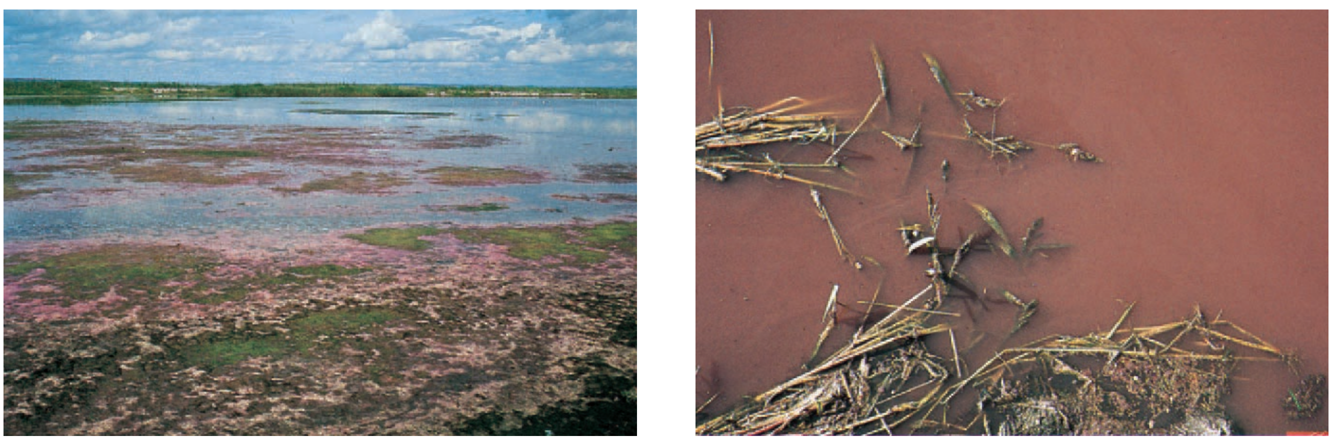


Figure 22.17 **Purple Bacteria.** *Ectothiorhodospira mobilis;* light micrograph. Bar = 10 μm.



(a) (b) (c)

Figure 22.18 Typical Purple Sulfur Bacteria. (**a**) *Chromatium vinosum* with intracellular sulfur granules. Bar = 10 μm. (**b**) Electron micrograph of *C. vinosum.* Note the intracytoplasmic vesicular membrane system. The large white areas are the former sites of sulfur globules. Bar = 0.3 μm. (**c**) *Thiocapsa roseopersicina.* Bar = 10 μm.



(a) (b)

Figure 22.19 Purple Photosynthetic Sulfur Bacteria. (**a**) Purple photosynthetic sulfur bacteria growing in a bog. (**b**) A sewage lagoon with a bloom of purple photosynthetic bacteria.

purple sulfur bacteria occur in bogs and lagoons under the proper conditions (**figure 22.19**).

Order *Thiotrichales*

The order *Thiotrichales* contains three families, the largest of which is the family *Thiotrichaceae.* This family has several genera that oxidize sulfur compounds (see the colorless sulfur bacteria [p. 496] and chapter 9 for sulfur oxidation and chemolithotrophy). Morphologically both rods and filamentous forms are present.

Two of the best-studied gliding genera in this family are *Beggiatoa* and *Leucothrix* (**figures 22.20** and **22.21**). *Beggiatoa* is microaerophilic and grows in sulfide-rich habitats such as sulfur springs, freshwater with decaying plant material, rice paddies, salt marshes, and marine sediments. Its filaments contain short, disklike cells and lack a sheath. *Beggiatoa* is very versatile metabolically. It oxidizes hydrogen sulfide to form large sulfur grains located in pockets formed by invaginations of the plasma membrane. *Beggiatoa* can subsequently oxidize the sulfur to sulfate. The electrons are used by the electron transport chain in energy production. Many strains also can grow heterotrophically with acetate as a carbon source, and some may incorporate CO_2 autotrophically.

Leucothrix (**figure 22.21**) is an aerobic chemoorganotroph that forms long filaments or trichomes up to 400 μm long. It is usually marine and is attached to solid substrates by a holdfast. *Leucothrix* has a complex life cycle in which dispersal is by the formation of gonidia. Rosette formation often is seen in culture (figure 22.21*d*). *Thiothrix* is a related genus that forms sheathed filaments and releases gonidia from the open end of

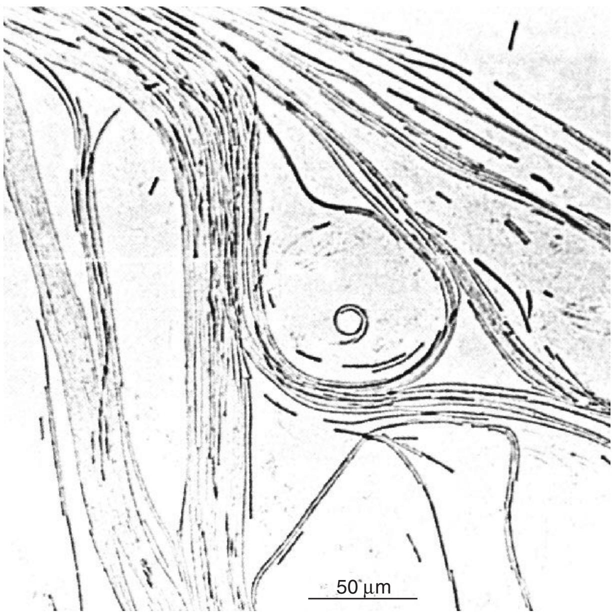


Figure 22.20 ***Beggiatoa.*** *Beggiatoa* sp. colony growing on agar.

the sheath (**figure 22.22**). In contrast with *Leucothrix, Thiothrix* is a chemolithotroph that oxidizes hydrogen sulfide and deposits sulfur granules internally. It also requires an organic compound for growth (i.e., it is a mixotroph). *Thiothrix* grows in sulfide-rich flowing water and activated sludge sewage systems.

Order *Methylococcales*

The family *Methylococcaceae* contains rods, vibrios, and cocci that use methane, methanol, and other reduced one-carbon compounds as their sole carbon and energy sources under aerobic or microaerobic (low oxygen) conditions. That is, they are methylotrophs (bacteria that use methane exclusively as their carbon and energy source often are called methanotrophs). The family contains six genera, two of which are *Methylococcus* (spherical, nonmotile cells) and *Methylomonas* (straight, curved, or branched rods with a single, polar flagellum). When oxidizing methane, the bacteria contain complex arrays of intracellular membranes. Almost all can form some type of resting stage, often a cyst somewhat like that of the azotobacteria. Methylotrophic growth depends on the presence of methane and related compounds. Methanogenesis from substrates such as H_2 and CO_2 is widespread in anaerobic soil and water, and methylotrophic bacteria grow above anaerobic habitats all over the world.

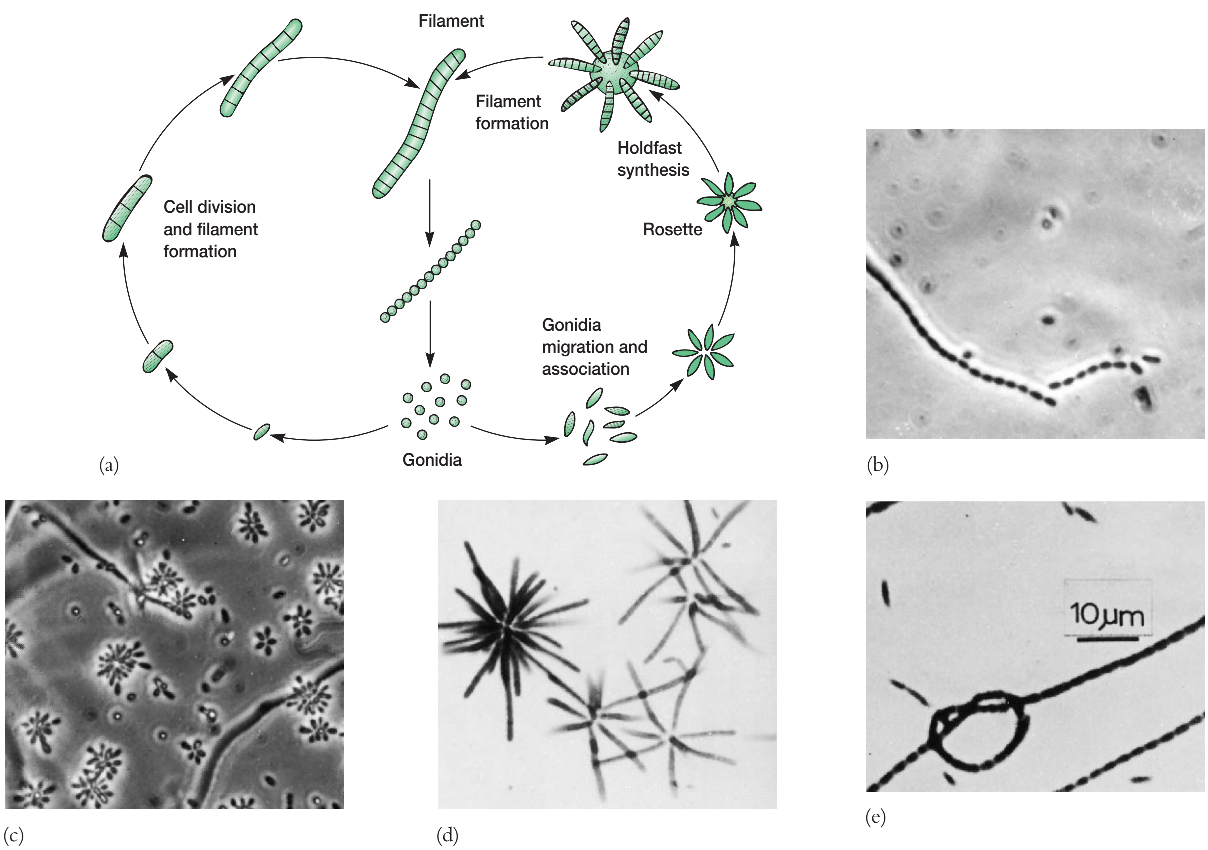


Figure 22.21 **Morphology and Reproduction of *Leucothrix mucor.*** (**a**) Life cycle of *L. mucor.* (**b**) Separation of gonidia from the tip of mature filament; phase contrast (×1,400). (**c**) Gonidia aggregating to form rosettes; phase contrast (×950). (**d**) Young developing rosettes (×1,500). (**e**) A knot formed by a *Leucothrix* filament.

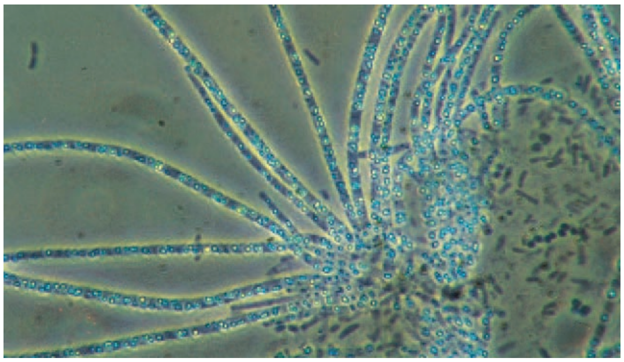
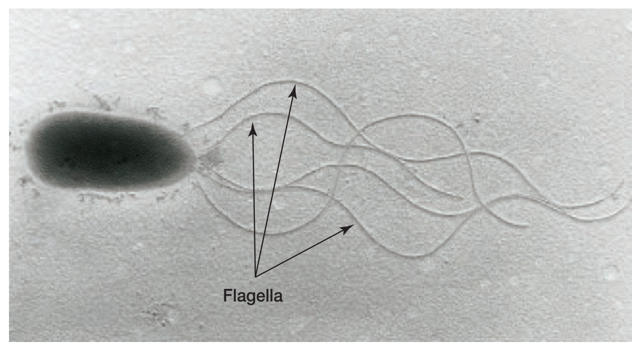


Figure 22.22 ***Thiothrix.*** A *Thiothrix* colony viewed with phase-contrast microscopy (×1,000).

(a)



(b)

Figure 22.23 **The Genus *Pseudomonas.*** (**a**) A phase-contrast micrograph of *Pseudomonas* cells containing PHB (poly-β-hydroxybutyrate) granules. (**b**) A transmission electron micrograph of *Pseudomonas putida* with five polar flagella, each flagellum about 5–7 μm in length.

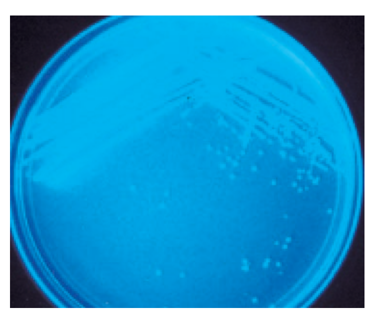


Figure 22.24 ***Pseudomonas* Fluorescence.** *Pseudomonas aeruginosa* colonies fluorescing under ultraviolet light.

Methane-oxidizing bacteria use methane as a source of both energy and carbon. Methane is first oxidized to methanol by the enzyme methane monooxygenase. The methanol is then oxidized to formaldehyde by methanol dehydrogenase, and the electrons from this oxidation are donated to an electron transport chain for ATP synthesis. Formaldehyde can be assimilated into cell material by the activity of either of two pathways, one involving the formation of the amino acid serine and the other proceeding through the synthesis of sugars such as fructose 6-phosphate and ribulose 5-phosphate.

Order *Pseudomonadales*

Pseudomonas is the most important genus in the order *Pseudomonadales,* family *Pseudomonadaceae.* The genus *Pseudomonas* contains straight or slightly gram-negative curved rods, 0.5 to 1.0 μm by 1.5 to 5.0 μm in length, that are motile by one or several polar flagella and lack prosthecae or sheaths (**figure 22.23;** *see figure 3.35*a). These chemoheterotrophs are aerobic and carry out respiratory metabolism with O_2 (and sometimes nitrate) as the electron acceptor. All pseudomonads have a functional tricarboxylic acid cycle and can oxidize substrates to CO_2. Most hexoses are degraded by the Entner-Doudoroff pathway rather than glycolytically. The Entner-Doudoroff pathway and tricarboxylic acid cycle (pp. 179, 183–84, appendix II)

In the first edition, the genus is an exceptionally heterogeneous taxon composed of 70 or more species. Many can be placed in one of five rRNA homology groups. The three best-characterized groups, RNA groups I–III, are subdivided according to properties such as the presence of poly-β-hydroxybutyrate (PHB), the production of a fluorescent pigment, pathogenicity, the presence of arginine dihydrolase, and glucose utilization. For example, the fluorescent subgroup does not accumulate PHB and produces a diffusible, water-soluble, yellow-green pigment that fluoresces under UV radiation (**figure 22.24**). *Pseudomonas aeruginosa, P. fluorescens, P. putida,* and *P. syringae* are members of this group. The second edition has reduced the size of the genus *Pseudomonas* drastically by transferring species to several newly named genera (e.g., *Burkholderia, Hydrogenophaga,* and *Methylobacterium*).

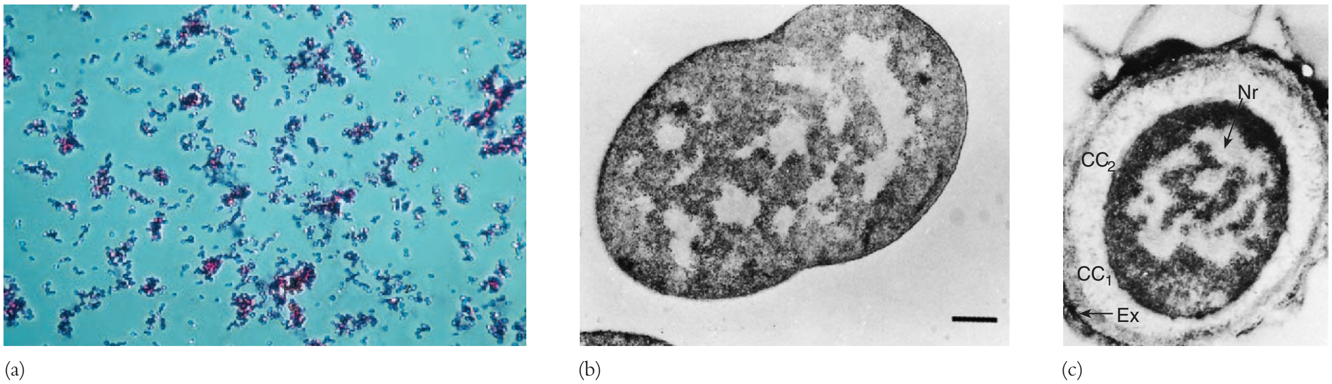


Figure 22.25 ***Azotobacter.*** (**a**) *A. chroococcum* (×270). (**b**) Electron micrograph of *A. chroococcum.* Bar = 0.2 μm. (**c**) *Azotobacter* cyst structure. Bar = 0.2 μm. The nuclear region (Nr), exine layers (CC_1 and CC_2), and exosporium (Ex) are visible.

The pseudomonads have a great practical impact in several ways, including the following:

1. Many can degrade an exceptionally wide variety of organic molecules. Thus they are very important in the **mineralization process** (the microbial breakdown of organic materials to inorganic substances) in nature and in sewage treatment. The fluorescent pseudomonads can use approximately 80 different substances as their carbon and energy sources; *Burkholderia cepacia* will degrade over 100 different organic molecules. Microbial decomposition of organic materials (pp. 611–14; 1010–14)
2. Several species (e.g., *P. aeruginosa*) are important experimental subjects. Many advances in microbial physiology and biochemistry have come from their study. For example, the genome of *Pseudomonas aeruginosa* has been sequenced. It is very big (about 6.3 million base pairs) and much more complex than the *E. coli* genome, although about 50% of its open reading frames are similar to those in *E. coli. P. aeruginosa* has an unusually large number of genes for catabolism, nutrient transport, the efflux of organic molecules, and metabolic regulation. This may explain its ability to grow in many environments and resist antibiotics.
3. Some pseudomonads are major animal and plant pathogens. *P. aeruginosa* infects people with low resistance such as cystic fibrosis patients and invades burn areas or causes urinary tract infections. *Burkholderia solanacearum* (*R. solanacearum*) causes wilts in many plants by producing pectinases, cellulases, and the plant hormones indoleacetic acid and ethylene. *P. syringae* and *B. cepacia* are also important plant pathogens.
4. Pseudomonads such as *P. fluorescens* are involved in the spoilage of refrigerated milk, meat, eggs, and seafood because they grow at 4°C and degrade lipids and proteins.

The genus *Azotobacter* also is in the family *Pseudomonadaceae.* The genus contains large, ovoid bacteria, 1.5 to 2.0 μm in diameter, that may be motile by peritrichous flagella. The cells are often pleomorphic, ranging from rods to coccoid shapes, and form cysts as the culture ages (**figure 22.25**). The genus is aerobic, catalase positive, and fixes nitrogen nonsymbiotically. *Azotobacter* is widespread in soil and water.

Order *Vibrionales*

Three closely related orders of the γ-proteobacteria contain a number of practically important bacterial genera. Each order has only one family of facultatively anaerobic gram-negative rods. **Table 22.6** summarizes the distinguishing properties of the families *Enterobacteriaceae, Vibrionaceae,* and *Pasteurellaceae.*

The order *Vibrionales* contains only one family, the *Vibrionaceae.* Members of the family *Vibrionaceae* are gram-negative, straight or curved rods with polar flagella (**figure 22.26**). Most are oxidase positive, and all use D-glucose as their sole or primary carbon and energy source (table 22.6). The majority are aquatic microorganisms, widespread in freshwater and the sea. There are six genera in the family: *Vibrio, Photobacterium, Enhydrobacter, Salinivibrio, Listonella,* and *Allomonas.*

Several vibrios are important pathogens. *V. cholerae* (*see figure 3.1*e) is the causative agent of cholera (*see section 39.4*), and *V. parahaemolyticus* sometimes causes gastroenteritis in humans following consumption of contaminated seafood. *V. anguillarum* and others are responsible for fish diseases.

The *Vibrio cholerae* genome has now been sequenced and found to contain 3,885 open reading frames distributed between two circular chromosomes, chromosome 1 (2.96 million base pairs) and chromosome 2 (1.07 million bp). The larger chromosome primarily has genes for essential cell functions such as DNA replication, transcription, and protein synthesis. It also has most of the virulence genes (e.g., the cholera toxin gene is located in an integrated CTXϕ phage on chromosome 1). Chromosome 2 also has essential genes such as transport genes and ribosomal protein genes. Copies of some genes are present on both chromosomes. Perhaps *V. cholerae* achieves faster genome duplication and cell division by distributing its genes between two chromosomes.

Several members of the family are unusual in being bioluminescent. *Vibrio fischeri* and at least two species of *Photobacterium* are among the few marine bacteria capable of **bioluminescence**

Table 22.6 Characteristics of Families of Facultatively Anaerobic Gram-Negative Rods

Characteristics	*Enterobacteriaceae*	*Vibrionaceae*	*Pasteurellaceae*
Cell dimensions	0.3–1.0 × 1.0–6.0 μm	0.3–1.3 × 1.0–3.5 μm	0.2–0.3 × 0.3–2.0 μm
Morphology	Straight rods; peritrichous flagella or nonmotile	Straight or curved rods; polar flagella	Coccoid to rod-shaped cells, sometimes pleomorphic; nonmotile
Physiology	Oxidase negative	Oxidase positive; all can use D-glucose as sole or principal carbon source	Oxidase positive; heme and/or NAD often required for growth; organic nitrogen source required
G + C content	38–60%	38–63%	38–47%
Symbiotic relationships	Some parasitic on mammals and birds; some species plant pathogens	Most not pathogens (with a few exceptions)	Parasites of mammals and birds
Representative genera	*Escherichia, Shigella, Salmonella, Citrobacter, Klebsiella, Enterobacter, Erwinia, Serratia, Proteus, Yersinia*	*Vibrio, Photobacterium*	*Pasteurella, Haemophilus*

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Box 22.1

Bacterial Bioluminescence

Several species in the genera *Vibrio* and *Photobacterium* can emit light of a blue-green color. The enzyme luciferase catalyzes the reaction and uses reduced flavin mononucleotide, molecular oxygen, and a long-chain aldehyde as substrates.

$$FMNH_2 + O_2 + RCHO \xrightarrow{\text{luciferase}} FMN + H_2O + RCOOH + \text{light}$$

The evidence suggests that an enzyme-bound, excited flavin intermediate is the direct source of luminescence. Because the electrons used in light generation are probably diverted from the electron transport chain and ATP synthesis, the bacteria expend considerable energy on luminescence. Luminescence activity is regulated and can be turned off or on under the proper conditions.

There is much speculation about the role of bacterial luminescence and its value to bacteria, particularly because it is such an energetically expensive process. Luminescent bacteria occupying the special luminous organs of fish do not emit light when they grow as free-living organisms in the seawater. Free-living luminescent bacteria can reproduce and infect young fish. Once settled in a fish's luminous organ, they begin emitting light, which the fish uses for its own purposes. Other luminescent bacteria growing on potential food items such as small crustacea may use light to attract fish to the food source. After ingestion, they could establish a symbiotic relationship in the host's gut.

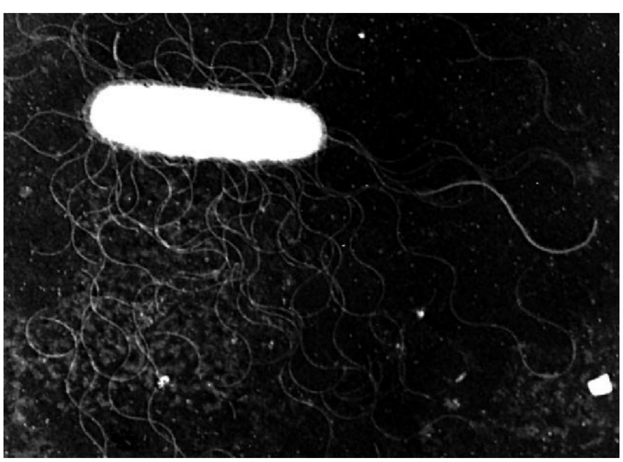


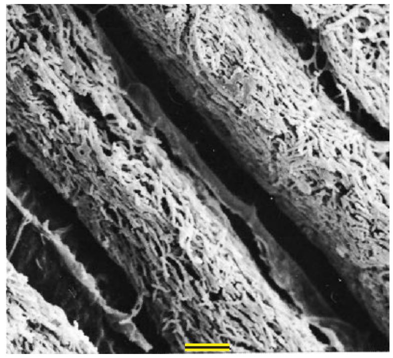
Figure 22.26 **The *Vibrionaceae*.** Electron micrograph of *Vibrio alginolyticus* grown on agar, showing a sheathed polar flagellum and unsheathed lateral flagella (×18,000).

and emit a blue-green light because of the activity of the enzyme luciferase (**Box 22.1**). The peak emission of light is usually between 472 and 505 nm, but one strain of *V. fischeri* emits yellow light with a major peak at 545 nm. Although many of these bacteria are free-living, *V. fischeri, P. phosphoreum,* and *P. leiognathi* live symbiotically in the luminous organs of fish (**figure 22.27**). Interestingly, *Vibrio fischeri* can degrade 3′, 5′-cyclic AMP and use it as a carbon, nitrogen, and phosphorus source for growth.

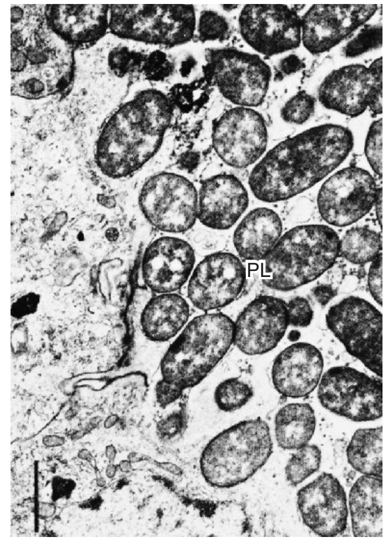
Order *Enterobacteriales*

The family *Enterobacteriaceae* is the largest of the families in table 22.6. It contains gram-negative, peritrichously flagellated or nonmotile, facultatively anaerobic, straight rods with simple nutritional requirements (*see figures 2.14*c, *3.27*a, *3.30,* and *3.31*c). In the second edition the order *Enterobacteriales* has only one family, *Enterobacteriaceae,* with 41 genera. The relationship between *Enterobacteriales* and the orders *Vibrionales* and *Pasteurellales* can be seen by inspecting figure 22.16.

(a)



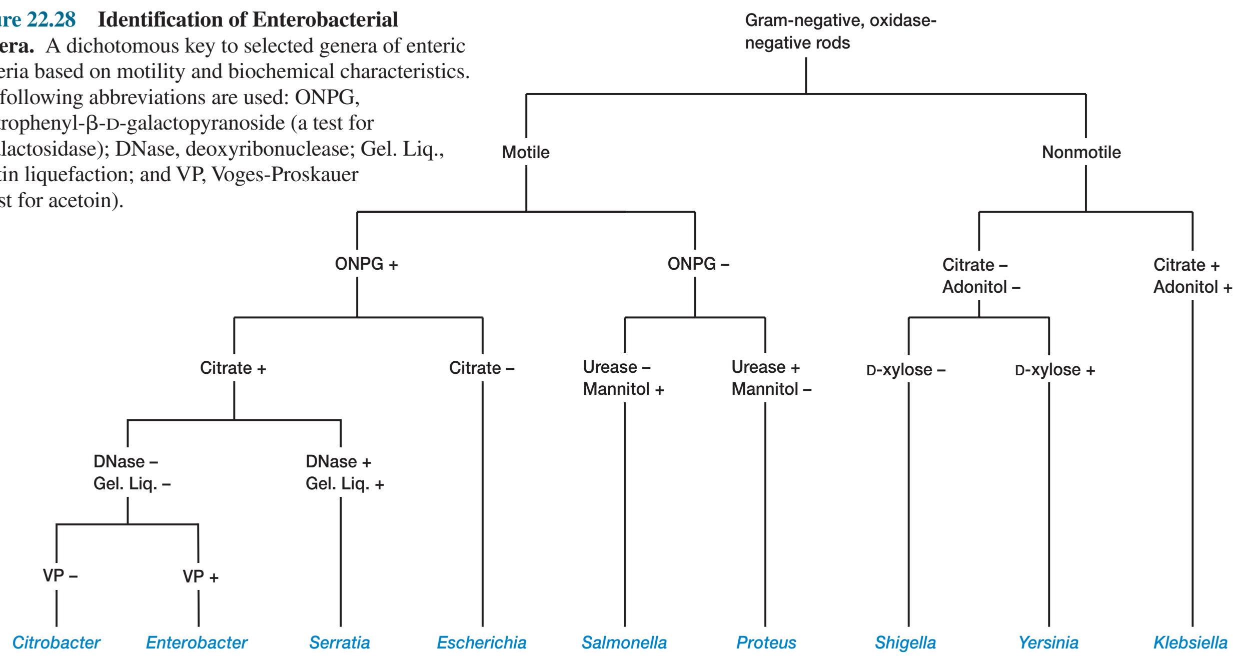
(b)



(c)

Figure 22.27 Bioluminescence. (**a**) A photograph of the Atlantic flashlight fish *Kryptophanaron alfredi.* The light area under the eye is the fish's luminous organ, which can be covered by a lid of tissue. (**b**) The masses of photobacteria in the SEM view are separated by thin epithelial cells. Line = 6.0 μm. (**c**) Ultrathin section of the luminous organ of a fish. *Equulities novaehollandiae,* with the bioluminescent bacterium *Photobacterium leiognathi,* PL. Bar = 2 μm.

Figure 22.28 Identification of Enterobacterial Genera. A dichotomous key to selected genera of enteric bacteria based on motility and biochemical characteristics. The following abbreviations are used: ONPG, *o*-nitrophenyl-β-D-galactopyranoside (a test for β-galactosidase); DNase, deoxyribonuclease; Gel. Liq., gelatin liquefaction; and VP, Voges-Proskauer (a test for acetoin).



The metabolic properties of the *Enterobacteriaceae* are very useful in characterizing its constituent genera. Members of the family, often called **enterobacteria** or **enteric bacteria** [Greek *enterikos,* pertaining to the intestine], all degrade sugars by means of the Embden-Meyerhof pathway and cleave pyruvic acid to yield formic acid in formic acid fermentations. Those enteric bacteria that produce large amounts of gas during sugar fermentation, such as *Escherichia* spp., have the formic hydrogenlyase complex that degrades formic acid to H_2 and CO_2. The family can be divided into two groups based on their fermentation products. The majority (e.g., the genera *Escherichia, Proteus, Salmonella,* and *Shigella*) carry out mixed acid fermentation and produce mainly lactate, acetate, succinate, formate (or H_2 and CO_2), and ethanol. In butanediol fermentation the major products are butanediol, ethanol, and carbon dioxide. *Enterobacter, Serratia, Erwinia,* and *Klebsiella* are butanediol fermenters. As mentioned previously (*see section 9.3*), the two types of formic acid fermentation are distinguished by the methyl red and Voges-Proskauer tests. Formic acid fermentation and the family Enterobacteriaceae (p. 181)

Because the enteric bacteria are so similar in morphology, biochemical tests are normally used to identify them after a preliminary examination of their morphology, motility, and growth responses (**figure 22.28** provides a simple example). Some more commonly used tests are those for the type of formic acid fermentation, lactose and citrate utilization, indole production from tryptophan, urea hydrolysis, and hydrogen sulfide production. For example, lactose fermentation occurs in *Escherichia* and *Enterobacter* but not in *Shigella, Salmonella,* or *Proteus.* **Table 22.7** summarizes a few of the biochemical properties useful in distinguishing between genera of enteric bacteria. The mixed acid fermenters are located on the left in this table and the butanediol fermenters on the right. The usefulness of biochemical tests in identifying enteric bacteria is shown by the popularity of commercial identification systems, such as the Enterotube and API 20-E systems, that are based on these tests. Commercial rapid identification systems (pp. 840–42); *E. coli* genome sequence (p. 349)

Members of the *Enterobacteriaceae* are so common, widespread, and important that they are probably more often seen in most laboratories than any other bacteria. *Escherichia coli* is undoubtedly the best-studied bacterium and the experimental organism of choice for many microbiologists. It is an inhabitant of the colon of humans and other warm-blooded animals, and it is quite useful in the analysis of water for fecal contamination (*see section 29.5*). Some strains cause gastroenteritis or urinary tract infections. Several enteric genera contain very important human pathogens responsible for a variety of diseases: *Salmonella* (**figure 22.29**), typhoid fever and gastroenteritis; *Shigella,* bacillary dysentery; *Klebsiella,* pneumonia; *Yersinia,* plague. Members of the genus *Erwinia* are major pathogens of crop plants and cause blights, wilts, and several other plant diseases. These and other members of the family are discussed in more detail at later points in the text (*see chapter 39*).

Order *Pasteurellales*

The second edition of *Bergey's Manual* places the family *Pasteurellaceae* in the order *Pasteurellales* and treats it much the same as the first edition does. The *Pasteurellaceae* differ from the other two families in several ways (table 22.6). Most notably, they are small (0.2 to 0.3 μm in diameter) and nonmotile, normally oxidase positive, have complex nutritional requirements of various kinds, and are parasitic in vertebrates. The family contains six genera: *Pasteurella, Haemophilus, Actinobacillus, Lonepinella, Mannheimia,* and *Phocoenobacter.*

As might be expected, members of this family are best known for the diseases they cause in humans and many animals. *Pasteurella multilocida* and *P. haemolytica* are important animal pathogens. *P. multilocida* is responsible for fowl cholera, which destroys many chickens, turkeys, ducks, and geese each year. *P. haemolytica* is at least partly responsible for pneumonia in cattle, sheep, and goats (e.g., "shipping fever" in cattle). *H. influenzae* type b is a major human pathogen that causes a variety of diseases, including meningitis in children (*see section 39.1*). *H. influenzae* genome sequence (p. 349)

1. Describe the general properties of the γ-proteobacteria.
2. What are the major characteristics of the purple sulfur bacteria? Contrast the families *Chromatiaceae* and *Ectothiorhodospira.*
3. Describe the genera *Beggiatoa, Leucothrix,* and *Thiothrix.*
4. In what habitats would one expect to see the *Methylococcaceae* growing and why?
5. What is a methylotroph? How do methane-oxidizing bacteria use methane as both an energy source and a carbon source?
6. Give the major distinctive properties of the genera *Pseudomonas* and *Azotobacter.* Briefly discuss the taxonomic changes that have occurred in the genus *Pseudomonas.*
7. Why are the pseudomonads such important bacteria? What is mineralization?
8. List the major distinguishing traits of the families *Vibrionaceae, Enterobacteriaceae,* and *Pasteurellaceae.*
9. What major human disease is associated with the *Vibrionaceae,* and what species causes it?
10. Briefly describe bioluminescence and the way it is produced.
11. Into what two groups can the enteric bacteria be placed based on their fermentation patterns?
12. Give two reasons why the enterobacteria are so important.

22.4 Class *Deltaproteobacteria*

Although the **δ-proteobacteria** are not a large assemblage of genera, they show considerable morphological and physiological diversity. These bacteria can be divided into two general groups, all of them chemoorganotrophs. Some genera are predators such as the bdellovibrios and myxobacteria. Others are anaerobes that generate sulfide from sulfate and sulfur while oxidizing organic nutrients. The class has seven orders and 17 families. **Figure 22.30** illustrates the phylogenetic relationships between selected δ-proteobacteria, and **table 22.8** summarizes the general properties of some representative genera.

Orders *Desulfovibrionales, Desulfobacterales,* and *Desulfuromonadales*

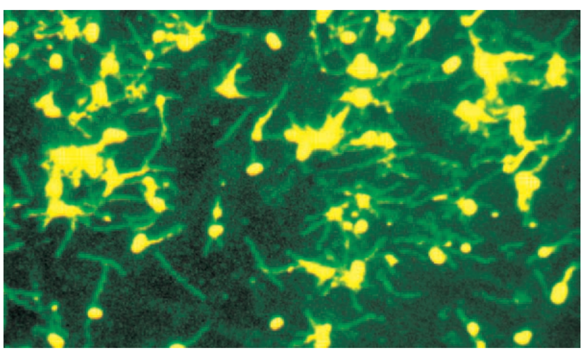
These sulfate- or sulfur-reducing bacteria are a diverse group united by their anaerobic nature and the ability to use elemental sulfur or sulfate and other oxidized sulfur compounds as electron acceptors during anaerobic respiration (**figure 22.31**). An electron transport chain generates ATP and reduces sulfur and sulfate to hydrogen sulfide. The best-studied sulfate-reducing genus is *Desulfovibrio; Desulfuromonas* uses only elemental sulfur as an acceptor. Anaerobic respiration (pp. 190–91)

These bacteria are very important in the cycling of sulfur within the ecosystem. Because significant amounts of sulfate are present in almost all aquatic and terrestrial habitats, sulfate-reducing bacteria are widespread and active in locations made anaerobic by microbial digestion of organic materials. *Desulfovibrio* and other sulfate-reducing bacteria thrive in habitats such as muds and sediments of polluted lakes and streams, sewage lagoons and digesters, and waterlogged soils. *Desulfuromonas* is most prevalent in anaerobic marine and estuarine sediments. It also can be isolated from methane digesters and anaerobic hydrogen-sulfide rich muds of freshwater habitats. It uses elemental sulfur, but not sulfate, as its electron acceptor. Often sulfate and sulfur reduction are apparent from the smell of hydrogen sulfide and the blackening of water and sediment by iron sulfide.

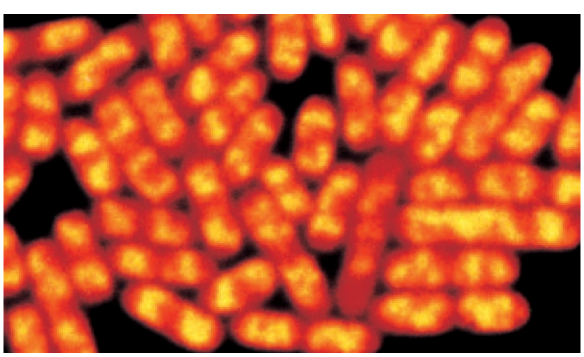
Table 22.7 Some Characteristics of Selected Genera in the *Enterobacteriaceae*

Characteristics	*Escherichia*	*Shigella*	*Salmonella*	*Citrobacter*	*Proteus*
Methyl red	+	+	+	+	+
Voges-Proskauer	–	–	–	–	d
Indole production	(+)	d	–	d	d
Citrate use	–	–	(+)	+	d
H_2S production	–	–	(+)	d	(+)
Urease	–	–	–	(+)	+
β-galactosidase	(+)	d	d	+	–
Gas from glucose	+	–	(+)	+	+
Acid from lactose	+	–	(–)	d	–
Phenylalanine deaminase	–	–	–	–	+
Lysine decarboxylase	(+)	–	(+)	–	–
Ornithine decarboxylase	(+)	d	(+)	(+)	d
Motility	d	–	(+)	+	+
Gelatin liquifaction (22°C)	–	–	–	–	+
% G + C	48–52	49–53	50–53	50–52	38–41
Other characteristics	1.1–1.5 × 2.0–6.0 µm; peritrichous when motile	No gas from sugars	0.7–1.5 × 2–5 µm; peritrichous flagella	1.0 × 2.0–6.0 µm; peritrichous	0.4–0.8 × 1.0–3.0 µm; peritrichous

[a](+) usually present
[b](–) usually absent
[c]d, strains or species vary in possession of characteristic



(a)



(b)

Figure 22.29 **The *Enterobacteriaceae*.** *Salmonella* treated with fluorescent stains. (**a**) *Salmonella enteritidis* with peritrichous flagella (×500). *S. enteritidis* is associated with gastroenteritis. (**b**) *Salmonella typhi* with acridine orange stain (×2,000). *S. typhi* causes typhoid fever.

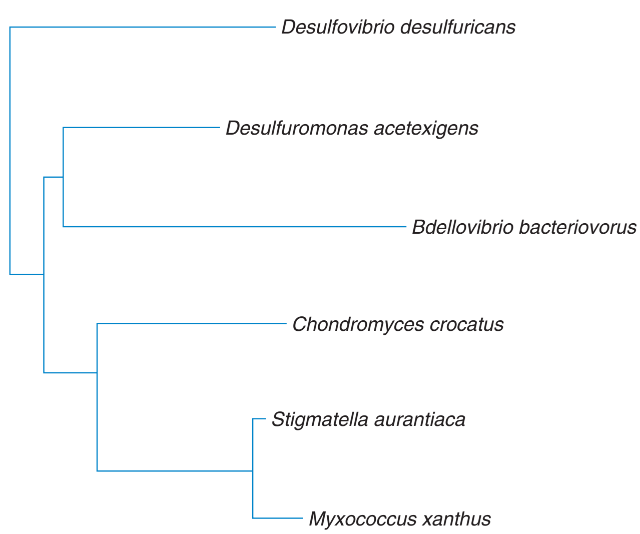
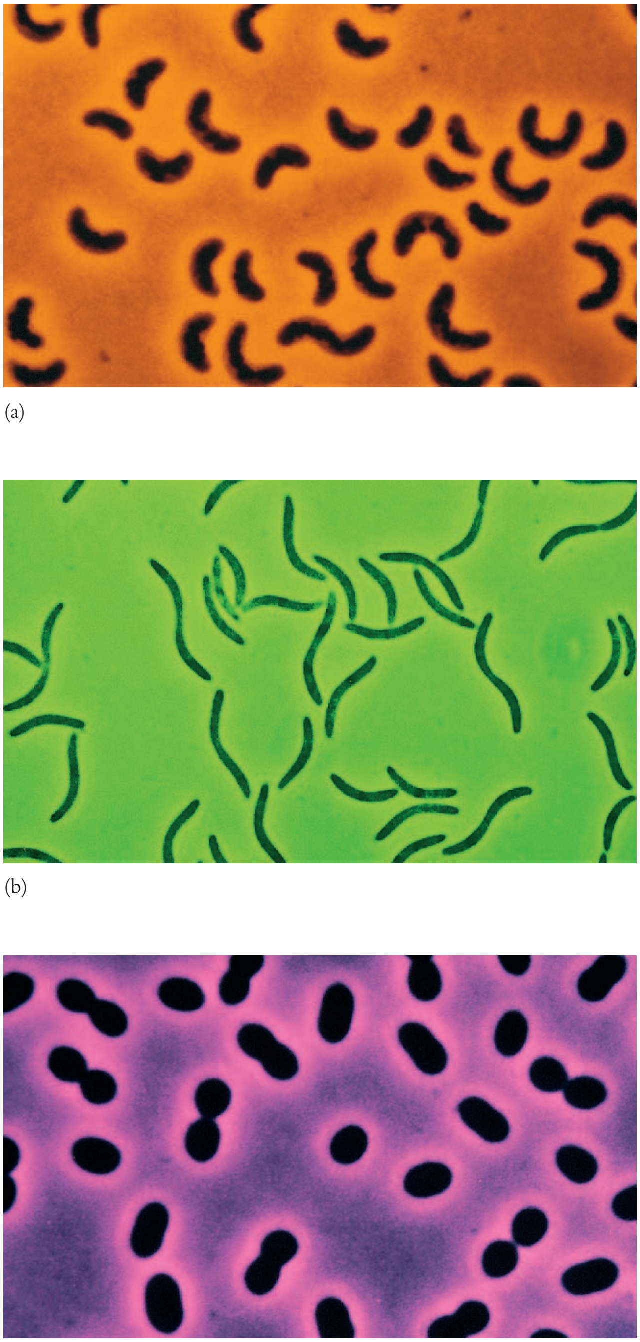


Figure 22.30 **Phylogenetic Relationships Among Selected δ-Proteobacteria.** The relationships of a few species based on 16S rRNA sequence data are presented. *Source: The Ribosomal Database Project.*

Yersinia	*Klebsiella*	*Enterobacter*	*Erwinia*	*Serratia*
+	(+)[a]	(–)[b]	+	d[c]
– (37°C)	(+)	+	(+)	+
d	d	–	(–)	(–)
(–)	(+)	+	(+)	+
–	–	–	(+)	–
d	(+)	(–)	–	–
+	(+)	+	+	+
(–)	(+)	(+)	(–)	d
(–)	(+)	(+)	d	d
–	–	(–)	(–)	–
(–)	(+)	d	–	d
d	–	(+)	–	d
– (37°C)	–	+	+	+
(–)	–	d	d	(+)
46–50	53–58	52–60	50–58	53–59
0.5–0.8 × 1.0–3.0 µm; peritrichous when motile	0.3–1.0 × 0.6–6.0 µm; capsulated	0.6–1.0 × 1.2–3.0 µm; peritrichous	0.5–1.0 × 1.0–3.0µm; peritrichous; plant pathogens and saprophytes	0.5–0.8 × 0.9–2.0 µm; peritrichous; colonies often pigmented

Table 22.8 Characteristics of Selected δ- and ϵ-Proteobacteria

Genus	Dimensions (µm) and Morphology	G + C Content (mol%)	Oxygen Requirement	Other Distinctive Characteristics
δ-Proteobacteria				
Bdellovibrio	0.2–0.5 × 0.5–1.4; comma-shaped rods with a sheathed polar flagellum	33.4–51.5	Aerobic	Preys on other gram-negative bacteria and grows in the periplasm, alternates between predatory and intracellular reproductive phases
Desulfovibrio	0.5–1.5 × 2.5–10; curved or sometimes straight rods, motile by polar flagella	46.1–61.2	Anaerobic	Oxidizes organic compounds to acetate and reduces sulfate or sulfur to H_2S
Desulfuromonas	0.4–0.9 × 1.0–4.0; straight or slightly curved or ovoid rods, lateral or subpolar flagella	50–63	Anaerobic	Reduces sulfur to H_2S, oxidizes acetate to CO_2, forms pink or peach-colored colonies
Myxococcus	0.4–0.7 × 2–10; slender rods with tapering ends, gliding motility	68–71	Aerobic	Forms fruiting bodies with microcysts not enclosed in a sporangium
Stigmatella	0.6–0.8 × 4–10; straight rods with tapered ends, gliding motility	68.5–68.7	Aerobic	Stalked fruiting bodies with sporangioles containing myxospores (0.9–1.2 × 2–4 µm)
ϵ-Proteobacteria				
Campylobacter	0.2–0.5 × 0.5–5; vibrioid cells with a single polar flagellum at one or both ends	30–38	Microaerophilic	Carbohydrates not fermented or oxidized; oxidase positive and urease negative; found in intestinal tract, reproductive organs, and oral cavity of animals
Helicobacter	0.5–1.0 × 2.5–5.0; helical, curved, or straight cells with rounded ends; multiple, sheathed flagella	33–42.5	Microaerophilic	Catalase and oxidase positive; urea rapidly hydrolyzed; found in the gastric mucosa of humans and other animals



(a)

(b)

(c)

Figure 22.31 The Dissimilatory Sulfate- or Sulfur-Reducing Bacteria. Representative examples. (**a**) Phase-contrast micrograph of *Desulfovibrio saprovorans* with PHB inclusions (×2,000). (**b**) *Desulfovibrio gigas;* phase contrast (×2,000). (**c**) *Desulfobacter postgatei;* phase contrast (×2,000).

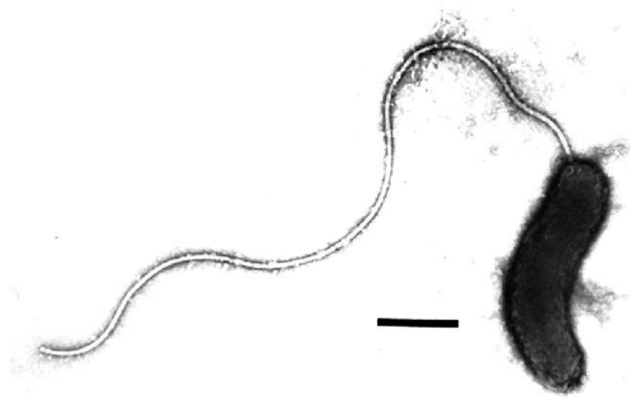


Figure 22.32 *Bdellovibrio* Morphology. Negatively stained *Bdellovibrio bacteriovorus* with its sheathed polar flagellum. Bar = 0.2 μm.

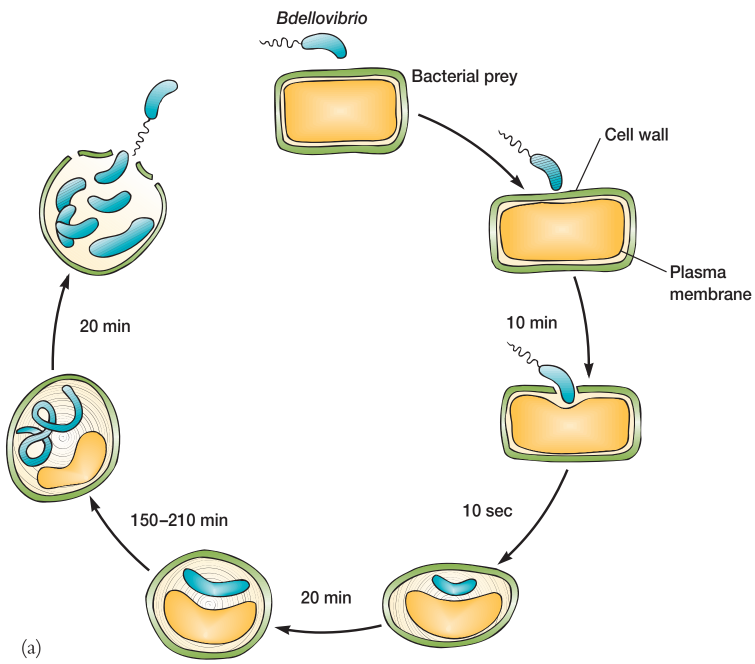
Hydrogen sulfide production in waterlogged soils can kill animals, plants, and microorganisms. Sulfate-reducing bacteria negatively impact industry because of their primary role in the anaerobic corrosion of iron in pipelines, heating systems, and other structures (*see chapter 42*). The sulfur cycle (pp. 614–15)

Order *Bdellovibrionales*

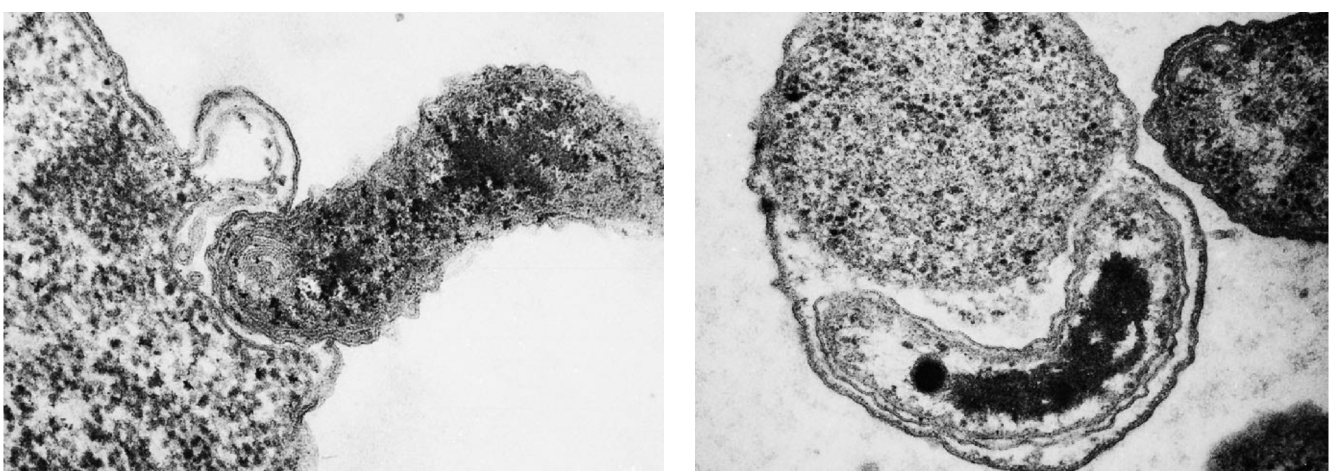
The order has only the family *Bdellovibrionaceae* and three genera. The genus *Bdellovibrio* [Greek *bdella,* leech] contains aerobic gram-negative, curved rods with polar flagella (**figure 22.32**). The flagellum is unusually thick due to the presence of a sheath that is continuous with the cell wall. *Bdellovibrio* has a distinctive life-style: it preys on other gram-negative bacteria and alternates between a nongrowing predatory phase and an intracellular reproductive phase.

The life cycle of *Bdellovibrio* is complex although it requires only 1 to 3 hours for completion (**figure 22.33**). The free bacterium swims along very rapidly (about 100 cell lengths per second) until it collides violently with its prey. It attaches to the bacterial surface, begins to rotate at rates as high as 100 revolutions per second, and bores a hole through the host cell wall in 5 to 20 minutes with the aid of several hydrolytic enzymes that it releases. Its flagellum is lost during penetration of the cell.

After entry, *Bdellovibrio* takes control of the host cell and grows in the space between the cell wall and plasma membrane while the host cell loses its shape and rounds up. The predator inhibits host DNA, RNA, and protein synthesis within minutes and disrupts the host's plasma membrane so that cytoplasmic constituents can leak out of the cell. The growing bacterium uses host amino acids as its carbon, nitrogen, and energy source. It employs fatty acids and nucleotides directly in biosynthesis, thus saving

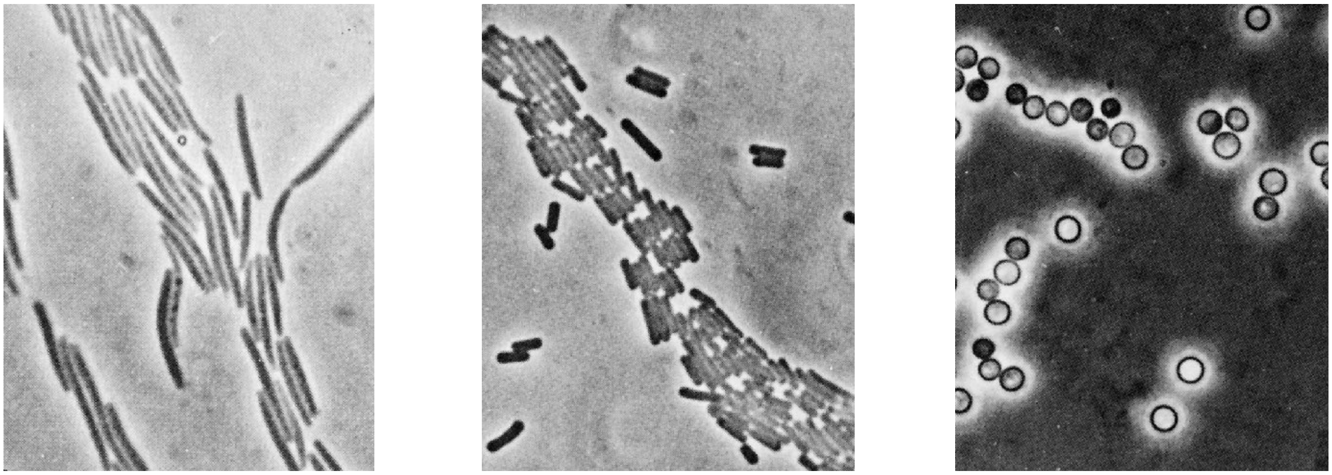


(a)



(b) (c)

Figure 22.33 The Life Cycle of *Bdellovibrio*. (**a**) A general diagram showing the complete life cycle (see text for details). (**b**) *Bdellovibrio bacteriovorus* penetrating the cell wall of *E. coli* (×55,000). (**c**) A *Bdellovibrio* encapsulated between the cell wall and plasma membrane of *E. coli* (×60,800).



(a) (b) (c)

Figure 22.34 Gliding, Fruiting Bacteria (Myxobacteria). Myxobacterial cells and myxospores. (**a**) *Stigmatella aurantiaca* (×1,200). (**b**) *Chondromyces crocatus* (×950). (**c**) Myxospores of *Myxococcus xanthus* (×1,100). All photographs taken with a phase-contrast microscope.

carbon and energy. The bacterium rapidly grows into a long filament under the cell wall and then divides into many smaller, flagellated progeny, which escape upon host cell lysis. Such multiple fission is rare in procaryotes.

The *Bdellovibrio* life cycle resembles that of bacteriophages in many ways. Not surprisingly, if a *Bdellovibrio* culture is plated out on agar with host bacteria, plaques will form in the bacterial lawn. This technique is used to isolate pure strains and count the number of viable organisms just as it is with phages.

Order *Myxococcales*

The **myxobacteria** are gram-negative, aerobic soil bacteria characterized by gliding motility, a complex life cycle with the production of fruiting bodies, and the formation of dormant myxospores. In addition, their G + C content is around 67 to 71%, significantly higher than that of most gliding bacteria. Myxobacterial cells are rods, about 0.6 to 0.9 by 3 to 8 μm long, and may be either slender with tapered ends or stout with rounded, blunt ends (**figure 22.34**). The order *Myxobacteriales* is divided into four families based on the shape of vegetative cells, myxospores, and sporangia.

Most myxobacteria are micropredators or scavengers. They secrete an array of digestive enzymes that lyse bacteria and yeasts. Many myxobacteria also secrete antibiotics, which may kill their prey. The digestion products, primarily small peptides, are absorbed. Most myxobacteria use amino acids as their major source of carbon, nitrogen, and energy. All are chemoheterotrophs with respiratory metabolism.

The myxobacterial life cycle is quite distinctive and in many ways resembles that of the cellular slime molds (**figure 22.35**). In the presence of a food supply, myxobacteria migrate along a solid surface, feeding and leaving slime trails. During this stage the cells often form a swarm and move in a coordinated fashion. Some species congregate to produce a sheet of cells that moves rhythmically to generate waves or ripples. When their nutrient supply is exhausted, the myxobacteria aggregate and differentiate into a **fruiting body.** This is a complex developmental process triggered by starvation and at least five different signals. Two of these signals have been characterized. Both the A factor, a mixture of peptides and amino acids, and the protein C factor are released and help trigger the process. At least 15 new proteins are synthesized during fruiting body formation. Fruiting bodies range in height from 50 to 500 μm and often are attractively colored red, yellow, or brown by carotenoid pigments. They vary in complexity from simple globular objects made of about 100,000 cells (*Myxococcus*) to the elaborate, branching, treelike structures formed by *Stigmatella* and *Chondromyces* (**figure 22.36**). Some cells develop into dormant **myxospores** that frequently are enclosed in walled structures called sporangioles or sporangia. Each species forms a characteristic fruiting body.

Myxospores are not only dormant but desiccation-resistant, and they may survive up to 10 years under adverse conditions. They enable myxobacteria to survive long periods of dryness and nutrient deprivation. The use of fruiting bodies provides further protection for the myxospores and assists in their dispersal. (The myxospores often are suspended above the soil surface.) Because myxospores are kept together within the fruiting body, a colony of myxobacteria automatically develops when the myxospores are released and germinate. This communal organization may be advantageous because myxobacteria obtain nutrients by secreting hydrolytic enzymes and absorbing soluble digestive products. A mass of myxobacteria can produce en-

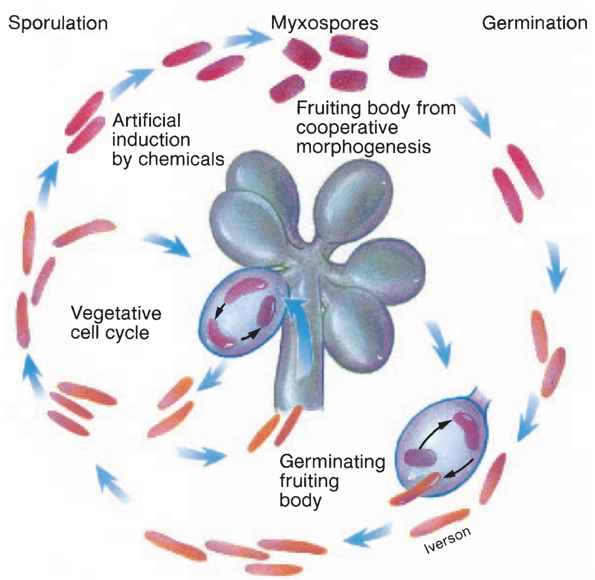


Figure 22.35 Myxobacterial Life Cycle. The outermost sequence depicts the chemical induction of myxospore formation followed by germination. Regular fruiting body production and myxospore germination are also shown.

zyme concentrations sufficient to digest their prey more easily than can an individual cell. Extracellular enzymes diffuse away from their source, and an individual cell will have more difficulty overcoming diffusional losses than a swarm of cells.

Myxobacteria are found in soils worldwide. They are most commonly isolated from neutral soils or decaying plant material such as leaves and tree bark, and from animal dung. Although they grow in habitats as diverse as tropical rain forests and the arctic tundra, they are most abundant in warm areas.

1. Briefly characterize the δ-proteobacteria.
2. Describe the metabolic specialization of the dissimilatory sulfate- or sulfur-reducing bacteria. Why are they important?
3. Characterize the genus *Bdellovibrio* and outline its life cycle in detail.
4. Give the major distinguishing characteristics of the myxobacteria. How do they obtain most of their nutrients?
5. Briefly describe the myxobacterial life cycle. What are fruiting bodies, myxospores, and sporangioles?

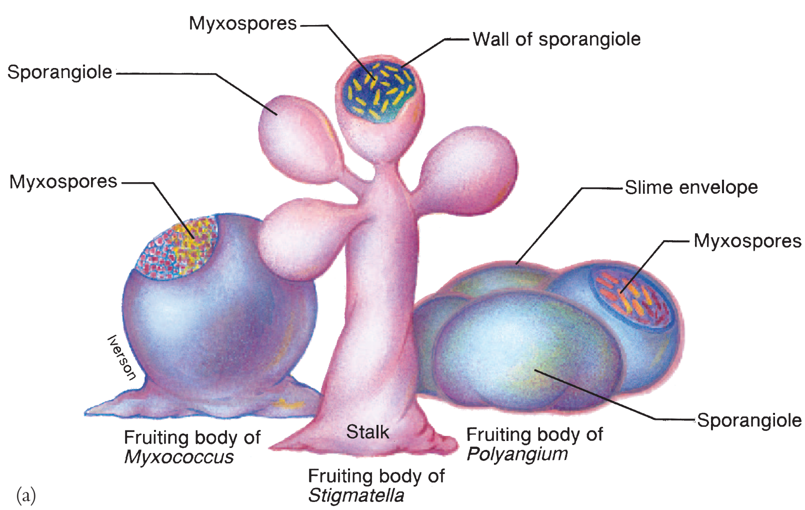
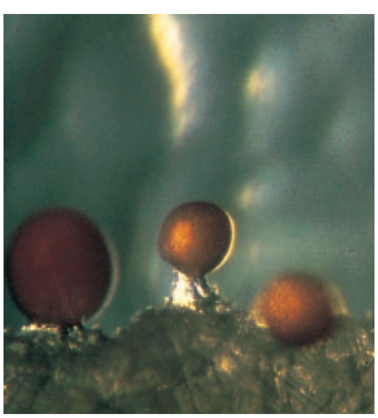
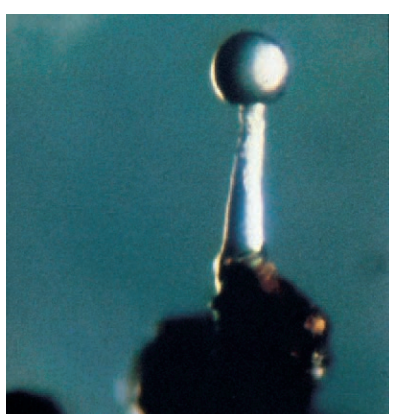


Figure 22.36 Myxobacterial Fruiting Bodies (**a**) An illustration of typical fruiting body structure. (**b**) *Myxococcus fulvus.* Fruiting bodies are about 150–400 μm high. (**c**) *Myxococcus stipitatus.* The stalk is as tall as 200 μm. (**d**) *Chondromyces crocatus* viewed with the SEM. The stalk may reach 700 μm or more in height.

(b)


(c)


(d)


22.5 Class *Epsilonproteobacteria*

The **ϵ-proteobacteria** are the smallest of the five proteobacterial classes. They all are slender gram-negative rods, which can be straight, curved, or helical. The ϵ-proteobacteria have one order, *Campylobacterales,* and two families, *Campylobacteraceae* and *Helicobacteraceae.* The two most important genera, *Campylobacter* and *Helicobacter,* are microaerophilic, motile, helical or vibrioid, gram-negative rods. Table 22.8 summarizes some of the characteristics of these two genera.

The genus *Campylobacter* contains both nonpathogens and species pathogenic for humans and other animals. *C. fetus* causes reproductive disease and abortions in cattle and sheep. It is associated with a variety of conditions in humans ranging from **septicemia** (pathogens or their toxins in the blood) to **enteritis** (inflammation of the intestinal tract). *C. jejuni* causes abortion in sheep and enteritis diarrhea in humans.

There are at least 14 species of *Helicobacter,* all isolated from the stomachs and upper intestines of humans, dogs, cats, and other mammals (*see figure 39.16*). In developing countries 70 to 90% of the population is infected; developed countries range from 25 to 50%. Most infections are probably acquired during childhood, but the precise mode of transmission is unclear. The major human pathogen is *Helicobacter pylori,* which is the cause of gastritis and peptic ulcer disease (*see pp. 918–19). H. pylori* produces large quantities of urease, and urea hydrolysis appears to be associated with its virulence.

The genomes of *Campylobacter jejuni* and *Helicobacter pylori* (both about 1.6 million base pairs in size) have been sequenced. They are now being studied and compared in order to understand the life styles and pathogenicity of these bacteria.

1. Briefly describe the properties of the ϵ-proteobacteria.
2. Give the general characteristics of *Campylobacter* and *Helicobacter.* What is their public health significance?

Summary

1. The proteobacteria are the largest and most diverse group of bacteria. On the basis of rRNA sequence data, they are divided into five classes: the α-, β-, γ-, δ-, and ϵ-proteobacteria.
2. The purple nonsulfur bacteria can grow anaerobically as photoorganoheterotrophs and often aerobically as chemoorganoheterotrophs (**figure 22.2**). They are found in aquatic habitats with abundant organic matter and low sulfide levels.
3. Rickettsias are obligately intracellular parasites responsible for many diseases (**figure 22.3**). They have plasma membrane carriers and make extensive use of host cell nutrients, coenzymes, and ATP.
4. Many bacteria have prosthecae, stalks, or reproduction by budding. These are often placed among the α-proteobacteria.
5. Two examples of budding and/or appendaged bacteria are *Hyphomicrobium* (budding bacteria that produce swarmer cells) and *Caulobacter* (bacteria with prosthecae and holdfasts) (**figures 22.4–22.7**).
6. *Rhizobium* carries out nitrogen fixation, whereas *Agrobacterium* causes the development of plant tumors. Both are in the family *Rhizobiaceae* of the α-proteobacteria.
7. Chemolithotrophic bacteria derive energy and electrons from reduced inorganic compounds. Nitrifying bacteria are aerobes that oxidize either ammonia or nitrite to nitrate and are responsible for nitrification (**table 22.2**).
8. The genus *Neisseria* contains nonmotile, aerobic, gram-negative cocci that usually occur in pairs. They colonize mucous membranes and cause several human diseases.
9. Members of the genus *Burkholderia* are aerobic gram-negative rods that almost always have polar flagella. They degrade a wide variety of organic molecules and can cause disease.
10. *Sphaerotilus, Leptothrix,* and several other genera have sheaths, hollow tubelike structures that surround chains of cells without being in intimate contact (**figure 22.13**).
11. The colorless sulfur bacteria such as *Thiobacillus* oxidize elemental sulfur, hydrogen sulfide, and thiosulfate to sulfate while generating energy chemolithotrophically.
12. The γ-proteobacteria are the largest subgroup of proteobacteria with great variety in physiological types (**table 22.5** and **figure 22.16**).
13. The purple sulfur bacteria are anaerobes and usually photolithoautotrophs. They oxidize hydrogen sulfide to sulfur and deposit the granules internally (**figure 22.18**).
14. Bacteria like *Beggiatoa* and *Leucothrix* grow in long filaments or trichomes (**figures 22.20** and **22.21**). Both genera have gliding motility. *Beggiatoa* is primarily a chemolithotroph and *Leucothrix,* a chemoorganotroph.
15. The *Methylococcaceae* are methylotrophs; they use methane, methanol, and other reduced one-carbon compounds as their sole carbon and energy sources.
16. The genus *Pseudomonas* contains straight or slightly curved, gram-negative, aerobic rods that are motile by one or several polar flagella and do not have prosthecae or sheaths (**figure 22.23**).
17. The pseudomonads participate in natural mineralization processes, are major experimental subjects, cause many diseases, and often spoil refrigerated food.
18. The most important facultatively anaerobic, gram-negative rods are found in three families: *Vibrionaceae, Enterobacteriaceae,* and *Pasteurellaceae* (**table 22.6**). These bacteria are in the γ-proteobacteria (orders *Vibrionales, Enterobacteriales,* and *Pasteurellales*).
19. The *Enterobacteriaceae,* often called enterobacteria or enteric bacteria, are gram-negative, peritrichously flagellated or nonmotile, facultatively anaerobic, straight rods with simple nutritional requirements.
20. The enteric bacteria are usually identified by a variety of physiological tests and are very important experimental organisms and pathogens of plants and animals (**table 22.7** and **figure 22.28**).
21. The δ-proteobacteria contain gram-negative bacteria that are anaerobic and can use elemental sulfur and oxidized sulfur compounds as electron acceptors in anaerobic respiration (**table 22.8**). They are very important in sulfur cycling in the ecosystem.
22. *Bdellovibrio* is an aerobic curved rod with sheathed polar flagellum that preys on other gram-negative bacteria and grows within their periplasmic space (**figures 22.32** and **22.33**).
23. Myxobacteria are gram-negative, aerobic soil bacteria with gliding motility and a complex life cycle that leads to the production of dormant myxospores held within fruiting bodies (**figures 22.34–22.36**).
24. The ϵ-proteobacteria are the smallest of the proteobacterial classes and contain two important pathogenic genera: *Campylobacter* and *Helicobacter.* These are microaerophilic, motile, helical or vibrioid, gram-negative rods.

Key Terms

α-proteobacteria *487*
β-proteobacteria *495*
binary fission *490*
bioluminescence *505*
budding *490*
colorless sulfur bacteria *496*
δ-proteobacteria *507*
enteric bacteria (enterobacteria) *505*
enteritis *514*
ε-proteobacteria *514*
fruiting body *512*
γ-proteobacteria *498*
holdfast *491*
methylotroph *491*
mineralization process *504*
myxobacteria *512*
myxospores *512*
nitrification *495*
nitrifying bacteria *493*
prostheca *490*
proteobacteria *487*
purple nonsulfur bacteria *488*
purple sulfur bacteria *500*
septicemia *514*
sheath *496*
stalk *490*

Questions for Thought and Review

1. Describe the general characteristics and organization of each of the five classes of proteobacteria.
2. Bacterial genera are distributed quite differently in the first and second editions of *Bergey's Manual.* Discuss the advantages and disadvantages of the phenetic and phylogenetic approaches to bacterial taxonomy shown by the first and second editions.
3. A number of bacterial groups are found in particular habitats to which they are well adapted. Give the habitat for each of the following groups and discuss the reasons for its preference: *Beggiatoa,* purple sulfur bacteria, *Hyphomicrobium,* and *Thiobacillus.*
4. Why might the ability to form dormant cysts be of great advantage to *Agrobacterium* but not as much to *Rhizobium?*
5. What advantage might a species gain by specializing in its nutrient requirements or habitat as methylotrophs do? Are there disadvantages to this strategy?
6. A pond with black water and a smell like rotten eggs might contain an excess of what kind of bacteria?
7. Why are biochemical tests of such importance in identifying enteric bacteria?
8. How does *Bdellovibrio* take advantage of its host's metabolism to grow rapidly and efficiently? In what ways does its reproduction resemble that of bacteriophages?
9. Why are gliding and budding and/or appendaged bacteria distributed among so many different sections in *Bergey's Manual?*
10. Discuss the ways in which the distinctive myxobacterial life cycle may be of great advantage to these organisms.

Critical Thinking Questions

1. *Helicobacter pylori* produces large quantities of urease. Urease catalyzes the reaction:

$$H_2N{-}\overset{\overset{\displaystyle O}{\|}}{C}{-}NH_2 \rightarrow CO_2 + 2NH_3$$

   Suggest why this allows *H. pylori* to inhabit the acidity of the gastric mucosa.
2. Methylotrophs oxidize methane to methanol, then to formaldehyde, and finally into acetate. Suggest mechanisms by which the bacterium protects itself from the toxic effects of the intermediates, methanol and formaldehyde.
3. *Bdellovibrio* is an intracellular predator. Once it invades the periplasm, it manages to inhibit many aspects of host metabolism. Suggest a mechanism by which this inhibition could occur so rapidly.

Additional Reading

General

Balows, A.; Trüper, H. G.; Dworkin, M.; Harder, W.; and Schleifer, K.-H. 1992. *The prokaryotes,* 2d ed. New York: Springer-Verlag.

Brun, Y. V. 2000. Developmental processes in bacteria. In *Encyclopedia of microbiology,* 2d ed., vol. 2. J. Lederberg, editor-in-chief, 15–28. San Diego: Academic Press.

Holt, J. G., editor-in-chief. 1984. *Bergey's Manual of systematic bacteriology,* vol. 1, N. R. Krieg, editor. Baltimore, Md.: Williams & Wilkins.

Holt, J. G., editor-in-chief. 1989. *Bergey's Manual of Systematic Bacteriology,* vol. 3, J. T. Staley, M. P. Bryant, and N. Pfennig, editors. Baltimore, Md.: Williams & Wilkins.

Holt, J. G., editor-in-chief. 1994. *Bergey's Manual of Determinative Bacteriology,* 9th ed. Baltimore, Md.: Williams & Wilkins.

Lederberg, J., editor. 1992. *Encyclopedia of microbiology.* San Diego, Calif.: Academic Press.

Mayer, F. 1986. *Cytology and morphogenesis of bacteria.* Berlin: Gebrüder Borntraeger.

Schlegel, H. G., and Bowien, B. 1989. *Autotrophic bacteria.* Madison, Wis.: Science Tech Publishers.

Shapiro, J. A. 1988. Bacteria as multicellular organisms. *Sci. Am.* 258(6):82–89.

Zavarzin, G. A.; Stackebrandt, E.; Murray, R. G. E. 1991. A correlation of phylogenetic diversity in the *Proteobacteria* with the influences of ecological forces. *Can. J. Microbiol.* 37:1–6.

22.1 Class *Alphaproteobacteria*

Brun, Y. V.; Marczynski, G.; and Shapiro, L. 1994. The expression of asymmetry during caulobacter cell differentiation. *Annu. Rev. Biochem.* 63:419–50.

Eremeeva, M. E., and Dasch, G. A. 2000. Rickettsiae. In *Encyclopedia of microbiology,* 2d ed., vol. 4, J. Lederberg, editor-in-chief, 140–80. San Diego: Academic Press.

Gober, J. W., and Marques, M. V. 1995. Regulation of cellular differentiation in *Caulobacter crescentus. Microbiol. Rev.* 59(1):31–47.

Hase, T. 1985. Developmental sequence and surface membrane assembly of rickettsiae. *Annu. Rev. Microbiol.* 39:69–88.

Hooykaas, P. J. J. 2000. Agrobacterium. In *Encyclopedia of microbiology,* 2d ed., vol. 1, J. Lederberg, editor-in-chief, 78–85. San Diego: Academic Press.

Moore, R. L. 1981. The biology of *Hyphomicrobium* and other prosthecate, budding bacteria. *Annu. Rev. Microbiol.* 35:567–94.

Østerås, M., and Jenal, U. 2000. Regulatory circuits in *Caulobacter. Curr. Opin. Microbiol.* 3:171–76.

Poindexter, J. S. 1981. The Caulobacters: Ubiquitous unusual bacteria. *Microbiol. Rev.* 45:123–79.

Winkler, H. H. 1990. Rickettsia species (as organisms). *Annu. Rev. Microbiol.* 44:131–53.

22.2 Class *Betaproteobacteria*

Gillis, M., et al. 1995. Polyphasic taxonomy in the genus *Burkholderia* leading to an emended description of the genus and proposition of

Burkholderia vietnamiensis sp. nov. for N_2-fixing isolates from rice in Vietnam. *Int. J. Syst. Bacteriol.* 45(2):274–89.

Tettelin, H., et al. 2000. Complete genome sequence of *Neisseria meningitidis* serogroup B strain MC58. *Science* 287(5459):1809–15.

22.3 Class *Gammaproteobacteria*

Burchart, R. P. 1981. Gliding motility of prokaryotes: Ultrastructure, physiology, and genetics. *Annu. Rev. Microbiol.* 35:497–529.

Dunlap, P. V. 1995. Making a living on cyclic AMP. *ASM News* 61(10):511–16.

Haber, C. L.; Allen, L. N.; Zhao, S.; and Hanson, R. S. 1983. Methylotrophic bacteria: Biochemical diversity and genetics. *Science* 221:1147–53.

Hanson, R. S., and Hanson, T. E. 1996. Methanotrophic bacteria. *Microbiol. Rev.* 60(2):439–71.

Heidelberg, J. F., et al. 2000. DNA sequence of both chromosomes of the cholera pathogen *Vibrio cholerae. Nature* 406:477–83.

Hill, S., and Sawers, G. 2000. *Azotobacter.* In *Encyclopedia of microbiology,* 2d ed., vol. 1, J. Lederberg, editor-in-chief, 359–71. San Dieg: Academic Press.

Kapatral, V.; Zago, A.; Kamath, S.; and Chugani, S. 2000. *Pseudomonas.* In *Encyclopedia of microbiology,* 2d ed., vol. 3, J. Lederberg, editor-in-chief, 876–92. San Diego: Academic Press.

Larkin, J. M., and Strohl, W. R. 1983. *Beggiatoa, Thiothrix,* and *Thioploca. Annu. Rev. Microbiol.* 37:341–67.

Low, B. K. 2000. *Escherichia coli* and *Salmonella,* Genetics. In *Encyclopedia of microbiology,* 2d ed., vol. 2, J. Lederberg, editor-in-chief, 270–82. San Diego: Academic Press.

MacDonell, M. T.; Swartz, D. G.; Ortiz-Conde, B. A.; Last, G. A.; and Colwell, R. T. 1986. Ribosomal RNA phylogenies for the vibrio-enteric group of eubacteria. *Microbiol. Sci.* 3:172–78.

Meighen, E. A. 1991. Molecular biology of bacterial bioluminescence. *Microbiol. Rev.* 55(1):123–42.

Neidhardt, F. C., editor-in-chief. 1996. *Escherichia coli* and *Salmonella typhimurium,* 2d ed. Washington, D.C.: ASM Press.

Perna, N. T., et al. 2001. Genome sequence of enterohaemorrhagic *Escherichia coli* O157:H7. *Nature* 409:529–33.

Reichenbach, H. 1981. Taxonomy of the gliding bacteria. *Annu. Rev. Microbiol.* 35:339–64.

Schaechter, M., and The View from Here Group. 2001. *Escherichia coli* and *Salmonella* 2000: The view from here. *Microbiol. Mol. Biol. Rev.* 65(1):119–30.

Schaechter, M. 2000. *Escherichia coli,* General biology. In *Encyclopedia of microbiology,* 2d ed., vol. 2, J. Lederberg, editor-in-chief, 260–69. San Diego: Academic Press.

Starr, M. P., and Chatterjee, A. K. 1972. The genus *Erwinia:* Enterobacteria pathogenic to plants and animals. *Annu. Rev. Microbiol.* 26:389–426.

Stover, C. K., et al. 2000. Complete genome sequence of *Pseudomonas aeruginosa* PAO1, an opportunistic pathogen. *Nature* 406:959–64.

22.4 Class *Deltaproteobacteria*

Diedrich, D. L. 1988. Bdellovibrios: Recycling, remodelling and relocalizing components from their prey. *Microbiol. Sci.* 5(4):100–103.

Dworkin, M. 1996. Recent advances in the social and developmental biology of the myxobacteria. *Microbiol. Rev.* 60(1):70–102.

Dworkin, M., and Kaiser, D. 1985. Cell interactions in myxobacterial growth and development. *Science* 230:18–24.

Shimkets, L. J. 1990. Social and developmental biology of the myxobacteria. *Microbiol. Rev.* 54(4):473–501.

Spormann, A. M. 1999. Gliding motility in bacteria: Insights from studies of *Myxococcus xanthus. Micro. Mol. Biol. Rev.* 63(3):621–41.

White, D. 2000. Myxobacteria. In *Encyclopedia of microbiology,* 2d ed., vol. 3, J. Lederberg, editor-in-chief, 349–62. San Diego: Academic Press.

22.5 Class *Epsilonproteobacteria*

Cover, T. L., and Blaser, M. J. 1995. *Helicobacter pylori:* A bacterial cause of gastritis, peptic ulcer disease, and gastric cancer. *ASM News* 61(1):21–26.

Dunn, B. E.; Cohen, H.; and Blaser, M. J. 1997. *Helicobacter pylori. Clin. Microbiol. Rev.* 10(4):720–41.

Marais, A.; Mendz, G. L.; Hazell, S. L.; and Mégraud, F. 1999. Metabolism and genetics of *Helicobacter pylori:* The genome era. *Microbiol. Mol. Biol. Rev.* 63(3):642–74.

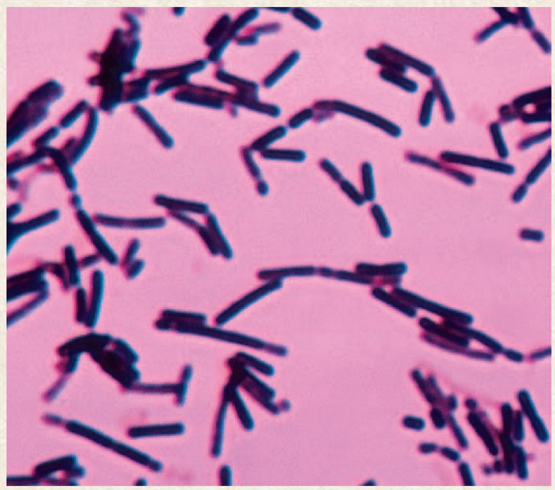
Parkhill, J., et al. 2000. The genome sequence of the food-borne pathogen *Campylobacter jejuni* reveals hypervariable sequences. *Nature* 403:655–68.

Tomb, J.-F., et al. 1997. The complete genome sequence of the gastric pathogen *Helicobacter pylori. Nature* 388:539–47.

Walker, R. I., et al. 1986. Pathophysiology of *Campylobacter* enteritis. *Microbiol. Rev.* 50(1):81–94.

CHAPTER 23

Bacteria: The Low G + C Gram Positives



Lactobacilli are indispensable to the food and dairy industry. They are not considered pathogens.

Outline

Concepts

1. Volume 2 of the first edition of *Bergey's Manual* contains six sections covering all gram-positive bacteria except the actinomycetes. Bacteria are distributed among these sections on the basis of their shape, the ability to form endospores, acid fastness, oxygen relationships, the ability to temporarily form mycelia, and other properties.
2. The second edition of *Bergey's Manual* groups the gram-positive bacteria phylogenetically into two major groups: the low G + C gram-positive bacteria and the high G + C gram-positive bacteria. The new classification is based primarily on nucleic acid sequences rather than phenotypic similarity.
3. The low G + C gram positives contain (1) clostridia and relatives, (2) the mycoplasmas, and (3) the bacilli and lactobacilli. Endospore formers, cocci, and rods are found among the clostridia and bacilli groups rather than being placed in separate sections as in the first edition. Thus common possession of a complex structure such as an endospore does not necessarily indicate close relatedness between the genera.
4. Peptidoglycan structure varies among different groups in ways that are often useful in identifying specific groups.
5. Although most gram-positive bacteria are harmless free-living saprophytes, some species from most major groups are pathogens of humans, other animals, and plants. Other gram-positive bacteria are very important in the food and dairy industries.

We noted, after having grown the bacterium through a series of such cultures, each fresh culture being inoculated with a droplet from the previous culture, that the last culture of the series was able to multiply and act in the body of animals in such a way that the animals developed anthrax with all the symptoms typical of this affection.

Such is the proof, which we consider flawless, that anthrax is caused by this bacterium.

—*Louis Pasteur*

Chapter 23 surveys many of the bacteria found in volume 2 of the first edition of *Bergey's Manual of Systematic Bacteriology.* Volume 2 contains all the major groups of gram-positive bacteria except the actinomycetes, which are covered in volume 4. The second edition of *Bergey's Manual* will divide the gram-positive bacteria differently for reasons discussed a little later. Thus this chapter describes the general differences between the first and second edition, and then focuses on the mycoplasmas, *Clostridium* and its relatives, and the bacilli and lactobacilli. The remaining groups in volume 2 will be described in chapter 24 along with the actinomycetes.

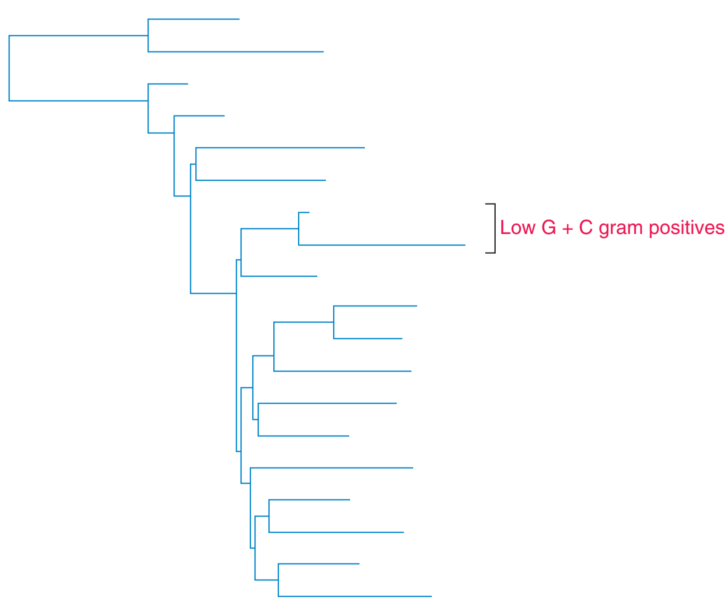
Volume 2 of the first edition of *Bergey's Manual* contains six sections that describe all gram-positive bacteria except the actinomycetes. Most of these bacteria are distributed among the first four sections on the basis of their general shape (whether they are rods, cocci, or irregular) and their ability to form endospores. The rod-shaped, acid-fast bacteria known as mycobacteria are placed in section 16 (the sections in the first edition of *Bergey's Manual* are numbered consecutively, beginning with the first section in volume 1). The last section describes the nocardioforms. These are gram-positive filamentous bacteria that form a branching network called a mycelium, which can break into rod-shaped or coccoid forms.

Analysis of the phylogenetic relationships within the gram-positive bacteria by comparison of 16S rRNA sequences *(see figure 19.15)* shows that they are divided into a low G + C group (see bottom dendrogram) and high G + C or actinobacterial group. The distribution of genera within and between the groups shown in figure 19.15 differs markedly from the classification in volume 2 of *Bergey's Manual.* Bacterial phylogeny and diversity (pp. 443–46); The structure of the gram-positive cell wall (pp. 56–58)

The third volume of the second edition of *Bergey's Manual* describes the low G + C gram-positive bacteria. These are placed in the phylum *Firmicutes* and divided into three classes: *Clostridia, Mollicutes,* and *Bacilli.* The phylum *Firmicutes* is large and complex; it has 10 orders and 33 families. It differs most obviously from the classification system of the first edition in containing the class *Mollicutes.* The mycoplasmas, class *Mollicutes,* are now placed with the low G + C gram positives rather than with gram-negative bacteria. Ribosomal RNA data indicate that the mycoplasmas are closely related to the clostridia, despite their lack of cell walls. **Figure 23.1** shows the phylogenetic relationships between some of the bacteria in this chapter.

23.1 Class *Mollicutes* (the Mycoplasmas)

The class *Mollicutes* has five orders and six families. The best-studied genera are found in the orders *Mycoplasmatales (Mycoplasma, Ureaplasma), Entomoplasmatales (Entomoplasma, Mesoplasma, Spiroplasma), Acholeplasmatales (Acholeplasma),*



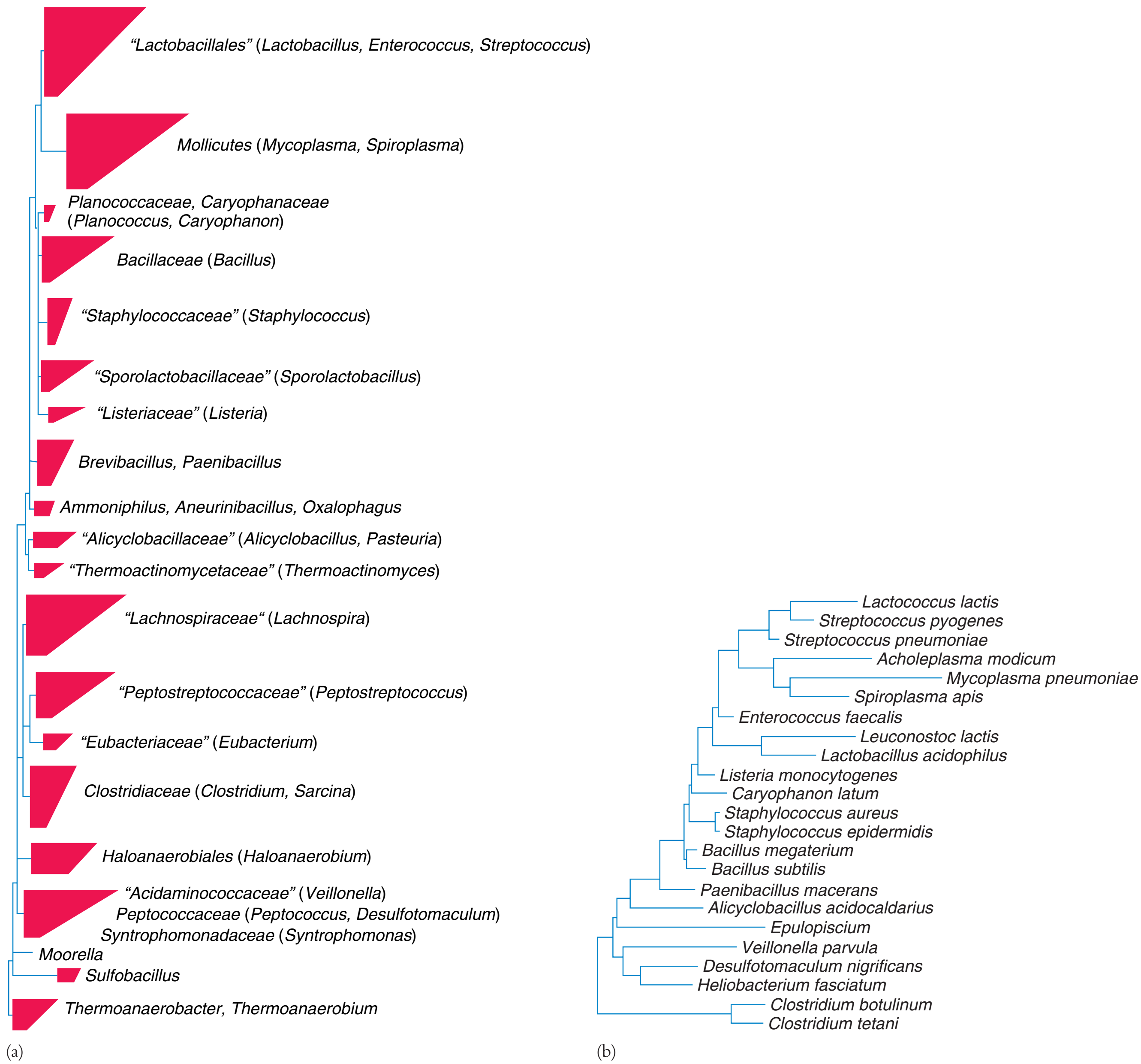


Figure 23.1 Phylogenetic Relationships in the Phylum *Firmicutes* (Low G + C Gram Positives). (**a**) The major phylogenetic groups with representative genera in parentheses. Each tetrahedron in the tree represents a group of related organisms; its horizontal edges show the shortest and longest branches in the group. Multiple branching at the same level indicates that the relative branching order of the groups cannot be determined from the data. The quotation marks around some names indicate that they are not formally approved taxonomic names. (**b**) The relationships of a few species based on 16S rRNA sequence date. *Source: The Ribosomal Database Project.*

Table 23.1 Properties of Some Members of the Class *Mollicutes*

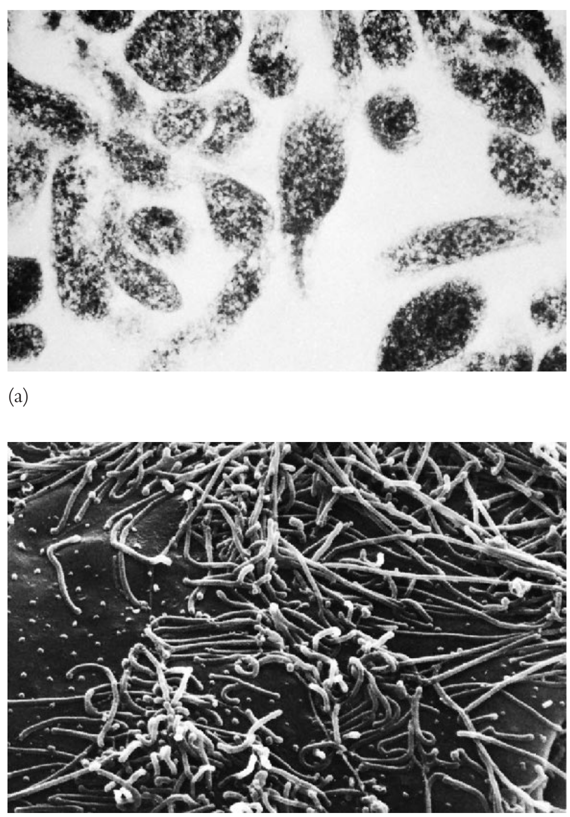
Genus	No. of Recognized Species	Guanine-Plus-Cytosine Content (mol%)	Genome Size (kpb)	Cholesterol Requirement	Habitat	Other Distinctive Features
Acholeplasma	13	26–36	1,500–1,650	No	Animals, some plants and insects	Optimum growth at 30–37°C
Anaeroplasma	4	29–34	1,500–1,600	Yes	Bovine or ovine rumen	Oxygen-sensitive anaerobes
Asteroleplasma	1	40	1,500	No	Bovine or ovine rumen	Oxygen-sensitive anaerobes
Entomoplasma	5	27–29	790–1,140	Yes	Insects, plants	Optimum growth, 30°C
Mesoplasma	12	27–30	870–1,100	No	Insects, plants	Optimum growth, 30°C; sustained growth in serum-free medium only with 0.04% Tween 80
Mycoplasma	104	23–40	600–1,350	Yes	Humans, animals	Optimum growth usually at 37°C
Spiroplasma	22	25–30	940–2,200	Yes	Insects, plants	Helical filaments; optimum growth at 30–37°C
Ureaplasma	6	27–30	760–1,170	Yes	Humans, animals	Urea hydrolysis

Adapted from J. G. Tully, et al., "Revised Taxonomy of the Class *Mollicutes*" in *International Journal of Systematic Bacteriology,* 43(2):378-85. Copyright © 1993 American Society for Microbiology, Washington, D.C. Reprinted by permission.

and *Anaeroplasmatales (Anaeroplasma, Asteroleplasma).* **Table 23.1** summarizes some of the major characteristics of these genera.

Members of the class *Mollicutes* are commonly called **mycoplasmas.** These bacteria lack cell walls and cannot synthesize peptidoglycan precursors. Thus they are penicillin resistant but susceptible to lysis by osmotic shock and detergent treatment. Because they are bounded only by a plasma membrane, these procaryotes are pleomorphic and vary in shape from spherical or pear-shaped organisms, about 0.3 to 0.8 μm in diameter, to branched or helical filaments (**figure 23.2**). Some mycoplasmas (e.g., *M. genitalium*) have a specialized terminal structure that projects from the cell and gives them a flask or pear shape. This structure aids in attachment to eucaryotic cells. They are the smallest bacteria capable of self-reproduction. Although most are nonmotile, some can glide along liquid-covered surfaces. Most species differ from the vast majority of bacteria in requiring sterols for growth. They usually are facultative anaerobes, but a few are obligate anaerobes. When growing on agar, most species will form colonies with a "fried-egg" appearance because they grow into the agar surface at the center while spreading outward on the surface at the colony edges (**figure 23.3**). Their genome is one of the smallest found in procaryotes, about 5 to 10 × 10^8 daltons; the G + C content ranges from 23 to 41%. Recently the complete genome of *Mycoplasma genitalium,* a parasite of the human genital and respiratory tracts, has been sequenced. The *M. genitalium* genome is only 580 kilobases long and appears to have 482 genes; it seems that not many genes are required to sustain a free-living existence. Mycoplasmas can be saprophytes, commensals, or parasites, and many are pathogens of plants, animals, or insects. *Mycoplasma genitalium* genome sequence (pp. 348–49)

The metabolism of mycoplasmas is not particularly unusual, although they are deficient in several biosynthetic sequences and often require sterols, fatty acids, vitamins, amino acids, purines, and pyrimidines. Those mycoplasmas needing sterols incorporate



(a)

(b)

Figure 23.2 The Mycoplasmas. Electron micrographs of *Mycoplasma pneumoniae* showing its pleomorphic nature. (**a**) A transmission electron micrograph of several cells (×47,880). The central cell appears flask or pear-shaped because of its terminal structure. (**b**) A scanning electron micrograph (×26,000).

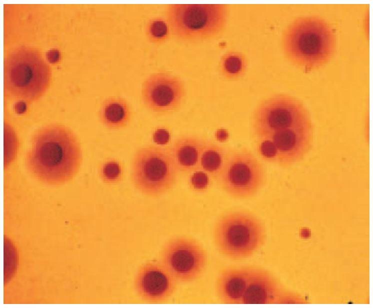


Figure 23.3 ***Mycoplasma* Colonies.** Note the "fried egg" appearance, colonies stained before viewing (×100).

them into the plasma membrane. Mycoplasmas are usually more osmotically stable than bacterial protoplasts, and their membrane sterols may be a stabilizing factor. Some produce ATP by the Embden-Meyerhof pathway and lactic acid fermentation. Others catabolize arginine or urea to generate ATP. The pentose phosphate pathway seems functional in at least some mycoplasmas; none appear to have the complete tricarboxylic acid cycle.

Mycoplasmas are remarkably widespread and can be isolated from animals, plants, the soil, and even compost piles. Indeed, about 10% of the mammalian cell cultures in use are probably contaminated with mycoplasmas, which seriously interfere with tissue culture experiments and are difficult to detect and eliminate. In animals, mycoplasmas colonize mucous membranes and joints and often are associated with diseases of the respiratory and urogenital tracts. Mycoplasmas cause several major diseases in livestock, for example, contagious bovine pleuropneumonia in cattle (*M. mycoides*), chronic respiratory disease in chickens (*M. gallisepticum*), and pneumonia in swine (*M. hyopneumoniae*). *M. pneumoniae* causes primary atypical pneumonia in humans, and there is increasing evidence that *M. hominis* and *Ureaplasma urealyticum* also are human pathogens. The 0.8 million base pair genome of *M. pneumoniae* has been sequenced. This pathogen has lost many biosynthetic pathways (e.g., amino acid biosynthesis pathways), which accounts for its parasitic lifestyle. Spiroplasmas have been isolated from insects, ticks, and a variety of plants. They cause disease in citrus plants, cabbage, broccoli, corn, honey bees, and other hosts. Arthropods probably often act as vectors and carry the spiroplasmas between plants. Presumably many more pathogenic mollicutes will be discovered as techniques for their isolation and study improve.

1. What morphological feature distinguishes the mycoplasmas? In what class are they found? Why have they been placed with the low G + C gram-positive bacteria?
2. Give other distinguishing properties of the class *Mollicutes.*
3. What might mycoplasmas use sterols for?
4. Where are mycoplasmas found in animals? List several animal and human diseases caused by them. What kinds of organisms do spiroplasmas usually infect?

23.2 Low G + C Gram-Positive Bacteria in *Bergey's Manual*

Because low G + C gram positives are treated so differently in the first and second editions of *Bergey's Manual,* it is necessary to compare the taxonomic approach used in the two editions before looking at individual groups of gram-positive bacteria. As we shall see, the fundamental difference in treatment is that of phenotypic versus phylogenetic approaches.

In the first edition of *Bergey's Manual,* most of the low G + C gram positives are located in the sections on gram-positive cocci (section 12), endospore-forming gram-positive rods and cocci (section 13), and regular, nonsporing, gram-positive rods (section 14) of volume 2. The first edition classifies these gram-positive organisms largely on the basis of observable characteristics such as cell shape, the clustering and arrangement of cells, the presence or absence of endospores, oxygen relationships, fermentation patterns, peptidoglycan chemistry, and so forth. Because of the importance of peptidoglycan and endospores in the taxonomy of these bacteria, it is best to briefly discuss these two important components of gram-positive bacteria.

Peptidoglycan structure varies considerably among different gram-positive groups. Most gram-negative bacteria have a peptidoglycan structure in which *meso*-diaminopimelic acid in position 3 is directly linked through its free amino group with the free carboxyl of the terminal D-alanine of an adjacent peptide chain (**figure 23.4*a*;** *see also figure 3.18*). This same peptidoglycan structure is present in many gram-positive genera, for example, *Bacillus, Clostridium, Lactobacillus, Corynebacterium, Mycobacterium,* and *Nocardia.* In other gram-positive bacteria, lysine is substituted for diaminopimelic acid in position 3, and the peptide subunits of the glycan chains are cross-linked by interpeptide bridges containing monocarboxylic L-amino acids or glycine, or both (figure 23.4*b*). Many genera including *Staphylococcus, Streptococcus, Micrococcus, Lactobacillus,* and *Leuconostoc* have this type of peptidoglycan. The genus *Streptomyces* and several other actinobacterial genera have replaced *meso*-diaminopimelic acid with L,L-diaminopimelic acid in position 3 and have one glycine residue as the interpeptide bridge. The plant pathogenic corynebacteria provide another example of peptidoglycan variation. In some of these bacteria, the interpeptide bridge connects positions 2 and 4 of the peptide subunits rather than 3 and 4 (figure 23.4*c*). Because the interpeptide bridge connects the carboxyl groups of glutamic acid and alanine, a diamino acid such as ornithine is used in the bridge. Many other variations in peptidoglycan structure are found, including other interbridge structures and large differences in the frequency of cross-linking between glycan chains. Bacilli and most gram-negative bacteria have fewer cross-links between chains than do gram-positive bacteria such as *Staphylococcus aureus* in which almost every muramic acid is cross-linked to another. These structural variants are often characteristic of particular groups and are therefore taxonomically useful.

Bacterial endospores are round or oval intracellular objects that have a complex structure with a spore coat, cortex, and

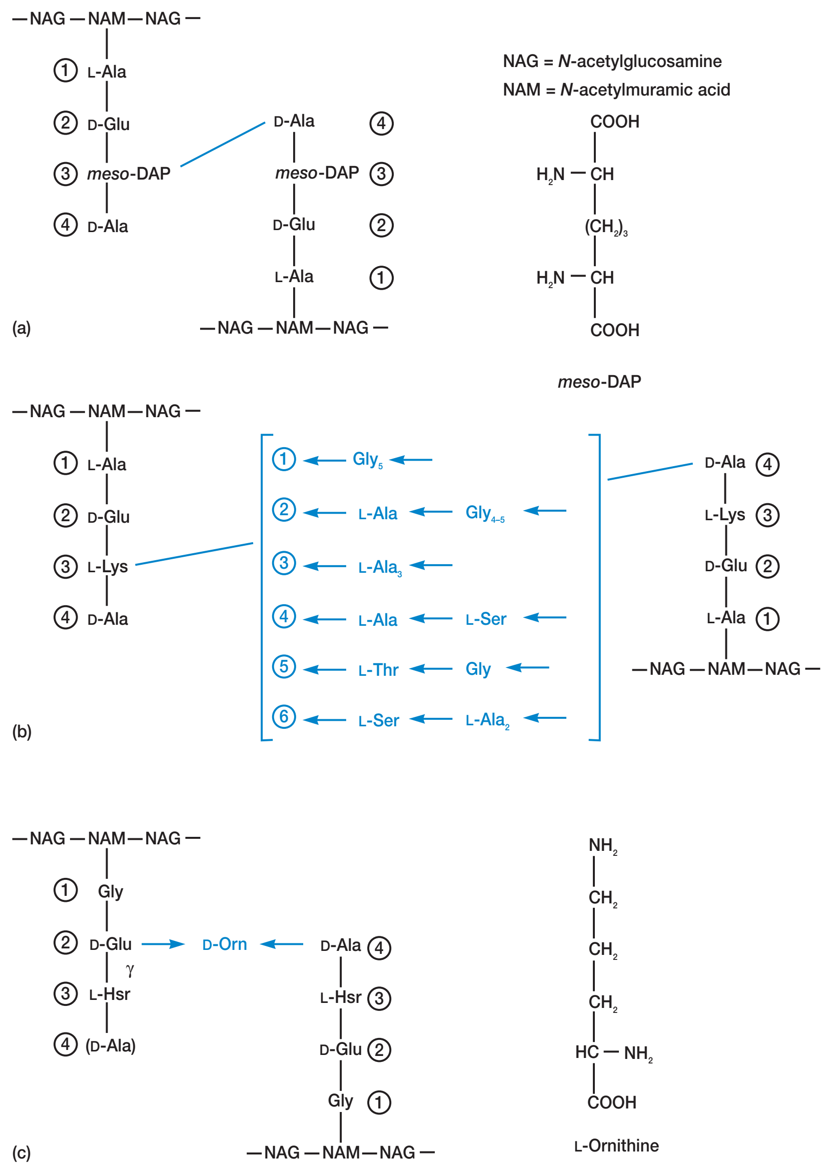


Figure 23.4 Representative Examples of Peptidoglycan Structure. (**a**) The peptidoglycan with a direct cross-linkage between positions 3 and 4 of the peptide subunits, which is present in most gram-negative and many gram-positive bacteria. (**b**) Peptidoglycan with lysin in position 3 and an interpeptide bridge. The bracket contains six typical bridges: (**1**) *Staphylococcus aureus,* (**2**) *S. epidermidis,* (**3**) *Micrococcus roseus* and *Streptococcus thermophilus,* (**4**) *Lactobacillus viridescens,* (**5**) *Streptococcus salvarius,* and (**6**) *Leuconostoc cremoris.* The arrows indicate the polarity of peptide bonds running in the C to N direction. (**c**) An example of the cross-bridge extending between positions 2 and 4 from *Corynebacterium poinsettiae.* The interbridge contains an L-diamino acid like ornithine, and L-homoserine (L-Hsr) is in position 3. The abbreviations and structures of amino acids in the figure are found in appendix I. See text for more detail.

inner spore membrane surrounding the protoplast (**figure 23.5**). They contain dipicolinic acid, are very heat resistant, and can remain dormant and viable for very long periods (**Box 23.1**). In one well-documented experiment, endospores remained viable for about 70 years. A recent research article reports that viable endospores have been recovered from Dominican bees that were encased in 25- to 40-million-year-old amber. If this result is confirmed, endospores from an ancestor of *Bacillus sphaericus* have survived for more than 25 million years! Similar reports have been made subsequently; all these studies will have to be reconfirmed. Usually endospores are observed either in the light microscope after spore staining or by phase-contrast microscopy of unstained cells (*see sections 2.2 and 2.3*). They also can be detected by heating a culture at 70 to 80°C for 10 minutes followed by incubation in the proper growth medium. Because only endospores and some thermophiles would survive such heating, bacterial growth tentatively confirms their presence. Bacterial endospore structure (pp. 68–71)

Although endospore-forming bacteria are distributed widely, they are primarily soil inhabitants. Soil conditions are often extremely variable, and endospores are an obvious advantage in surviving periods of dryness or nutrient deprivation.

The second edition takes a phylogenetic approach primarily based on 16S rRNA data rather than phenotypic similarity. The traditional low G + C gram-positive bacteria are divided into two classes: *Clostridia* (the clostridia and relatives), and *Bacilli* (the bacilli and lactobacilli) (figure 23.1). Each class has endospore-forming bacteria and both rods and cocci. Thus the new group-

Box 23.1

Spores in Space

During the nineteenth-century argument over the question of the evolution of life, the panspermia hypothesis became popular. According to this hypothesis, life did not evolve from inorganic matter on earth but arrived as viable bacterial spores that had escaped from another planet. More recently, the British astronomer Fred Hoyle has revived the hypothesis based on his study of the absorption of radiation by interstellar dust. Hoyle maintains that dust grains were initially viable bacterial cells that have been degraded, and that the beginning of life on earth was due to the arrival of bacterial endospores that had survived their trip through space.

Even more recently Peter Weber and J. Mayo Greenberg from the University of Leiden in the Netherlands have studied the effect of very high vacuum, low temperature, and UV radiation on the survival of *Bacillus subtilis* endospores. Their data suggest that endospores within an interstellar molecular cloud might be able to survive between 4.5 to 45 million years. Molecular clouds move through space at speeds sufficient to transport spores between solar systems in this length of time. Although these results do not prove the panspermia hypothesis, they are consistent with the possibility that bacteria might be able to travel between planets capable of supporting life.

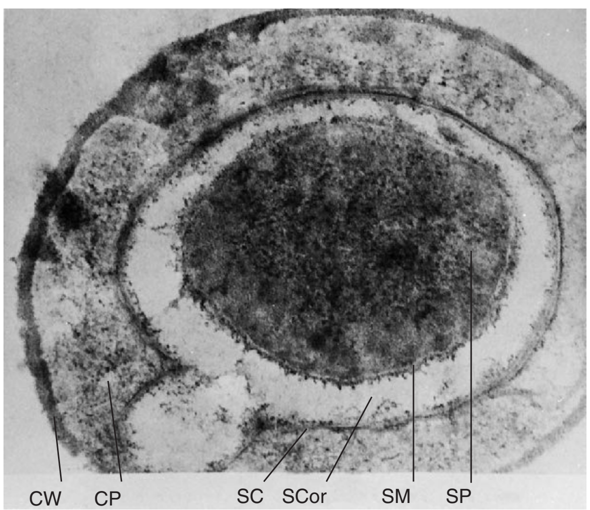


Figure 23.5 **Bacterial Endospores.** A cross section of *Bacillus megaterium* and its endospore within the vegetative cell wall, CW; cell protoplast, CP; spore coat, SC; spore cortex, SCor; spore membrane, SM; and spore protoplast, SP (×120,000).

ings of gram-positive bacteria do not at all resemble those in the first edition. Phylogenetically coherent groups do not necessarily contain bacteria that closely resemble one another phenotypically. The following survey of low G + C gram-positive bacteria will follow the general organization of the second edition, but will refer to the first edition where appropriate.

23.3 Class *Clostridia*

The class *Clostridia* has a very wide variety of gram-positive bacteria distributed into three orders and 11 families. The characteristics of some of the more important genera are summarized in **table 23.2.** Phylogenetic relationships are shown in figure 23.1.

By far the largest genus is *Clostridium.* It includes obligately anaerobic, gram-positive bacteria that form endospores and do not carry out dissimilatory sulfate reduction. The genus contains well over 100 species in several distinct phylogenetic clusters. It is quite likely that the *Clostridium* will be subdivided into several genera in the future.

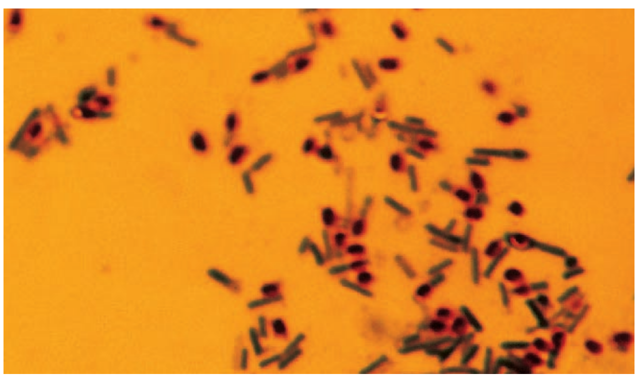
Members of the genus *Clostridium* have great practical impact. Because they are anaerobic and form heat-resistant endospores, they are responsible for many cases of food spoilage, even in canned foods. *C. botulinum* (**figure 23.6*a*;** *see also figure 2.8*d) is the causative agent of botulism (*see section 39.4*). Clostridia often can ferment amino acids to produce ATP by oxidizing one amino acid and using another as an electron acceptor in a process called the Stickland reaction (*see figure 9.11*).

This reaction generates ammonia, hydrogen sulfide, fatty acids, and amines during the anaerobic decomposition of proteins. These products are responsible for many unpleasant odors arising during putrefaction. Several clostridia produce toxins and are major disease agents. *C. tetani* (figure 23.6*b*) is the causative agent of tetanus, and *C. perfringens* (*see figure 2.15*a), of gas gangrene and food poisoning. Clostridia also are industrially valuable; for example, *C. acetobutylicum* is used to manufacture butanol in some countries. Microbiology of food (chapter 41)

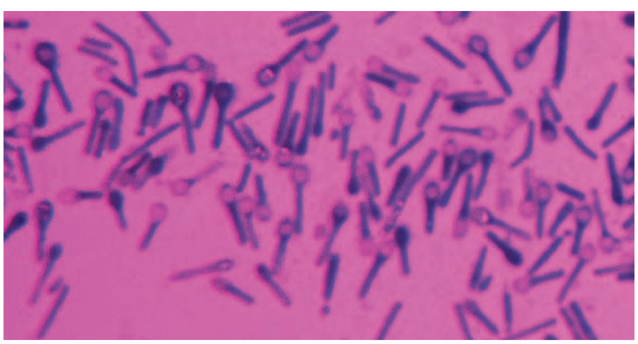
In addition to the genus *Clostridium,* this section has several other genera of interest. *Desulfotomaculum* is an anaerobic, endospore-forming genus that reduces sulfate and sulfite to hydrogen sulfide during anaerobic respiration (**figure 23.7**). Although it stains gram negative, electron microscopic studies have shown that *Desulfotomaculum* has a gram-positive type cell wall despite its staining characteristics and really is a member of the low G + C gram positives. An excellent example of the diversity in this section are the heliobacteria. The heliobacteria, genera *Heliobacterium* and *Heliophilum,* are a group of unusual anaerobic photosynthetic bacteria characterized by the presence of bacteriochlorophyll *g.* They have a photosystem I type reaction center like the green sulfur bacteria, but have no intracytoplasmic photosynthetic membranes; pigments are contained in the plasma membrane. Like *Desulfotomaculum,* they have a gram-positive

Table 23.2 Characteristics of Clostridia and Relatives

Genus	Dimensions (μm) and Morphology	G + C Content (mol%)	Oxygen Requirement	Other Distinctive Characteristics
Clostridium	0.3–2.0 × 1.5–20; rod-shaped, often pleomorphic, nonmotile or peritrichous	22–55	Anaerobic	Does not carry out dissimilatory sulfate reduction; usually chemoorganotrophic, fermentative, and catalase negative; forms oval or spherical endospores
Desulfotomaculum	0.3–1.5 × 3–9; straight or curved rods, peritrichous or polar flagella	37–50	Anaerobic	Reduces sulfate to H_2S, forms subterminal to terminal endospores; stains gram negative but has gram-positive wall, catalase negative
Heliobacterium	1.0 × 4–10; rods that are frequently bent, gliding motility	52–55	Anaerobic	Photoheterotrophic with bacteriochlorophyll *g;* stains gram negative but has gram-positive wall, some form endospores
Veillonella	0.3–0.5; cocci in pairs, short chains, and masses; nonmotile	36–43	Anaerobic	Gram negative; pyruvate and lactate fermented, but not carbohydrates; acetate, propionate, CO_2, and H_2 produced from lactate; parasitic in mouths, intestines, and respiratory tracts of animals



(a)



(b)

Figure 23.6 The Clostridia. (**a**) *C. botulinum,* spores elliptical and subterminal, cells slightly swollen (×1,100). (**b**) *C. tetani,* spores round and terminal (×1,100).

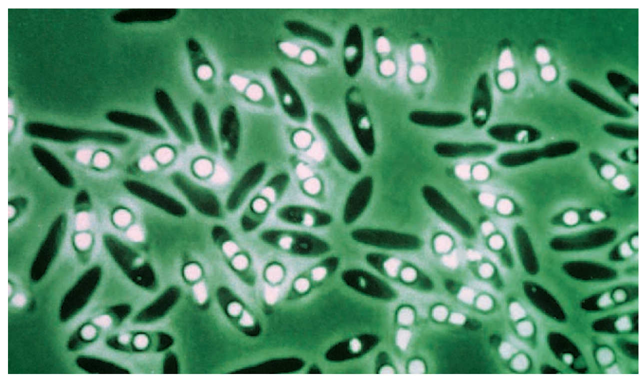


Figure 23.7 ***Desulfotomaculum.*** *Desulfotomaculum acetoxidans* with spores; phase contrast (×2,000).

type cell wall with lower than normal peptidoglycan content, and they stain gram negative. Some heliobacteria form endospores.

The genus *Veillonella* provides another good example of the difference between the first and second editions with respect to placement of genera. The second edition of *Bergey's Manual* locates the genus in the low G + C gram positives. In the first edition *Veillonella* is placed with other anaerobic gram-negative cocci in section 8 and assigned to the family *Veillonellaceae.* The family *Veillonellaceae* contains anaerobic, chemoheterotrophic cocci ranging in diameter from about

0.3 to 2.5 μm. Usually they are diplococci (often with their adjacent sides flattened), but they may exist as single cells, clusters, or chains. All have complex nutritional requirements and ferment substances such as carbohydrates, lactate and other organic acids, and amino acids to produce gas (CO_2 and often H_2) plus a mixture of volatile fatty acids. They are parasites of homeothermic (warm-blooded) animals.

Like many groups of anaerobic bacteria, members of this family have not been thoroughly studied. Some species are part of the normal biota of the mouth, the gastrointestinal tract, and the urogenital tract of humans and other animals. For example, *Veillonella* is plentiful on the tongue surface and dental plaque of humans (*see figure 39.25*) and it can be isolated about 10 to 20% of the time from the vagina. *Veillonella* is unusual in growing well on organic acids such as lactate, pyruvate, and malate while normally being unable to ferment glucose and other carbohydrates. It is well adapted to the oral environment because it can use the lactic acid produced from carbohydrates by the streptococci and other oral bacteria. Gram-negative, anaerobic cocci are found in infections of the head, lungs, and the female genital tract, but their precise role in such infections is unclear.

1. Briefly discuss the ways in which low G + C bacteria are treated differently in the first and second editions of *Bergey's Manual.*
2. Describe, in a diagram, the chemical composition and structure of the peptidoglycan found in gram-negative bacteria and many gram-positive genera.
3. Briefly discuss the ways in which three other peptidoglycan types differ from the gram-negative peptidoglycan.
4. How do bacilli and most gram-negative bacteria differ from gram-positive bacteria such as *S. aureus* with respect to cross-linking frequency?
5. What is a bacterial endospore? Give its most important properties and two ways to demonstrate its presence.
6. Give the general characteristics of *Clostridium, Desulfotomaculum,* the heliobacteria, and *Veillonella.* Briefly discuss why each is interesting or of practical importance.

23.4 Class *Bacilli*

The second edition of *Bergey's Manual* gathers a large variety of gram-positive bacteria into one class, *Bacilli,* and two orders, *Bacillales* and *Lactobacillales.* Within these orders are 16 families and around 59 gram-positive genera representing cocci, endospore-forming rods and cocci, and nonsporing rods. The biology of some members of the order *Bacillales* will be described first; then important representatives of the order *Lactobacillales* will be considered. The phylogenetic relationships between some of these organisms are pictured in figure 23.1, and the characteristics of selected genera are summarized in **table 23.3.**

Order Bacillales

The genus *Bacillus,* family *Bacillaceae,* is the largest in the order (**figure 23.8;** *see also figures 3.1*c *and 3.11*). The genus contains gram-positive, endospore-forming, chemoheterotrophic rods that are usually motile and peritrichously flagellated. It is aerobic, or sometimes facultative, and catalase positive. In the first edition the genus is clearly diverse phenotypically and genotypically. More recently, rRNA sequence data have been used to divide the genus *Bacillus* into at least five separate lines. Several species already have been moved to two new genera. The genus *Alicyclobacillus* contains acidophilic, sporing, gram-positive or gram-variable rods that have ω-alicyclic fatty acids with 6- or 7-carbon rings in their membranes. Members are aerobic or facultative and have a G + C content of about 51 to 60%. The second new genus is *Paenibacillus* [Latin *paene,* almost, and bacillus]. This genus contains gram-positive rods from rRNA group 3 that are facultative, motile by peritrichous flagella, have ellipsoidal endospores and swollen sporangia, produce acid and sometimes gas from glucose and various sugars, and have a G + C content of 40 to 54%. Some examples of organisms that were formerly in the genus *Bacillus* are *Paenibacillus alvei, P. macerans,* and *P. polymyxa.*

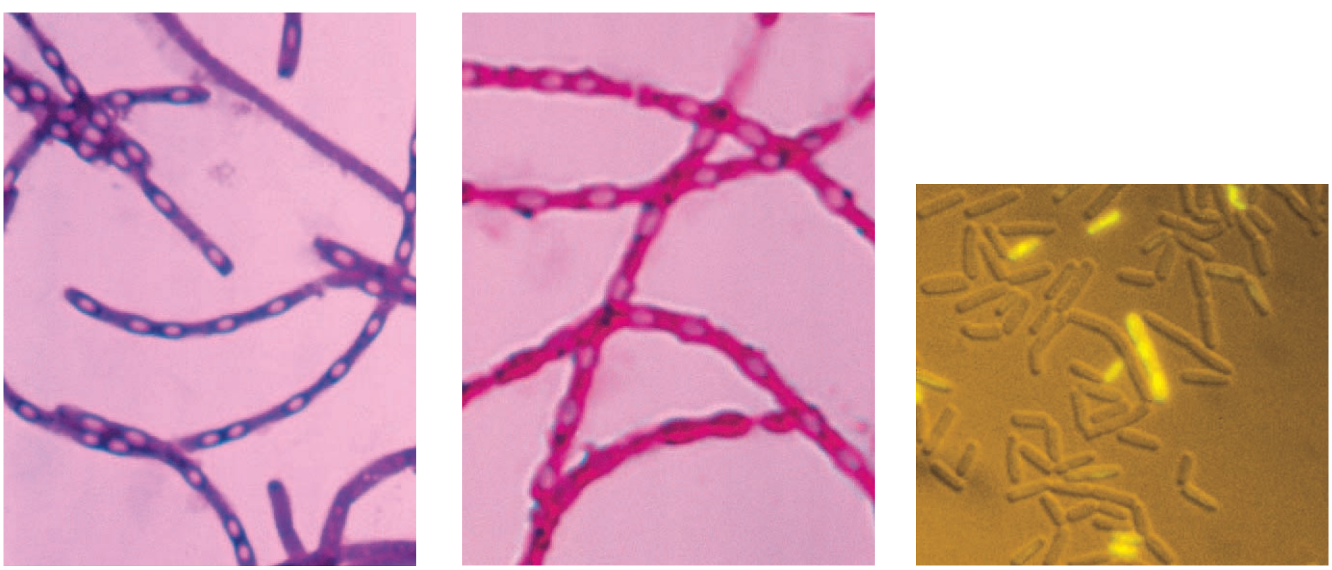
The 4.2 million base pair genome of *Bacillus subtilis,* the type species of the genus *Bacillus,* has been sequenced. Several families of genes have been expanded by gene duplication; the largest such family encodes ABC transporters (*see p. 101*), which are the most frequent type of protein in *B. subtilis.* The genome contains genes for the catabolism of many diverse carbon sources, and for protein secretion and antibiotic synthesis. There are at least 10 integrated prophages or remnants of prophages.

Many species of *Bacillus* are of considerable importance. For example, members of the genus *Bacillus* produce the antibiotics bacitracin, gramicidin, and polymyxin. *B. cereus* (figure 23.8*c*) causes some forms of food poisoning and can infect humans. *B. anthracis* is the causative agent of the disease anthrax, which can affect both farm animals and humans (*see section 39.3*). Several species are used as insecticides. For example, *B. thuringiensis* and *B. sphaericus* form a solid protein crystal, the **parasporal body,** next to their spores during endospore formation (**figure 23.9**). The *B. thuringiensis* parasporal body contains protein toxins that will kill over 100 species of moths by dissolving in the alkaline gut contents of caterpillars and destroying their gut epithelium (*see section 42.6*). The solubilized toxin proteins are cleaved by midgut proteases to smaller toxic polypeptides that attack the epithelial cells. The alkaline gut contents escape into the blood, causing paralysis and death. One of these toxins has been isolated and shown to form pores in the plasma membrane. These channels allow monovalent cations such as potassium to pass through. Most of these *B. thuringiensis* toxin genes are carried on large plasmids. The *B. sphaericus* parasporal body contains proteins toxic for mosquito larvae and may be useful in controlling the mosquitos that carry the malaria parasite *Plasmodium.* Microbial insecticides (chapter 42)

The genus *Thermoactinomyces* is placed with the actinomycetes in the first edition of *Bergey's Manual.* The second edition moves the genus to the family *Thermoactinomycetaceae* in

Table 23.3 Characteristics of Members of the Class *Bacilli*

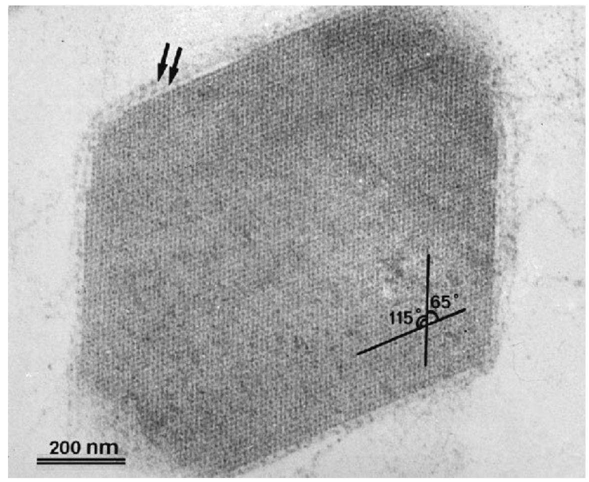
Genus	Dimensions (µm) and Morphology	G + C Content (mol%)	Oxygen Requirement	Other Distinctive Characteristics
Bacillus	0.5–2.5 × 1.2–10; straight rods, peritrichous	32–69	Aerobic or facultative	Forms endospores; catalase positive; chemoorganotrophic
Caryophanon	1.5–3.0 × 10–20; multicellular rods with rounded ends, peritrichous	41–46	Aerobic	Acetate only major carbon source; catalase positive; trichome cells have greater width than length, trichomes can be in short chains
Enterococcus	0.6–2.0 × 0.6–2.5; spherical or ovoid cells in pairs or short chains, nonsporing, sometimes motile	34–42	Facultative	Ferments carbohydrates to lactate with no gas; complex nutritional requirements; catalase negative; occurs widely, particularly in fecal material
Lactobacillus	0.5–1.2 × 1.0–10; usually long, regular rods, nonsporing, rarely motile	32–53	Facultative or microaerophilic	Fermentative, at least half the end-product is lactate; requires rich, complex media; catalase and cytochrome negative
Lactococcus	0.5–1.2 × 0.5–1.5; spherical or ovoid cells in pairs or short chains, nonsporing, nonmotile	38–40	Facultative	Chemoorganotrophic with fermentative metabolism; lactate and no gas produced; catalase negative; complex nutritional requirements; in dairy and plant products
Leuconostoc	0.5–0.7 × 0.7–1.2; cells spherical or ovoid, in pairs or chains; nonmotile and nonsporing	38–44	Facultative	Requires fermentable carbohydrate and nutritionally rich medium for growth; fermentation produces lactate, ethanol, and gas; catalase and cytochrome negative
Staphylococcus	0.9–1.3; spherical cells occurring singly and in irregular clusters, nonmotile and nonsporing	30–39	Facultative	Chemoorganotrophic with both respiratory and fermentative metabolism, usually catalase positive, associated with skin and mucous membranes of vertebrates
Streptococcus	0.5–2.0; spherical or ovoid cells in pairs or chains, nonmotile and nonsporing	34–46	Facultative	Fermentative, producing mainly lactate and no gas; catalase negative; commonly attack red blood cells (α- or β-hemolysis); complex nutritional requirements; commensals or parasites on animals
Thermoactinomyces	0.4–1.0 in diameter; branched, septate mycelium typical of actinomycetes	52.0–54.8	Aerobic	Usually thermophilic; true endospores form singly on hyphae; numerous in decaying hay, vegetable matter, and compost



(a) (b) (c)

Figure 23.8 ***Bacillus*** (**a**) *B. anthracis,* spores elliptical and central (×1,600). (**b**) *B. subtilis,* spores elliptical and central. (**c**) *B. cereus* stained with SYTOX Green nucleic acid stain and viewed by epifluorescence and differential interference contrast microscopy. The cells that glow green are dead.

(a)



(b)

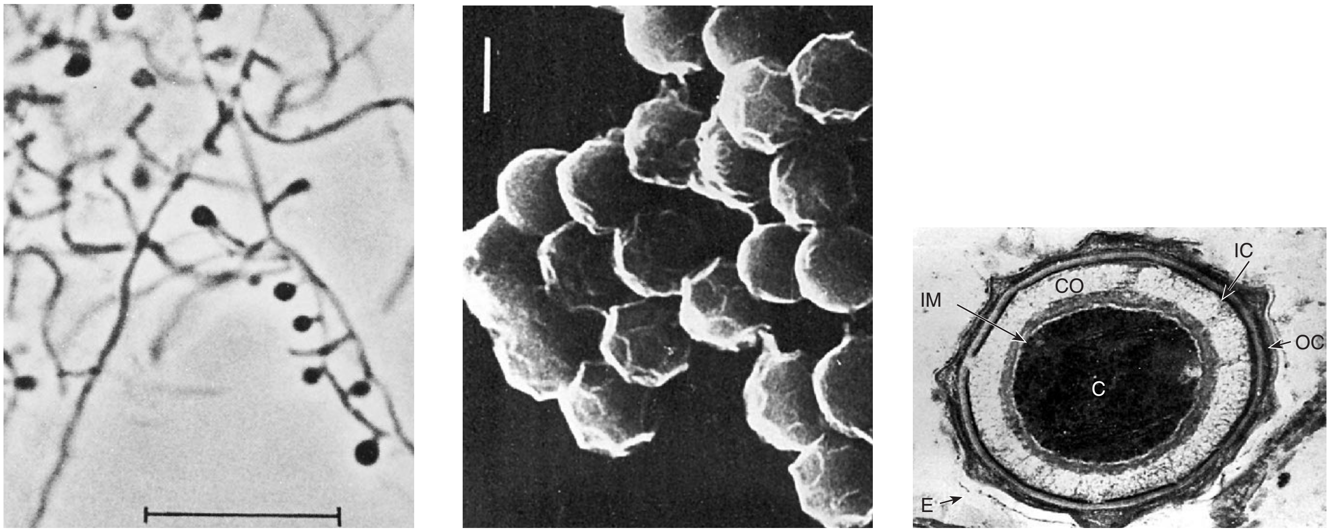
Figure 23.9 The Parasporal Body. (**a**) An electron micrograph of a *B. sphaericus* sporulating cell containing a parasporal body just beneath the endospore. Bar = 400 nm. (**b**) The crystalline parasporal body at a higher magnification. The crystal is surrounded by a two-layered envelope (arrows). Bar = 200 nm.

the order *Bacillales.* This genus is thermophilic and grows between 45 and 60°C; it forms single spores on both its aerial and substrate mycelia (**figure 23.10**). Its G + C content is lower than that of the actinobacteria (52 to 55 mol%), and its 16S rRNA sequence suggests a relationship with the genus *Bacillus. Thermoactinomyces* is commonly found in damp haystacks, compost piles, and other high-temperature habitats. *Thermoactinomyces* spores (figure 23.10*b,c*) are true endospores and very heat resistant; they can survive at 90°C for 30 minutes. They are formed within hyphae and appear to have typical endospore structure, including the presence of calcium and dipicolinic acid. *Thermoactinomyces vulgaris* (figure 23.10*a*) from haystacks, grain storage silos, and compost piles is a causative agent of farmer's lung, an allergic disease of the respiratory system in agricultural workers. Recently spores from *Thermoactinomyces vulgaris* were recovered from the mud of a Minnesota lake and found to be viable after about 7,500 years of dormancy.

One of the more unusual genera in this order is *Caryophanon.* This gram-positive bacterium is strictly aerobic, catalase positive, and motile by peritrichous flagella. Its normal habitat is cow dung. *Caryophanon* morphology is distinctive. Individual cells are disk-shaped (1.5 to 2.0 μm wide by 0.5 to 1.0 μm long) and joined to form rods about 10 to 20 μm long (**figure 23.11**).

The family *Staphylococcaceae* contains four genera, the most important of which is the genus *Staphylococcus.* Members of this genus are facultatively anaerobic, nonmotile, gram-positive cocci that usually form irregular clusters (**figure 23.12,** *see also figure 3.1*a). They are catalase positive, oxidase negative, ferment glucose, and have teichoic acid in their cell walls. Staphylococci are normally associated with the skin, skin glands, and mucous membranes of warm-blooded animals.

Staphylococci are responsible for many human diseases. *S. epidermidis* is a common skin resident that is sometimes responsible for endocarditis and infections of patients with lowered resistance (e.g., wound infections, surgical infections, urinary tract infections). *S. aureus* is the most important human staphylococcal pathogen and causes boils, abscesses, wound infections, pneumonia, toxic shock syndrome, and other diseases. Recently strains of multiple drug resistant *S. aureus* have appeared and proven very difficult to treat medically (*see section 35.7*).



(a) (b) (c)

Figure 23.10 ***Thermoactinomyces.*** (**a**) *Thermoactinomyces vulgaris* aerial mycelium with developing endospores at tips of hyphae. Bar = 10 μm. (**b**) Scanning electron micrograph of *T. sacchari* spores. Bar = 1 μm. (**c**) Thin section of a *T. sacchari* endospore. Bar = 0.1 μm. E, exosporium; OC, outer spore coat; IC, inner spore coat; CO, cortex; IM, inner forespore membrane; C, core.

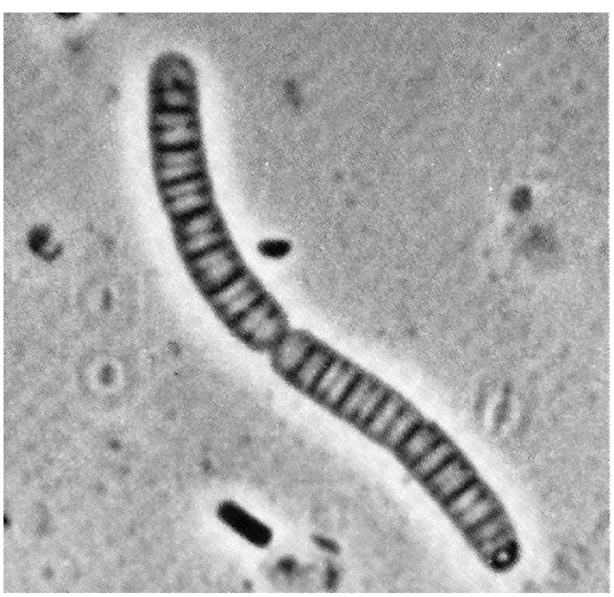
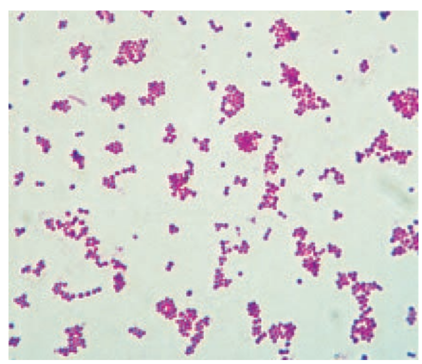
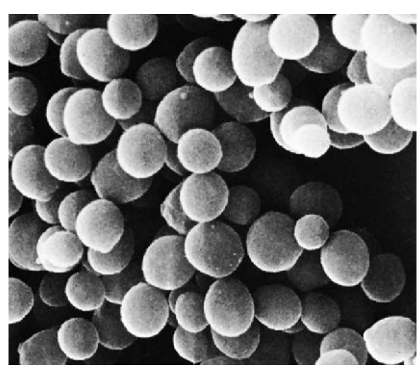


Figure 23.11 ***Caryophanon*** **Morphology.** *Caryophanon latum* in a trichome chain. Note the disk-shaped cells stacked side by side; phase contrast (×3,450).



(a)



(b)

Figure 23.12 ***Staphylococcus.*** (**a**) *Staphylococcus aureus,* Gram-stained smear (×1,500). (**b**) *Staphylococci* arranged like a cluster of grapes; scanning electron micrograph (×34,000).

(a) (b) (c)

Figure 23.13 ***Lactobacillus.*** (**a**) *L. acidophilus* (×1,000). (**b**) *L. lactis.* Gram stain (×1,000). (**c**) *L. bulgaricus;* phase contrast (×600).

It also is a major cause of food poisoning as the following case illustrates. In March 1986 an outbreak of acute gastrointestinal illness occurred following a buffet supper served to 855 people at a New Mexico country club. At least 67 people were ill with diarrhea, nausea, or vomiting, and 24 required emergency medical treatment or hospitalization. The problem was *S. aureus* growing in the turkey and dressing served at the buffet. One or more of the food handlers carried *S. aureus* and had contaminated the food. Because the turkey was cooled for 3 hours after cooking, there was sufficient time for the bacterium to grow and produce toxins (*see sections 39.4 and 41.4*). This case is not uncommon. Turkey accounts for 10 to 21% of all bacterial food poisoning cases in which the source of poisoning is known, and it must be cooked and handled carefully.

Unlike other common staphylococci, *S. aureus* produces the enzyme **coagulase,** which causes blood plasma to clot. Growth and hemolysis patterns on blood agar are also useful in identifying these staphylococci (figure 23.17). Recently the structure of staphylococcal α-hemolysin has been determined. The toxin lyses a cell by forming solvent-filled channels in its plasma membrane. Water-soluble toxin monomers bind to the cell surface and associate with each other to form pores. The hydrophilic channels then allow free passage of water, ions, and small molecules. *S. aureus* usually grows on the nasal membranes and skin; it also is found in the gastrointestinal and urinary tracts of warm-blooded animals. Staphylococcal diseases (pp. 919–23)

Listeria, family *Listeriaceae,* is another medically important genus in this order. The genus contains short rods that are aerobic or facultative, catalase positive, and motile by peritrichous flagella. It is widely distributed in nature, particularly in decaying matter. *Listeria monocytogenes* is a pathogen of humans and other animals and causes listeriosis, an important food infection (*p. 931*).

Order *Lactobacillales*

Many members of the order *Lactobacillales* produce lactic acid as their major or sole fermentation product and are sometimes collectively called **lactic acid bacteria.** *Streptococcus, Enterococcus, Lactococcus, Lactobacillus,* and *Leuconostoc* are all members of this group. Lactic acid bacteria are nonsporing and usually nonmotile. They lack cytochromes and obtain energy by substrate-level phosphorylation rather than by electron transport and oxidative phosphorylation. They normally depend on sugar fermentation for energy. Nutritionally they are fastidious and many vitamins, amino acids, purines, and pyrimidines must be supplied because of their limited biosynthetic capabilities. Lactic acid bacteria usually are categorized as facultative anaerobes, but some classify them as aerotolerant anaerobes. Oxygen relationships (pp. 127–29).

The largest genus in this order is *Lactobacillus* with almost 80 species. *Lactobacillus* contains nonsporing rods and sometimes coccobacilli that lack catalase and cytochromes, are usually facultative or microaerophilic, produce lactic acid as their main or sole fermentation product, and have complex nutritional requirements (**figure 23.13**). Lactobacilli carry out either a homolactic fermentation using the Embden-Meyerhof pathway or a heterolactic fermentation with the pentose phosphate pathway (*see section 9.2*). They grow optimally under slightly acidic conditions, when the pH is between 4.5 to 6.4. The genus is found on plant surfaces and in dairy products, meat, water, sewage, beer, fruits, and many other materials. Lactobacilli also are part of the normal flora of the human body in the mouth, intestinal tract, and vagina. They usually are not pathogenic.

Lactobacillus is indispensable to the food and dairy industry (*see chapter 41*). Lactobacilli are used in the production of fermented vegetable foods (sauerkraut, pickles, silage), beverages (beer, wine, juices), sour dough, Swiss cheese and other hard cheeses, yogurt, and sausage. Yogurt is probably the most popular fermented milk product in the United States and is produced

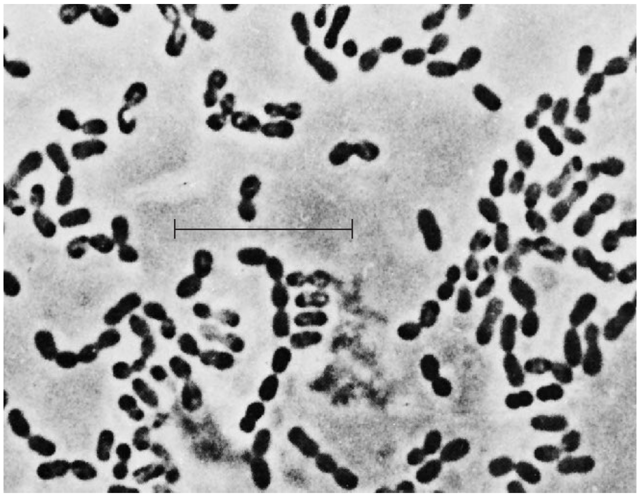


Figure 23.14 ***Leuconostoc.*** *Leuconostoc mesenteroides;* phase-contrast micrograph. Bar = 10 μm.

commercially and by individuals using yogurt-making kits. In commercial production, nonfat or low-fat milk is pasteurized, cooled to 43°C or lower, inoculated with *Streptococcus thermophilus* and *Lactobacillus bulgaricus. S. thermophilus* grows more rapidly at first and renders the milk anaerobic and weakly acidic. *L. bulgaricus* then acidifies the milk even more. Acting together, the two species ferment almost all the lactose to lactic acid and flavor the yogurt with diacetyl (*S. thermophilus*) and acetaldehyde (*L. bulgaricus*). Fruits or fruit flavors to be added are pasteurized separately and then combined with the yogurt.

Lactobacilli also create problems. They sometimes are responsible for spoilage of beer, milk, and meat because their metabolic end products contribute undesirable flavors and odors.

Leuconostoc, family *Leuconostocaceae,* contains facultative gram-positive cocci, which may be elongated or elliptical and arranged in pairs or chains (**figure 23.14**). Leuconostocs lack catalase and cytochromes and carry out **heterolactic fermentation** (*see section 9.3*) by converting glucose to D-lactate and ethanol or acetic acid by means of the phosphoketolase pathway (**figure 23.15**). They can be isolated from plants, silage, and milk. The genus is used in wine production, in the fermentation of vegetables such as cabbage (sauerkraut; *see figure 41.23*) and cucumbers (pickles), and in the manufacture of buttermilk, butter, and cheese. *L. mesenteroides* synthesizes dextrans from sucrose and is important in industrial dextran production. *Leuconostoc* species are involved in food spoilage and tolerate high sugar concentrations so well that they grow in syrup and are a major problem in sugar refineries.

Several important genera have chemoheterotrophic, mesophilic, nonsporing, gram-positive cocci and are placed in the families *Enterococcaceae (Enterococcus)* and *Streptococcaceae (Streptococcus, Lactococcus).* In practice, they are often distinguished primarily based on phenotypic properties such as oxygen relationships, cell arrangement, the presence of catalase and cytochromes, and peptidoglycan structure. The most important of these genera is *Streptococcus,* which is facultatively anaerobic and catalase negative. The streptococci and their close relatives, the enterococci and lactococci, occur in pairs or chains when grown in liquid media (**figure 23.16;** *see also figure 3.1*b), do not form endospores, and usually are nonmotile. They are all chemoheterotrophs that ferment sugars with lactic acid, but no gas, as the major product—that is, they carry out homolactic fermentation (*see section 9.3*). A few species are anaerobic rather that facultative.

The genus *Streptococcus* is large and complex. The first edition of *Bergey's Manual* lists 38 species clustered in four groups; pyogenic streptococci, oral streptococci, anaerobic streptococci, and other streptococci. Many bacteria that were placed within the

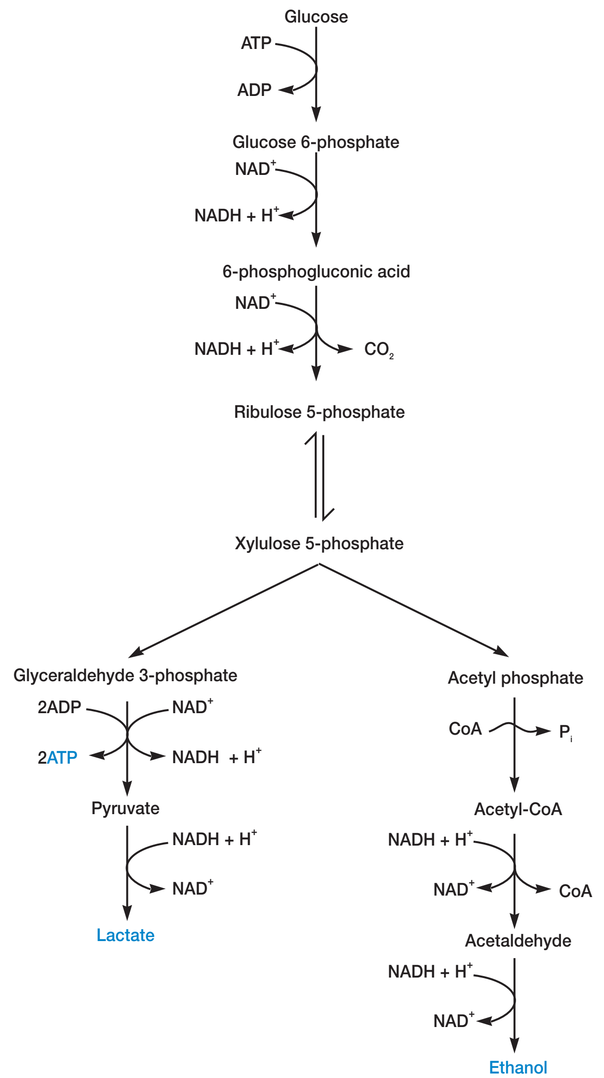
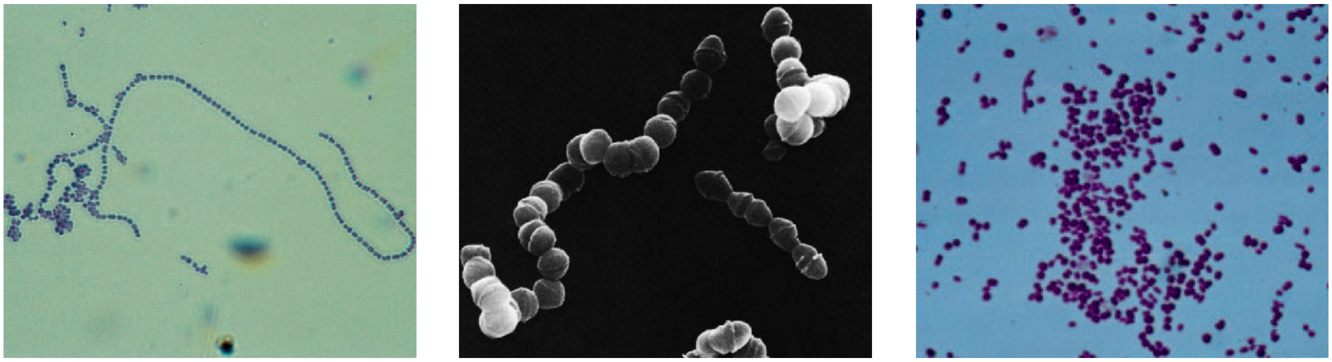


Figure 23.15 **Heterolactic Fermentation and the Phosphoketolase Pathway.** The phosphoketolase pathway converts glucose to lactate, ethanol, and CO_2.



(a) (b) (c)

Figure 23.16 Streptococcus. (**a**) *Streptococcus pyogenes* (×900). (**b**) Scanning electron micrograph of *Streptococcus* (×33,000). (**c**) *Streptococcus pneumoniae* (×900).

Table 23.4 Classification of the Streptococci, Enterococci, and Lactococci

Characteristics	*Streptococcus*	*Enterococcus*	*Lactococcus*
Predominant arrangement (most common first)	Chains, pairs	Pairs, chains	Pairs, short chains
Capsule/slime layer	+	–	–
Habitat	Mouth, respiratory tract	Gastrointestinal tract	Dairy products
Growth at 45°C	Variable	+	–
Growth at 10°C	Variable	Usually +	+
Growth at 6.5% NaCl broth	Variable	+	–
Growth at pH 9.6	Variable	+	–
Hemolysis	Usually β (pyogenic) or α (oral)	α, ß, –	Usually –
Serological group (Lancefield)	Variable (A–O)	Usually D	Usually N
Mol% G + C (normal range)	34–46	34–42	38–40
Representative species	Pyogenic streptococci *S. agalactiae* *S. pyogenes* *S. equi* *S. dysgalactiae* Oral streptococci *S. gordonii* *S. salvarius* *S. sanguis* *S. oralis* *S. pneumoniae* *S. mitis* *S. mutans* Other streptococci *S. bovis* *S. thermophilus*	*E. faecalis* *E. faecium* *E. avium* *E. durans* *E. gallinarum*	*L. lactis* *L. raffinolactis* *L. plantarum*

genus have been moved to two new genera, *Enterococcus* and *Lactococcus*. Undoubtedly the second edition will still have many streptococcal species. Some major characteristics of these three closely related genera are summarized in **table 23.4. Table 23.5** lists a few properties of selected genera.

Many characteristics are used to identify these cocci. One of their most important taxonomic characteristics is the ability to lyse erythrocytes when growing on blood agar, an agar medium containing 5% sheep or horse blood (**figure 23.17**). In **α-hemolysis** a 1 to 3 mm greenish zone of incomplete hemolysis forms around the